Data as a Networked Asset*

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What is the value of a firm's data? Because data is non-rival, its value extends beyond the originating firm. We uncover inter-firm data conduits embedded in mobile apps, constructing the first measure of data-sharing linkages. Data sharing induces comovement in firms' operational, financial, and stock-market performance, propagates shocks such as cyberattacks, and generates herding in product design. These patterns weaken after Apple's privacy policy disrupts data sharing. Embedding these linkages in a dynamic model, we derive a formula that values data as a networked asset generating cash flows at both originating and connected firms, and identify systemically important firms.

Keywords: Data, data sharing, network, systemic importance, customer capital, intangible capital, privacy, cyberattack, SDKs, digital economy

JEL Codes: D62, D85, E22, E23, G12, G14, L51, L86, O33

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1 Introduction

Data has emerged as a highly productive asset. According to the Bureau of Economic Analysis (BEA), US investment in data assets reached $875 billion in 2024, on par with R&D and software (Santiago Calderón, 2026). Data has several distinct attributes. It is non-rival: one firm's use of data does not diminish its availability for others, allowing data to be used by multiple firms simultaneously (Jones and Tonetti, 2020). Data also exhibits externalities: information collected by one firm on its customers can benefit other firms by revealing consumer preferences, underlying economic forces, and serving as training data for prediction models (Choi, Jeon, and Kim, 2019; Ichihashi, 2021; Acemoglu, Makhdoumi, Malekian, and Ozdaglar, 2022). Data non-rivalry, combined with externality, expands the uses of a firm's data, often transcending industry boundaries.

In this paper, we explore several questions: How is data collected by one firm shared with others? What is the scope and network structure of data sharing? What are the economic implications, and in particular, how data sharing affects the value of data and firms' decision-making? What are the impact and unintended consequences of data-sharing regulations? Finally, we investigate which firms are systemically important in the data economy. These firms likely contribute significant amounts of data, but their systemic importance cannot be simply determined by size alone; the network topology of inter-firm data flows plays a critical role.

We trace inter-firm data flows originating from mobile applications (apps), which have become the primary channels for data collection in the economy.[1] Our sample contains 1,031 app-owning public firms that account for more than 60% of the total assets of Compustat firms. These apps collect data and transmit it to Software Development Kits (SDKs) that specialize in data analytics, which we refer to as "data-sharing SDKs". They merge data collected from different firms' apps about the same consumer, creating profiles that cover consumers' behavior across various categories. By sharing customer data with these SDKs, firms in return gain access to consumer insights that combine not only their own data on consumers but also that of other firms.

The data infrastructure built on these SDKs is economically significant: among firms whose revenues come from data-reliant segments, the intensity of data-sharing SDK usage positively predicts firms' revenues as strongly as R&D intensity.[2] In addition, when SDK providers release major privacy-driven updates that restrict data collection and sharing, the data-reliant firms exposed to these SDKs through their apps experience a significantly negative impact on operational and stock-market performances. Finally, firms' regulatory filings, earnings calls, and other public

[1] Mobile apps host roughly 60–80% of the time consumers spend on digital media, and they collect data that websites cannot, including precise location, device sensors, and identifiers that link a consumer's activity across apps (see Section 2 and Internet Appendix C.4). Over 60% of iOS apps track users across apps, websites, and offline stores (Bian, Ma, and Tang, 2021), and nearly 90% of Android apps collect user data and enable sharing with third parties (Binns, Lyngs, Van Kleek, Zhao, Libert, and Shadbolt, 2018).

[2] In Section 2, we introduce our classification of Compustat business segments by the degree of data reliance.

records confirm the economic significance of SDK-mediated data sharing across industries.

Using granular information on app-level SDK installations, we construct measures of data connectedness based on firms' overlap in the usage of data-sharing SDKs. Specifically, two firms are considered connected when a common set of such SDKs is installed on their apps, with the degree of connectedness increasing in the SDK overlap. This approach enables us to construct the first measure of data-sharing conduits between firms and to map out the entire network structure.

The signals that data-connected firms receive from the SDKs contain data from one another. When one firm gains customers, it collects more data, which is shared through the SDKs with its connected peers, making them more informed about their customers as well. Being able to profile customers more effectively in turn allows these connected firms to acquire customers and improve profitability. Our empirical analysis confirms this dynamic. We find that data sharing induces comovements in firms' operational performances. The economic magnitude is large, more than doubling that of the product similarity measure from Hoberg and Phillips (2016) in explaining the comovement in operational performances. Additionally, data-connected firms exhibit stock-return correlations that cannot be explained by standard asset-pricing factors or other common exposures, such as product similarity, supply-chain linkages, or common analyst coverage. The strength of comovement depends intuitively on the network topology: firms that are indirectly connected through common data-sharing peers comove over and above the effect of any direct connectedness, and comovement is strongest where the data-sharing network is locally dense.

Our paper is the first to uncover this new form of economic linkage. We provide direct evidence on a mechanism behind firms' performance comovement induced by this data linkage. A central channel through which shared consumer data translates into firm performance is targeted advertising. We document that data-connected firms comove in their digital advertising spending, in how they allocate advertisements across apps and web channels, and in advertising effectiveness—what advertising expenses achieve in customer acquisition. Two features distinguish data sharing from the alternative explanation that common shocks move firms' overall advertising spending together. First, the comovement is absent in non-digital advertising, which relies less on SDK-mediated data. Second, as we discuss next, the advertising comovement of data-connected firms is weakened by Apple's privacy policy that disrupts inter-firm data sharing.

Recently, with rising consumer awareness of privacy concerns, data privacy regulations, such as GDPR in Europe and CCPA in the U.S. (California), have imposed strict rules on data collection and sharing. We find that one unintended consequence of these policies is a weakening of performance comovement between data-connected firms. In a difference-in-differences framework, we explore the introduction of Apple's user privacy framework, App Tracking Transparency (ATT). ATT allows app users to block identity tracking, thus making it difficult for the SDKs to merge data from different apps. By examining the impact of ATT, we not only shed light on the consequences

of privacy initiatives from the private sector but also validate that firms' overlap in data-sharing SDKs is fundamentally about linkages in data. Importantly, ATT does not influence how the other forms of firms' overlap (e.g., product similarity) contribute to their performance comovements.

The comovement in firms' performances emerges ultimately from the propagation of shocks from one firm to another. Using cyberattacks as our empirical setting, we provide direct evidence of how data sharing facilitates shock propagation. Intuitively, cyberattacks reduce firms' data stock and access, impair their ability to collect data, and more broadly, diminish operational efficiency that underpins customer engagement and data acquisition. We find that the focal firm's data-connected peers experience significantly greater deterioration in operational and stock-market performances than those that are not data-connected. These findings underscore the importance of examining firms' interconnectedness in the data economy for tracing the ripple effects of shocks.

The empirical findings raise a measurement question at the heart of the data economy: what is the value of a firm's data? Because data is shared, this value is not confined to the firm where the data originates; it includes the present value of the cash flows that the data generates at the connected firms. Quantifying this value requires accounting not only for direct spillover effects but also for higher-order spillovers. Furthermore, data spillovers have persistent effects over time: data influences firms' ability to engage with customers, and customer engagement, in turn, drives further data accumulation, resulting in self-perpetuating growth that is interconnected across firms.

To value data as a networked asset, we develop a dynamic network model where data plays the role of productive capital. The model takes as input the empirical network of data-sharing conduits based on firms' SDK overlap, and simulations of the calibrated model replicate the data-induced comovement in firm performances and the ATT effects. In the model, each firm seeks to forecast customer preferences, with forecasting precision depending on both its own data and data from connected peers.[3] More accurate forecasts enhance customer engagement, which in turn boosts customer capital, product demand, and cash flow.[4] Customer engagement also creates new data, improving forecasting precision for both the firm itself and its connected peers.

Building on the dynamic model, we derive a closed-form formula for data valuation. The value of a firm's data is the present value of the cash flows that the data generates both at the firm where it originates and at the other firms that receive them through the data-sharing network. Our valuation framework embeds the entire network structure of data sharing, the propagation of data across the network through higher-order externalities, and the self-perpetuating data growth

[3] In Buera and Oberfield (2020), there is randomness in a firm's adoption of insights from other industries. In our model, a firm's information on its customers depends on a weighted sum of other firms' data, with the weight corresponding to the empirically measured data-sharing conduits. While the amount of data transmitted via the conduits varies over time and carries randomness (shocks), the data-sharing conduits are deterministic.

[4] Customer capital accumulates, for example, as data improves firm-consumer match (Gourio and Rudanko, 2014b). Hsieh and Rossi-Hansberg (2023) point out that intangible capital lowers the cost of entering new markets and acquiring customers. Data, as a particular form of intangibles, serves a similar purpose as it facilitates customer acquisition.

(Farboodi, Mihet, Philippon, and Veldkamp, 2019; Farboodi and Veldkamp, 2025) that feeds on cross-firm data propagation over multiple time horizons. In the absence of data sharing, our formula collapses to the standard Gordon growth model for data valuation.

We use our calibrated model to compute the value of data as a networked asset. Before ATT, Meta's data is valued at around $780 billion, and all firms' data in our sample approximately $1.984 trillion, which is 8.4% of US GDP in 2021. For comparison, Bhutani, Ordoñez, and Veldkamp (2026) estimate the value of data that US firms acquire in a year at about 1.5% of GDP; our number is larger because it capitalizes the full data stock rather than annual data flow and reflects the value of spillover effects on the data-sharing network. ATT weakened inter-firm data sharing, reducing data value. In terms of network externality, the ratio of the total value of Meta's data to its own value decreased from 3 to 1.15, and that of Alphabet from 6 to 1.33. In summary, our framework quantifies the value of a firm's data for the whole data economy, its network externality and systemic importance, and how such externality varies with privacy policies.

Finally, our model features a trade-off that is distinctive to firms in the data economy. When designing products, a firm balances monetization and customer engagement. For instance, a software company offers services for free to boost customer engagement, but doing so reduces current cash flows. We characterize an intertemporal trade-off: prioritizing customer engagement and data collection contributes to future knowledge on customers and future profitability but compromises current profits. Data from connected peers informs the firm about its customers and thereby alleviates the tension. In the example, being more informed about its customers allows the software company to target those who are willing to pay for the premium features and extract more profits.

Intriguing product-design dynamics emerge in our model. When one firm prioritizes customer engagement, it generates more data that informs its connected peers about their customers, thus allowing these peer firms to prioritize customer engagement (and generate more data as well) without significantly sacrificing current profits. Conversely, when a firm downplays customer engagement (prioritizes monetization), its reduced data collection and diminished data spillover make it harder for the connected peers to balance profitability and customer engagement. For any given level of profitability, these connected peer firms, now being less informed about their customers, have to sacrifice more customer engagement in their product design and collect less data as well.

In equilibrium, firms' product-design decisions exhibit "herding" behavior. Empirically, we find that a firm's product-design choices are strongly influenced by those of its data-connected peers, even after accounting for other common exposures, such as product overlap and supply-chain linkages. Furthermore, herding in product design among data-connected firms is weakened by the introduction of ATT, indicating that this empirical pattern is indeed driven by data sharing.[5]

[5]We focus on how a firm's incentive to generate data depends on other firms' choices. Relatedly, Farboodi and Veldkamp (2020) study how an agent's production of different types of data depends on others' information choices.

These results provide an out-of-sample validation of our model: the herding is a prediction of the same mechanism that generates the performance comovement—data sharing interconnects not only firms' performances but also their data-investment incentives—and the model's equilibrium dynamics align with the empirical findings even though these moments are not targeted in the calibration. These results also complete the account of ATT's impact: beyond reducing the value of data as a networked asset, the policy weakened the complementarity in firms' data investment.

In our model, data functions as productive capital, analogous to the role of capital in classic investment theories (Hayashi, 1982; Abel and Eberly, 1994), with a firm's product-design choices mirroring investment decisions. Specifically, the marginal q of a firm's data—the derivative of its value function with respect to its data stock—drives the decision on whether to prioritize customer engagement and data collection in product design. However, there are two critical distinctions. First, data investment features a positive externality, in contrast to the traditional investment dynamics where one firm's investment often crowds out others' investment.[6] Second, firms' investment decisions are directly interconnected through data sharing in our model. A firm's data marginal q incorporates the expected trajectories of data inflows from other firms (indegree network externality) but disregards the data outflows to other firms (outdegree network externality). This internalization of indegree externality creates strategic complementarity, or herding behavior, among firms. Meanwhile, the failure to internalize outdegree externality leads to under-investment.

Literature. To the best of our knowledge, our paper is the first to systematically measure the data-sharing network among firms and explore its economic implications. Data is non-rival. It can be used by multiple firms at the same time (e.g., Jones and Tonetti, 2020). The data conduits based on shared SDKs in firms' mobile applications have become the primary channels for inter-firm data sharing. We embed this empirically measured network of data-sharing conduits into an otherwise canonical model of data-driven firm growth (e.g., Farboodi and Veldkamp, 2025).

Data is an intangible asset. Our paper contributes to the literature on the scope of intangible asset usage.[7] While the existing papers focus on the scope within firms (e.g., Argente, Moreira, Oberfield, and Venkateswaran, 2025; Crouzet, Eberly, Eisfeldt, and Papanikolaou, 2024), our paper characterizes the usage of data across firms, supported by the network of data-sharing conduits, and explores the implications on firm growth—a central topic in the literature—and systemic risk, which is a new question that we connect to the literature: data sharing facilitates growth but also synchronizes firms' behavior by propagating shocks, thereby amplifying aggregate fluctuations. Prior studies on the transmission of information shocks through networks focus on production and

[6]For example, investment by one firm may increase the cost of investment inputs and financing or intensify product-market competition that other firms face (e.g., Asriyan, Laeven, Martin, Van der Ghote, and Vanasco, 2024).

[7]More broadly, our paper contributes to the literature on the non-rival nature of intangible capital (e.g., McGrattan and Prescott, 2009, 2010; Cong, Xie, and Zhang, 2021; Crouzet, Eberly, Eisfeldt, and Papanikolaou, 2022).

trade connections (e.g., Bui, Huo, Levchenko, and Pandalai-Nayar, 2022). In our setting, the data-sharing conduits are directly for firms to send and receive information on their customers. Our mechanism also differs from prior studies on information frictions and aggregate fluctuations that emphasize costly information acquisition (e.g., Veldkamp and Wolfers, 2007).

When it comes to data usage, our paper emphasizes data access rather than data ownership, following Varian (2019). In our setting, data is generated as a by-product of firms' operations (e.g., Bergemann, Bonatti, and Gan, 2022; Farboodi and Veldkamp, 2025).[8] Beyond the scope of our paper, data ownership is often discussed in association with market-based approaches to data sale (e.g., Acemoglu, Makhdoumi, Malekian, and Ozdaglar, 2022; Liu, Ma, and Veldkamp, 2025) and related to compensating data contributors, particularly in response to privacy concerns.

Our paper contributes to the literature on intangible asset valuation.[9] Valuing data as a productive asset has become increasingly important (Veldkamp, 2023). Empirically, we show that firms' stock-return comovements can be attributed to data-sharing linkages. Theoretically, our network-augmented Gordon growth formula adds to that discussion by providing a structural valuation approach that internalizes higher-order data spillovers at multiple time horizons. Our paper is the first to incorporate the network structure of data sharing into the value-function approach of data valuation. Our network-based estimate of the aggregate value of data stock also complements the valuation approach of Bhutani et al. (2026), who capitalize the value of the data newly acquired in a given year. Baley and Veldkamp (2025) provide a comprehensive summary of different methods. Traditional cost-based methods face clear limitations as the cost of data acquisition is not well defined when data is a by-product of firms' operations. Prior studies measure the value of data indirectly by focusing on complementary labor inputs (e.g., Abis and Veldkamp, 2023; Corhay, Hu, Li, Tong, and Tsou, 2024) or the outcome of data usage (e.g., Eeckhout and Veldkamp, 2022; Farboodi, Singal, Veldkamp, and Venkateswaran, 2024; Dong, Hu, Li, and Liu, 2025).

In our model, firms' decision-making and valuation are interconnected through a network adjacency matrix of data conduits, and the equilibrium conditions have a spatial structure (Comin, Dmitriev, and Rossi-Hansberg, 2012; de Paula, 2017; Redding and Rossi-Hansberg, 2017). Following Diebold and Yılmaz (2014), Ballester, Calvo-Armengol, and Zenou (2006), and Denbee, Julliard, Li, and Yuan (2021), we decompose aggregate valuation of data-generated cash flows into individual firms' contributions and develop the first metric of firms' systemic importance in the data economy.[10] This approach of modeling systemically important agents has been adopted in

[8]The notion that data is a by-product of economic activity was well established in the information economics literature (e.g., Veldkamp, 2005; Ordoñez, 2013; Fajgelbaum, Schaal, and Taschereau-Dumouchel, 2017).

[9]There is a growing literature on measuring and valuing intangible capital (e.g., Eisfeldt and Papanikolaou, 2013; Gourio and Rudanko, 2014b; Kogan, Papanikolaou, Seru, and Stoffman, 2017; Peters and Taylor, 2017; Kelly, Papanikolaou, Seru, and Taddy, 2021; Bhandari and McGrattan, 2021; Dou, Ji, Reibstein, and Wu, 2021; Ewens, Peters, and Wang, 2024). The cost-based approach has been a primary method in the literature.

[10]Our paper adds to the systemic risk literature (Billio, Getmansky, Lo, and Pelizzon, 2012; Acharya, Pedersen,

studies on social connections (e.g., Graham, 2008; Calvó-Armengol, Patacchini, and Zenou, 2009; Fogli and Veldkamp, 2021) and financial markets (e.g., Ozdagli and Weber, 2017; Herskovic, 2018; Eisfeldt, Herskovic, Rajan, and Siriwardane, 2022; Eisfeldt, Herskovic, and Liu, 2023).

Our findings on the economic implications of data sharing, the ATT impact, and cyberattacks contribute to several strands of the empirical literature. Data regulations restrict the scope of data usage, thereby impeding firm growth. The associated negative impact on firms' valuations is in line with the direct evidence on how ATT affects firms' stock prices (Bian et al., 2021). Our paper contributes by uncovering an unintended benefit of data regulations: they reduce systemic risk by weakening the comovement of firms' performance and herding in firms' product design.

Prior studies have examined how data regulations affect firms, including outcomes such as web traffic (Goldberg, Johnson, and Shriver, 2024), revenues (Aridor, Che, and Salz, 2023), innovation and venture investment (Bessen, Impink, Reichensperger, and Seamans, 2020; Janssen, Kesler, Kummer, and Waldfogel, 2022; Jia, Jin, and Wagman, 2021), SDK usage in Android mobile apps (Jin, Liu, and Wagman, 2024), data reliance (Demirer, Jiménez Hernández, Li, and Peng, 2024), and firms' technology choice (Peukert, Bechtold, Batikas, and Kretschmer, 2022).[11] Among these studies, Jin et al. (2024) examine how the GDPR changed apps' usage of SDKs in the Android ecosystem, treating SDK adoption as an outcome of privacy regulation. Relatedly, Wu (2023) finds that launching mobile apps predicts firm growth and lower risk, treating the app as a firm-level technology. Our focus is complementary to and distinct from both: we treat the data-sharing SDKs that firms have in common as links that connect firms into a network, and we study the performance comovement and cross-firm shock propagation that these links generate—phenomena that are defined at the level of firm pairs and of the network rather than of individual apps or firms.

Few empirical studies examine spillover effects in the data economy. Aridor, Che, and Salz (2023) document consumer-side spillovers: consumers' privacy decisions enable firms to infer other consumers' types. Using ATT as a shock to the ease of merging data across firms, our paper complements this literature by documenting firm-side data externalities, demonstrating that firms' data collection and product-design choices affect other firms within the data-sharing network.

Our paper also provides new evidence on the economic implications of cyberattacks. Jamilov, Rey, and Tahoun (2021) find that the impact of cyberattacks is propagated within industries and countries. Crosignani, Macchiavelli, and Silva (2023) document the propagation of cyberattacks through supply chains. Akey, Lewellen, Liskovich, and Schiller (2023) examine the impact of cy-

Philippon, and Richardson, 2016; Adrian and Brunnermeier, 2016; Benoit, Colliard, Hurlin, and Pérignon, 2016; Bai, Krishnamurthy, and Weymuller, 2018; Duarte and Eisenbach, 2021; Greenwood, Landier, and Thesmar, 2015).

[11] Specifically about ATT and ban on the use of third-party cookies, the literature has examined outcomes including firms' product monetization decisions (Kesler, 2023; Aridor, Che, Hollenbeck, Kaiser, and McCarthy, 2026), advertising effectiveness (Alcobendas, Kobayashi, Shi, and Shum, 2023; Aridor et al., 2026; Wernerfelt, Tuchman, Shapiro, and Moakler, 2024), app market concentration (Li and Tsai, 2022), financial forecasting (Abis, Tang, and Bian, 2025), and fraud (Bian, Pagel, Tang, and Raval, 2023). About GDPR, Johnson (2022) provides a review of the literature.

berattacks on firm value. Our paper adds to this literature by documenting how the negative impact of cyberattacks is propagated across firms through the data-sharing network.

In summary, our empirical analysis reveals that firms are not only interconnected through the conventional economic relationships—such as supply chains and production networks (e.g., Cohen and Frazzini, 2008; Menzly and Ozbas, 2010; Barrot and Sauvagnat, 2016; Auer, Levchenko, and Sauré, 2019; Boehm, Flaaen, and Pandalai-Nayar, 2019; Carvalho, Nirei, Saito, and Tahbaz-Salehi, 2021; Huo, Levchenko, and Pandalai-Nayar, 2025), product-market overlap (e.g., Hoberg and Phillips, 2010, 2016, 2018), locations (e.g., Parsons, Sabbatucci, and Titman, 2020; Comin, Dmitriev, and Rossi-Hansberg, 2012), technological proximity (e.g., Bloom, Schankerman, and Van Reenen, 2013; Liu and Ma, 2021), or shared analyst coverage (e.g., Ali and Hirshleifer, 2020)—but also by the data-sharing conduits enabled by the SDKs embedded in firms' mobile applications. Our contribution lies in documenting how this new type of economic linkage impacts firms' performance comovement, corporate valuation, and shock propagation across firms.

2 Data Sharing Network: Measurement and Evidence

2.1 The empirical setting

Background: data economy and mobile applications. The digital economy grew at an annualized rate of 7.1% from 2017 to 2022.[12] A key driver of the digital economy's rise is the growing use of mobile devices—a trend that accelerated during the COVID-19 pandemic when demand for digital services across work, entertainment, and communication skyrocketed. In 2022, the average US adult spent over 4.5 hours per day on mobile devices.[13]

Data is an essential productive asset in the digital economy. Companies rely on data collected from mobile devices to understand consumer preferences, customize product offerings, and guide innovation choices. Bian et al. (2021) find that over 60% of mobile applications (apps) track users' activities across websites, apps, and offline stores. Binns et al. (2018) report that nearly 90% of Android apps collect user data. Companies leveraging consumer data for targeted advertising—such as Google and Meta—generated approximately $780 billion in revenue in 2023.[14]

Mobile applications are where most consumer data is generated: roughly 60–80% of the time consumers spend on digital media is spent inside native apps rather than web browsers, and apps observe forms of consumer behavior that websites cannot, such as precise location, device sensors, and activity linked across apps.[15] Another key advantage of mobile device data is its high

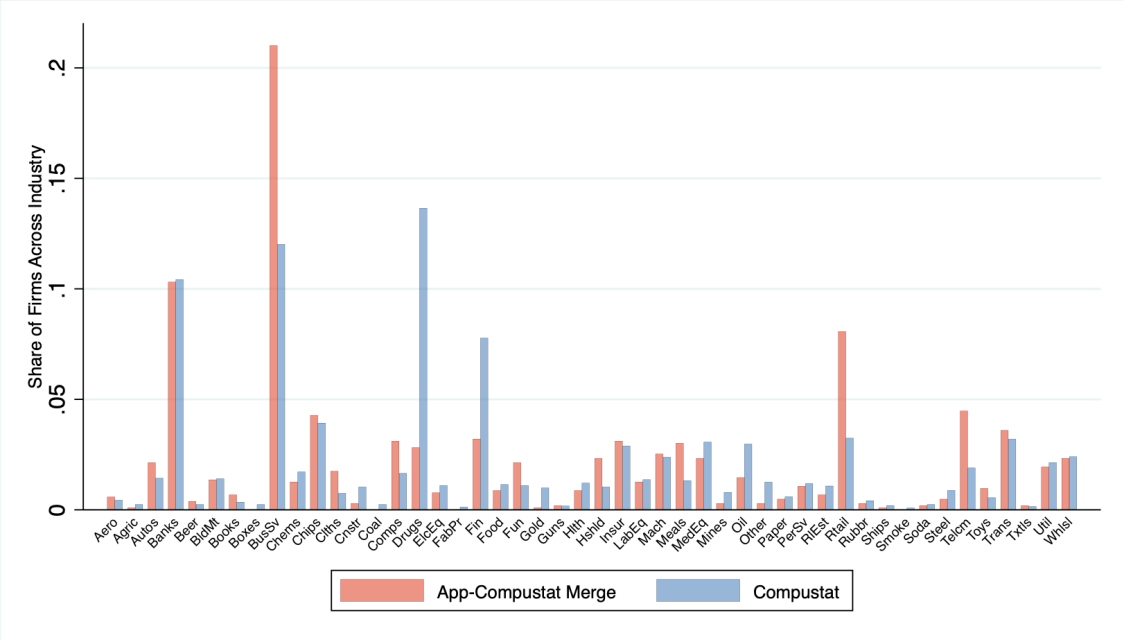
[12]Source: https://www.bea.gov/data/special-topics/digital-economy

[13]Source: https://www.emarketer.com/content/us-time-spent-with-connected-devices-2022

[14]Source: https://www.marknteladvisors.com/research-library/digital-marketing-market.html

[15]Mobile devices account for about 70% of total digital time, and 85–90% of mobile time is spent in native apps

FIGURE 1: **Firms in Our Sample vs. Compustat Universe: Industry Distribution**



NOTE.—Figure 1 compares the industry distribution of firms in the data economy ("App-Compustat Merge") with those in the broader Compustat universe ("Compustat"). It shows the distribution over the Fama–French 48 industries.

connectivity, enabled by consistent and universal identifiers such as the Identifier for Advertisers (IDFA). This device-level identifier is assigned by Apple and allows app developers, advertisers, analytics platforms, and ad networks to track user behavior across iOS apps, providing a consistent framework for linking activity and building user profiles from diverse data sources.[16] In contrast, web-based data relies on fragmented cookie systems managed by individual websites and advertising networks, which are often blocked by browsers like Safari. In addition, users can delete or block cookies in their browser settings. Overall, IDFA offers greater consistency, making mobile data a valuable asset for firms. Consistent with mobile apps being value-relevant corporate assets, Wu (2023) estimates the market value of individual apps from stock-price reactions to app releases.

Firms in the data economy. Motivated by the growing importance of app-based data collection and utilization, we analyze firms with at least one mobile app. Our empirical exercises require information on firms' mobile apps and standard accounting information from Compustat. A total of 1,031 firms meet the criteria. They cover over 60% of the total assets of Compustat firms. Figure 1 compares the industry distribution of firms in our sample and that of all Compustat firms.[17]

rather than mobile browsers, which together explain why native apps host 60–80% of total digital engagement. Internet Appendix C.4 details this calculation, an alternative calibration based on web-traffic data, and the underlying sources.

[16]The Android counterpart is the Google Advertising ID (GAID), also called the Android Advertising ID (AAID).

[17]In the Internet Appendix, we provide additional information on firms in our sample. Figure A.1 shows the share of firms that our sample covers relative to the broader Compustat universe, as measured by both firm count and total

To quantify the importance of data as a productive asset, we use Compustat business-segment data from 2014 to 2024 and construct a score, ranging from 0 to 100, for how intensively firms in each segment's industry rely on consumer data. A segment is data-reliant if its score is at least 70.[18] In Internet Appendix C.1, we provide details on the methodology. In Figure C.1, we report that data-reliant segments generate 15.4% of the aggregate revenue of US public firms in 2014, and the share reached 22.9%—$6.7 trillion—in 2024.[19] These magnitudes echo the BEA's national accounts of data assets discussed in the introduction.

Panel B of Table A.1 in the Internet Appendix A shows where data as a productive asset resides across industries. Our industry ranking lines up closely with the industries that the BEA identifies as the largest contributors to data investment—an external validation of our business-segment classification. Data-reliant revenue is concentrated in consumer-facing and information industries—banking, business services, computers, publishing, entertainment, telecommunications, and retail—and is nearly absent in upstream industries such as oil, mining, chemicals, and agriculture. This concentration delineates the industry scope of our analysis. The data-revenue shares of firms in our sample are roughly 1.5 to 2 times the economy-wide figures (Panel A of Table A.1). The industry boundary of data economy expands. Firms in traditional upstream sectors increasingly operate customer-facing apps—among them automakers (GM, Ford), energy majors (Shell, Chevron), airlines (Delta, United), and equipment manufacturers (AGCO, John Deere).

SDKs and the data-sharing conduit. Software Development Kits (SDKs) are pre-built app components ("building blocks") that firms integrate into their mobile apps. SDKs support various functionalities, such as messaging and payment. A subset of SDKs facilitates data sharing. They collect user data from host apps. By using unique identifiers such as IDFA, these SDKs link user activity across multiple apps, enabling the construction of comprehensive consumer profiles. Firms integrate such SDKs in their apps to access these customer profiles, which provide valuable insights for marketing, product development, and other strategic decisions.[20]

SDK-mediated data sharing naturally gives rise to a network structure of the data economy.

assets. In 24 out of the 48 Fama–French industries, our sample firms account for more than 50% of total assets. In Figure A.2, we examine the distribution of firm size (logarithm of total assets) in Panel A and asset turnover (sales-to-asset ratio) in Panel B. Firms in our sample are larger and more efficient in generating revenues.

[18] A large language model (GPT-5.1) is instructed to assign 0–10 when data use is incidental; 11–40 when data supports the business but is not central to value creation; 41–70 when data is important and routinely used in operations or decision-making; and 71–100 when the business model is data-centric and data is a core input into production or service delivery. For all regressions, we use a more conservative threshold of 80 for data-reliant segments.

[19] The revenue share climbs through 2020, dips in 2021–2022 following ATT, and resumes growing thereafter in Figure C.1—an aggregate counterpart to the firm-level negative effects of ATT that we document throughout the paper and use to validate our measure of data connectedness and its economic effects.

[20] SDK providers often generate indirect revenue through advertising. By supplying customer profiles to companies that use their data-sharing SDKs, they allow these firms to engage in more effective targeted advertising, and a portion of the resulting advertising expenditures, paid to ad publishers, ultimately flows back to the SDK provider.

Two firms that share data with the same SDKs are connected as the customer profiles they obtain from the SDKs contain data from both firms.

We use Amazon and General Motors (GM) to illustrate such data linkages. According to our measure introduced below in Section 2.2, GM is connected with Amazon, because both install a similar set of data-sharing SDKs on their apps—that is, they share the *data conduits*.

GM collects extensive data on consumer behavior through vehicle purchases, financing, and usage of its IoT (internet of things) software on the vehicles.[21] By sharing data with the SDKs, GM enables the SDKs to form more informative consumer profiles that contribute to Amazon's advertisement targeting. For instance, GM's data on consumers' car-purchasing patterns helps Amazon focus ads on individuals likely to buy complementary products, such as home EV chargers, smart car accessories, or high-tech gadgets. The improved precision in targeting enhances Amazon's ability to convert high-value customers while improving its inventory planning for automotive-related products, boosting overall sales and profitability.

Similarly, data collected by Amazon flows through the SDKs and gets merged into consumer profiles that GM leverages to predict consumer preferences and market trends. As disclosed in the privacy labels of Amazon's iOS mobile app, Amazon collects user data that facilitates third-party advertising. Data on consumer purchasing trends, such as demand for EV-related products among specific demographic groups, helps GM tailor its marketing strategies and product offerings.

Several institutional features of how firms adopt SDKs are relevant for interpreting our methodology and, in particular, our use of firms' SDK overlap as an explanatory variable for firms' comovement in performance metrics. First, there is no bilateral contract between connected firms: each firm contracts separately with the SDK provider. Second, a firm cannot join a specific peer's SDK in order to share data with that peer; the customer profiles it receives are formed at the SDK-provider level, and the firm has neither control over nor visibility into which other firms use the same SDK. Third, within the data-sharing SDK category, adoption decisions are made on an app-by-app basis by technical teams during development, rather than by executives that focus on strategic decisions. Fourth, the market for data-sharing SDKs is also highly concentrated around a few de facto standards, so firms' SDK overlap largely reflects independent choices from the same short menu of dominant software kits. Finally, SDK integration is sticky: switching requires re-engineering an app's analytics and advertising infrastructure, so the SDK overlap we measure reflects adoption decisions accumulated over years rather than firms' responses to their swiftly changing operational and financial metrics.

[21] GM's app in the Apple App Store discloses in its privacy labels that it collects and shares location data, contact information, and user and device identifiers with third parties.

The economic importance of SDKs. Despite their central role in data sharing, these SDKs have received little attention in the literature. We provide direct evidence that they are economically important. Table A.2 in the Internet Appendix shows that data-sharing SDK usage predicts revenues, particularly where data is central to the business. Among firms whose revenues come entirely from highly data-reliant segments, a one-standard-deviation increase in the number of data-sharing SDKs installed is associated with 0.254 log points higher quarterly sales, a magnitude similar to that of R&D intensity (0.258) and larger than that of physical investment (0.098). In contrast, the association is close to zero among the other firms. The same pattern holds when we scale the number of data-sharing SDKs by the total number of SDKs. Since the denominator captures firms' general propensity to adopt more or fewer SDKs, our finding is not about firms' general technological infrastructure but their reliance on data-sharing SDKs given the technological infrastructure.

The evidence above is not causal, as firms still choose which data-sharing SDKs to install. We therefore exploit major privacy-driven updates released by SDK providers to their entire user bases, which restrict firms' access to consumer data but are outside individual firms' control. We identify 18 such updates and examine their effects in two ways. First, a firm-level event study in Figure C.6 shows that exposed firms earn average cumulative abnormal returns of -3.2% relative to the Fama–French three-factor benchmark over the three months following an update; the decline is concentrated among retail and consumer-service firms, which lose 7.0%, while other exposed firms are largely unaffected.[22] Second, an app-level difference-in-differences analysis in Table C.1 shows that exposed apps experience 18% fewer downloads and 12% lower daily active users (DAU) relative to matched controls.[23] Figure C.7 shows no pre-trends and a persistent decline in DAU, reaching roughly 26% after six months. These results provide quasi-experimental evidence that disruptions to data-sharing SDKs reduce customer engagement and firm value. Internet Appendix C.5 details the SDK updates and empirical designs.

The ATT shock. Our empirical analysis leverages a policy shock introduced by Apple's App Tracking Transparency (ATT), implemented on April 26, 2021, which requires app developers to obtain explicit user consent before tracking user activity. Specifically, the policy requires apps to display a prompt asking for permission to track user activity. For users who do not opt in, apps cannot access the IDFA for tracking, preventing user activity from being linked across apps through this common identifier. Opt-in rates were low, ranging from 4% to 18% in the first 12

[22] The stock-price effects are economically large: across exposed firms, they imply an aggregate market-value loss of roughly \$0.6–\$0.7 trillion.

[23] Because the coefficients are sizable, we use the exact transformation $100(e^{\beta}-1)\%$ rather than the usual $100\beta\%$ approximation. This implies declines of 18% in downloads and 12% in DAU for US-listed treated apps, and 14% and 11% in the worldwide sample.

months following the rollout of ATT in the United States.[24] As a result, ATT substantially reduced the SDKs' ability to link data on consumer activities across apps through IDFA. The disruption was economically large: Meta projected a revenue reduction exceeding $10 billion for 2022 from Apple's privacy changes, and industry analysts estimated the combined 2022 revenue impact on Meta, YouTube, Snap, and Twitter at nearly $16 billion.[25]

SDKs and ATT in firms' disclosures. Firms' own communications provide direct evidence on the importance of SDK-mediated data sharing. Because firms rarely have occasion to discuss the SDKs embedded in their apps, we organize the evidence around ATT in 2021—the event that prompted firms to discuss data-sharing SDKs explicitly. We analyze SEC filings (10-Ks and 10-Qs), earnings calls, and public web sources, including archived privacy policies, over 2020–2022: 9.2% of firms explicitly mention ATT or the IDFA in regulatory filings, 24.9% disclose a broader dependence on third-party data, 16.1% discuss ATT on earnings calls, and roughly nine in ten leave a public record of reliance on cross-app data sharing. Note that these percentages are conservative lower bounds, as filings rarely name specific technologies. Across all three sources, 373 disclosures provide quantitative estimates linked to ATT or the IDFA. These estimates are predominantly negative; in SEC filings, for example, 99.1% of the quantitative disclosures describe adverse effects. Among firms explicitly stating the revenue impact, the median reported effect is 15.0% of revenue.[26] Firms' disclosures shift from prospective risk-factor language before ATT enforcement to realized dollar impacts afterward. Internet Appendix C.6 provides more details.

Data sharing outside the SDK ecosystem. Other channels of data sharing—through direct firm-to-firm agreements and purchased data—differ from the SDK-based sharing in important ways. First, direct data exchanges between firms are rare: Elsaify and Hasan (2021) identify roughly 1,600 publicly reported exchanges among 1,285 of the large firms in their sample over 2017–2018—about 0.2% of all possible firm pairs—whereas 58% of firm pairs in our sample are connected through data-sharing SDKs. Second, such exchanges are negotiated bilaterally and are heterogeneous in content, format, and contractual terms; they therefore lack the standardized, scalable network structure of SDK-mediated sharing, in which a single SDK embedded in numerous apps pools data from many firms that do not have direct bilateral contracts with one another. Third,

[24] See more details at `https://www.flurry.com/blog/att-opt-in-rate-monthly-updates/`. Kraft, Skiera, and Koschella (2023) document a 17% opt-in rate as of March 2022.

[25] See "Facebook Feels $10 Billion Sting from Apple's Privacy Push," *Wall Street Journal*, February 3, 2022, and the estimates by Lotame, a data-management and advertising-analytics firm, reported by *Business Insider*, April 26, 2022. The latter are modeled industry estimates rather than realized accounting losses.

[26] Stitch Fix attributes a $42.6 million revenue loss to the disruption of iOS-based marketing, Snap attributes missing revenue guidance to the iOS changes, and press reports link the decline of Oracle's advertising business from $2 billion to $300 million to the loss of third-party data access. The affected firms extend to broader industries, such as mortgage lenders (Rocket), retail (Procter & Gamble, Home Depot), and healthcare firms (Labcorp).

customer data is difficult to merge across sources: the same consumer appears under different identifiers across devices, browsers, and operating systems, and this identifier fragmentation impedes integration (Lin and Misra, 2022), whereas the SDKs collect and link information across apps through a common identifier (e.g., IDFA). Finally, as discussed above, app-based data dominates web-based data in magnitude and collection efficiency. These considerations support our treatment of shared SDKs as the primary conduits for inter-firm data flows.

2.2 Measuring data connectedness and performance comovement

Data on mobile apps and SDKs. We begin by introducing our primary data source on firms' app design and usage. We use two products from Apptopia, a leading provider of mobile app intelligence that has systematically tracked the universe of mobile applications since 2014. First, we obtain information about app characteristics (e.g., category, age) and performance (e.g., downloads, active users, session length). Second, we use Apptopia's SDK Intelligence, which provides data on the installation and removal dates of SDKs for each app. More specifically, Apptopia tracks the history of apps' SDK installations and removals in mobile apps by analyzing the publicly available app installation package. When a new version of an app is released, Apptopia reanalyzes the update package to maintain an accurate and current record of the app's SDK profile.

Apptopia classifies SDKs into several categories. We identify the data-sharing SDKs as those that are related to advertising networks, marketing campaigns, data analytics, and other activities that analyze and improve customers' willingness to pay. We verify the accuracy of SDKs' functionality by reviewing the documentation files in their GitHub repositories. We focus on the 50 most popular data-sharing SDKs in iOS apps, ranked by worldwide net installations as of April 26, 2021, when ATT was introduced. The most popular SDK accumulated 262,209 net installations (i.e., installations minus uninstallations) since 2014, while the 50th most popular SDK accumulated only 5,096 net installations in the same period, indicating a high level of concentration. In the Internet Appendix, we conduct robustness analysis by considering the top 20 SDKs.

Our SDK-based connectedness measure is constructed from iOS apps because ATT—the policy shock central to our identification—is a disruption to data sharing specific to the iOS ecosystem and thus its dampening effect on the data-sharing outcomes validates our iOS-based data connectedness measure. For app-performance outcomes such as app downloads and daily active users, we combine activity across iOS and Android because we are ultimately interested in firm-level performance and firms' access to consumer data helps stimulate customer engagement for both iOS and Android users. In the Internet Appendix, we show that the results are robust to controlling for an analogous connectedness measure constructed from firms' Android apps and its ATT interaction.

The remaining data sources are standard. We use Compustat and CRSP for financial in-

formation, and draw on the Text-based Network Industry Classifications (TNIC), FactSet Revere, USPTO, and I/B/E/S to construct various measures of firm linkages introduced below. Details of these data sources are provided in the Internet Appendix B.1.

Measuring data connectedness. To characterize the data-sharing network, we develop a pairwise measure for data connectedness. This methodology is akin to the approach taken by Hoberg and Phillips (2010, 2016), which develop a product overlap measure to study competition among firms; likewise, Bloom et al. (2013) introduce an R&D space overlap measure to explore the impact of technological proximity on innovation and firm performance. We compute the cosine similarity between pairs of firms based on their usage of data-sharing SDKs in their iOS apps.[27]

Specifically, we denote app a's (from firm i) data conduit via SDK k at time (quarter) t as: $s_{iakt} = m_{iat} \times d_{iakt}$, where m_{iat} is the number of monthly active users (MAU) averaged within quarter t of firm i's app a; $d_{iakt} = 1$ if this app installs SDK k and 0 otherwise. The installation indicator is scaled by the size of the app's user base to capture how relevant this data conduit is.

Aggregating across all apps owned by firm i at time t, we define firm i's data conduit via SDK k at firm-SDK-quarter level as: $S_{ikt} = \sum_{a \in \mathcal{A}_{it}} s_{iakt}$ where $\mathcal{A}_{it}$ is the set of apps owned by firm i at time t. We then stack all of firm i's data conduits at time t into a $K \times 1$ vector: $\mathbf{S_{it}} = \left[S_{i1t}, S_{i2t}, ..., S_{iKt}\right]'$, where K is the total number of SDKs. K is 50 in the baseline version of our measure. The cosine similarity between firm i's and j's data conduits at time t is given by:

$$\rho_{ijt}^{data} = \frac{\mathbf{S_{it}'} \cdot \mathbf{S_{jt}}}{|\mathbf{S_{it}}| \cdot |\mathbf{S_{jt}}|}, \tag{1}$$

where $|\cdot|$ is the Euclidean norm. Data connectedness is similar within industry and across industries (see Figure 3). This new type of inter-firm linkage is distinct from the traditional ones within an industry or along the supply chain that we control for in our analysis.

Performance comovement. In our empirical analysis, we show that data connectedness manifests itself in the comovement between firms across different performance metrics. We compute the correlation between any pair of firms across various metrics, including the logarithm of app downloads, average daily active users (DAU), earnings growth, and asset turnover (sales/assets) at a quarterly frequency.[28] DAU is the number of unique users who open at least one of the firm's apps on a given day; firm-level DAU aggregates across all apps owned by the firm and is averaged within the quarter. MAU (monthly active users) is defined analogously at the monthly frequency.

[27]The SDKs used in firms' Android and iOS apps can differ. Since ATT only affects data sharing in the iOS environment, we focus exclusively on data connectedness based on overlaps in SDK usage across iOS apps.

[28]Quarterly earnings growth rate is calculated as $2 \cdot (\textit{Net Income}_t - \textit{Net Income}_{t-1}) / (\textit{Net Income}_t + \textit{Net Income}_{t-1})$, instead of $(\textit{Net Income}_t - \textit{Net Income}_{t-1}) / \textit{Net Income}_{t-1}$ to smooth out volatility.

For each performance metric, we compute one correlation for quarters before the implementation of ATT in 2021Q2 and another for those after the implementation. Our full-sample period is 2014Q3–2023Q2. We also consider comovement in firms' stock returns. For each firm pair, comovement is measured as the correlation of monthly returns within non-overlapping 12-month windows. We construct seven pre-ATT and three post-ATT windows, using April 2021 as the cutoff.[29] We consider three types of returns: raw returns, abnormal returns based on CAPM, and DGTW-adjusted returns (Daniel, Grinblatt, Titman, and Wermers, 1997).

Other firm linkages. Our focus is on the impact of data connectedness on firms' performances. We control for other inter-firm linkages. In the Internet Appendix B.1, we list data sources and provide information on variable construction for other measures of firms' overlap, such as product-market similarity, app user-base overlap, supply-chain linkages, geographic overlap, technology proximity (based on patent information), stock-market analyst overlap, etc.

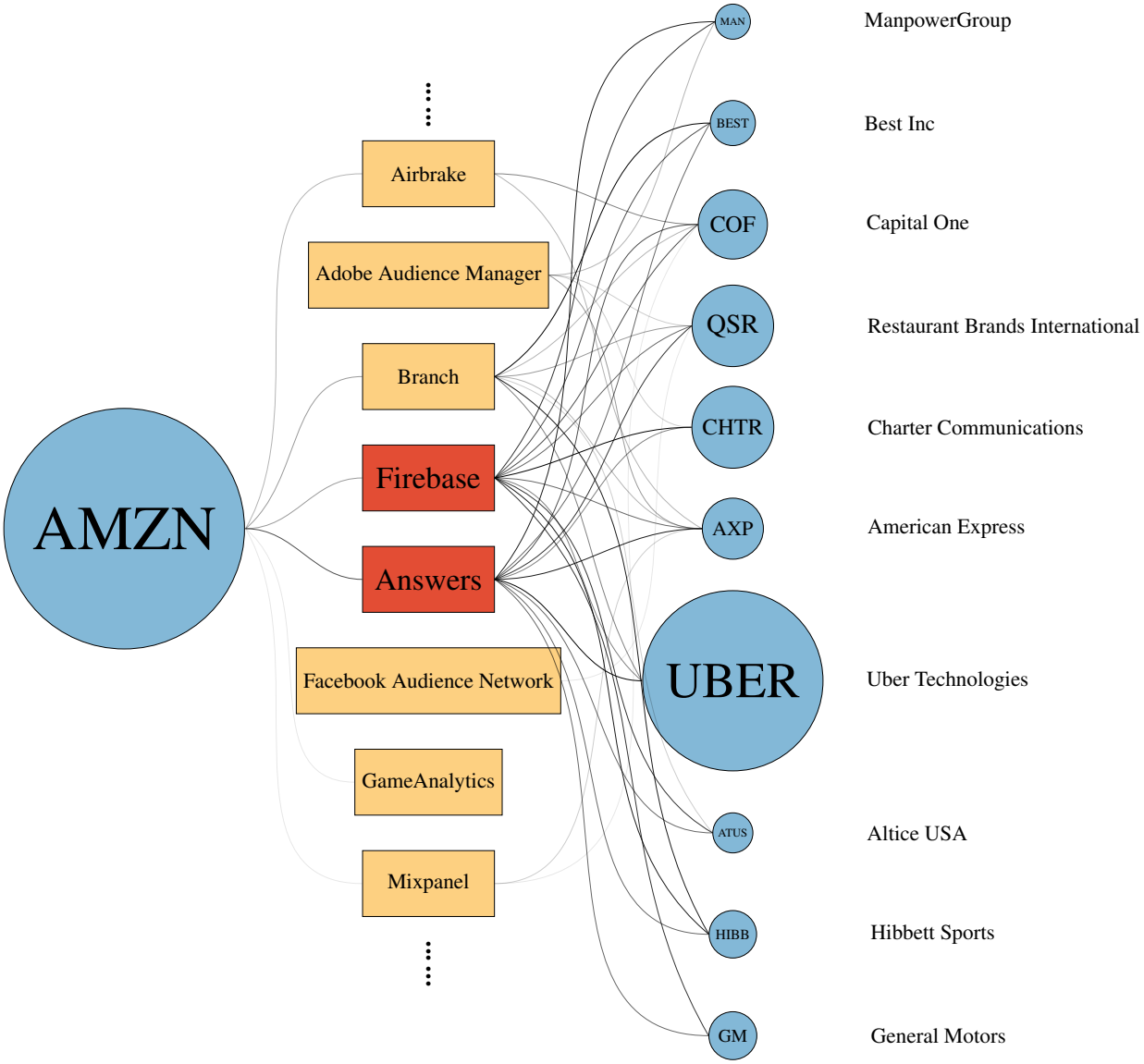
Firms' product-design choices. To test a unique prediction of our model in Section 4, we measure firms' usage of functionality SDKs that reflect firms' product-design choices rather than data sharing. For each of the following functionality SDK categories—payment, security, customer support, review & feedback—we calculate: 1) the number of unique SDKs used by a firm, 2) the change in the number of unique SDKs used by a firm, and 3) the weighted sum of the number of unique SDKs used by peer (data-connected) firms, where the weight is the pairwise data connectedness described above. The third measure captures the average product-design decisions made by the focal firm's data-connected peers, which, according to our model, affect the focal firm's product-design choices. These functionality SDKs are pre-built software components (just as the data-sharing SDKs are), which firms can integrate into their mobile apps. Apptopia tracks these app features by analyzing the installation packages of each new app version.

2.3 Descriptive statistics

The data-sharing conduits. In Section 2.1, we use Amazon as an example of how a firm shares data with other firms (e.g., GM). In Figure 2, we map out the major data-sharing conduits that connect Amazon with other firms. Data-sharing SDKs that Amazon installed on its apps are shown as red or yellow rectangles. The red-colored SDKs are the most important as they connect Amazon with *all* the other firms in the figure. The thickness of lines connecting firms and SDKs represents the proportion of a firm's monthly active users (MAU) linked to an SDK.

[29] The post-ATT windows cover April 2021–March 2022, April 2022–March 2023, and April 2023–March 2024.

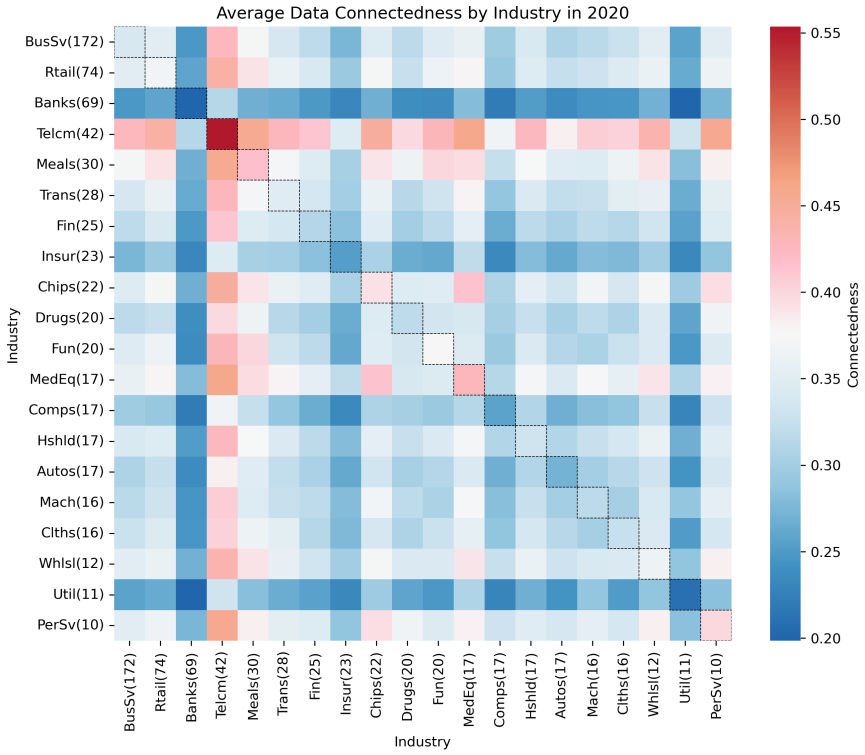
FIGURE 2: **Amazon's Data-Sharing Conduits and Peers**



NOTE.— Figure 2 illustrates Amazon and the 10 firms most connected to it in the data space. Firms are depicted as blue circles, with the circle size corresponding to the firm's monthly active users (MAU). Data-related SDKs are shown as red or yellow rectangles. The red-colored SDKs are those that connect Amazon with all the other firms in the figure. Lines connecting firms to SDKs indicate SDK usage, with the thickness of the line representing the relative importance of the SDK to the firm, measured by the proportion of the firm's MAU linked to the SDK.

Figure A.3 in the Internet Appendix visualizes the entire network structure. The graph contains a vast amount of information, including firms' sizes, the strength of pairwise connections, the formation of clusters (i.e., groups of firms whose overlap in the data-sharing SDKs is high), and importantly, which firms are the most well-connected and thus centrally located in the graph. Google (ticker: GOOGL) and Meta (ticker: META) appear to be the two most influential firms in the data network, along with telecom firms such as AT&T (ticker: T) and Comcast (ticker: CM-

FIGURE 3: **Data Connectedness within and across Industries**



NOTE.—Figure 3 presents the average pairwise data connectedness across Fama–French 48 industry combinations in 2020, focusing on the 20 industries with the largest number of firms in our sample. Warmer (cooler) colors indicate higher (lower) levels of connectedness.

CSA). Several firms from other industries, including Sony (ticker: SONY) from entertainment, DoorDash (ticker: DASH) from transportation, and McDonald's (ticker: MCD) from restaurants, also take central positions. Importantly, a firm's size does not always correlate with its network centrality. For example, large firms like Nike, Walmart, and Netflix remain on the periphery, while several smaller firms occupy central positions in the network due to their interconnectedness.

While these patterns are intuitive, centrality visualized in Figure A.3 is only based on the direct linkages. Data sharing induces interconnectedness of higher orders—firm A's data is shared through SDKs with firm B that in turn may share data with firm C. Moreover, data spillover has persistent impact over multiple time horizons as data affects firms' ability to stimulate customer engagement which in turn contributes to further data accumulation, resulting in self-perpetuating dynamics. In Section 4, we develop a framework to identify systemically important firms in the data economy that accounts for indirect linkages and data spillover effects across multiple horizons.

Data sharing within and across industries. In the case study of Amazon (see Figure 2), it is apparent that firms across industries share data with one another. Next, we compare cross- and within-industry data sharing. One might expect that within an industry, firms would avoid adopting the same SDKs as they are reluctant to share data with competitors. However, a countervailing force exists: data from industry peers can be more informative about a firm's own customers than data from different industries. Empirically, these opposing forces appear to offset each other.

In Figure 3, firms within the same industry share data to a degree very similar to that observed between firms in different categories. The diagonal elements represent data connectedness among firms within the same industry, while the off-diagonal elements capture cross-industry interconnectedness. Data sharing along the diagonal is neither systematically stronger nor weaker than that off the diagonal. One exception is the telecom industry, which exhibits stronger data connectedness with all other sectors. This is because telecom firms' apps typically install more SDKs.[30] Another interesting pattern is that heavily regulated industries, such as banks, insurance, and utilities, share less data with other industries. To demonstrate that our results are not specific to within- or cross-industry sharing, our analysis in the main text considers all firm pairs, and in the Internet Appendix D, we only include firm pairs that have below-median product-space overlap.

Summary statistics. Our measure of data connectedness for each firm pair is a cosine similarity score, which has an average of 0.171 and 90th percentile at 0.45. Note that data connectedness is stable over time. The average year-over-year Spearman correlation of our pairwise measure is 0.80, which is comparable to average correlation of 0.78 in the product space (Hoberg and Phillips, 2010, 2016; Frésard, Hoberg, and Phillips, 2020) in our sample period. Our analysis thus treats the data-sharing conduits as given and focuses on their impact on firms' performances and decision-making. Firms' SDK installation decision lies beyond the scope of this paper. While we do not analyze changes in the data conduits, we verify that our results remain robust when we average our measure of data connectedness in different sample periods (see the Internet Appendix D).

We examine how data connectedness affects firms' correlations in app and financial performances. These correlations vary substantially across firm pairs. For example, the 10th and 90th percentiles of log-downloads correlation are -0.67 and 0.73, respectively. For stock returns, there is stronger comovement in raw returns (0.247) compared to CAPM abnormal returns (0.034) or DGTW-adjusted returns (0.008). This is because raw returns capture exposure to common risk factors. All three measures of return comovement exhibit significant variations across firm pairs, with the 90th percentile at 0.67, 0.47, and 0.44, respectively.[31] Table B.3 in the Internet Appendix

[30] Internet Service Providers are the Internet gatekeepers and in a unique position to observe and capture rich cross-app and cross-device data—they install most of the popular SDKs to monetize from data mainly through advertisement.

[31] The pairwise return correlations have more observations than the performance correlations: 10 per pair from non-overlapping 12-month windows versus 2 per pair for the pre- and post-ATT periods as described in Section 2.

TABLE 1: **Summary Statistics**

Panel A. Pairwise variables: data connectedness and performance comovement

	mean	sd	p10	p25	p50	p75	p90	count
Pairwise connections								
data connectedness	0.171	0.19	0.00	0.00	0.11	0.29	0.45	1,401,082
Performance comovement								
downloads corr.	0.042	0.51	-0.67	-0.38	0.06	0.47	0.73	1,401,082
DAU corr.	0.073	0.57	-0.73	-0.42	0.11	0.59	0.81	1,395,990
earnings growth corr.	0.002	0.38	-0.49	-0.22	0.00	0.23	0.50	1,372,544
sales/assets corr.	0.095	0.46	-0.56	-0.25	0.12	0.46	0.71	1,223,256
Return comovement								
raw return	0.247	0.33	-0.21	0.02	0.27	0.50	0.67	5,748,100
return - CAPM	0.034	0.33	-0.41	-0.21	0.03	0.27	0.47	5,645,298
return - DGTW	0.008	0.33	-0.42	-0.23	0.01	0.24	0.44	4,993,880

Panel B. Levels of the comovement outcome variables

	mean	sd	p10	p25	p50	p75	p90	count
downloads (millions)	4.850	22.13	0.00	0.02	0.15	1.06	5.65	26,566
daily active users (millions)	2.919	16.39	0.00	0.00	0.04	0.39	2.19	26,566
earnings growth	-0.013	4.30	-1.53	-0.40	0.02	0.43	1.54	37,582
sales/assets	0.223	0.18	0.06	0.10	0.17	0.29	0.46	35,939
raw return	0.009	0.12	-0.12	-0.05	0.01	0.07	0.14	119,106
return - CAPM	-0.002	0.10	-0.11	-0.05	-0.00	0.04	0.10	119,106
return - DGTW	-0.000	0.09	-0.10	-0.04	0.00	0.04	0.09	119,106

NOTE.—Table 1 reports the summary statistics on key variables. Panel A lists variables constructed at the firm-pair level. The first row reports statistics of data connectedness defined in (1). The other rows are for the outcome variables (i.e., firms' performance comovements). Downloads and DAU are measured in logs. The comovement of app and financial performances is calculated separately for the periods before and after the introduction of ATT (2021Q2); return comovement is calculated as the correlation between firms' monthly returns over non-overlapping 12-month windows, defined relative to the introduction of ATT in April 2021. The data on app performance, financial performance, and returns spans from 2014Q3 to 2023Q2. Panel B reports firm-quarter statistics for the levels of the four outcome variables whose pairwise correlations form the comovement analysis. Downloads and daily active users (DAU) are in raw units (millions; quarterly total downloads and average DAU across the firm's apps) rather than the log units used in the correlations. Earnings growth is the symmetric quarterly growth rate, $2(NI_t - NI_{t-1})/(NI_t + NI_{t-1})$, and sales/assets is quarterly asset turnover. Returns are measured monthly. All levels are winsorized at 0.5% in both tails.

reports summary statistics of all variables, including the other inter-firm linkages, firm characteristics, and firms' choices of (non-data) functionality SDKs that reflect their product design.

3 Data Sharing and Firm Performances

3.1 Performance comovement

Data is generated as a by-product of business operations (e.g., Bergemann et al., 2022; Jones and Tonetti, 2020; Farboodi and Veldkamp, 2025). When a firm expands its operation, it collects more data through increased interaction with customers. The new data is shared with its SDK-connected peers. Being more informed about customers allows the peer firms to grow. We now test the hypothesis that two data-connected firms exhibit performance comovement. We also examine the impact of ATT that allows customers to sever the link between their data and their universal identifiers (i.e., IDFA discussed in Section 2.1). By disrupting data aggregation based on universal IDs, ATT can weaken firms' performance comovement induced by data sharing.

Our empirical specification follows the gravity model in the international trade literature (e.g., Imbs, 2004; Baxter and Kouparitsas, 2005; Di Giovanni and Levchenko, 2010). Specifically, we estimate the following regression:

$$Corr_{ijt}^{Perf} = \alpha + \beta_1 \rho_{ij}^{data} + \beta_2 ATT_t \times \rho_{ij}^{data} + \beta_3' \rho_{ij}^{other} + \beta_4' ATT_t \times \rho_{ij}^{other} + \theta_{it} + \iota_{jt} + \varepsilon_{ijt}\,. \quad (2)$$

The correlation metrics, $Corr_{ijt}^{Perf}$, capture firm pairwise comovement across multiple dimensions, where *Perf* can be firms' app performance, financial performance, or stock returns. For each pair ij and comovement in app and financial performance, we have two observations, one before ATT and one after, with $t \in \{\text{pre-ATT, post-ATT}\}$. For return comovement, we have one observation per 12-month window and a total of 10 observations (7 before ATT and 3 after ATT). We include firm-by-time fixed effects (θ_{it} and ι_{jt}) and double-cluster standard errors by firm-i and firm-j.

To ensure that our results are not confounded by other firm linkages, we control for eight alternative linkages, denoted by ρ_{ij}^{other} (an 8-by-1 vector), which have been shown in prior literature to impact firms' comovement. These linkages include product-market overlap, supply chain relationships, technological proximity, common analyst coverage, etc. For all the firm linkages, including our measure of data connectedness, ρ_{ij}^{data}, we use the pre-shock (ATT) average values.[32]

The coefficients of interest are β_1 and β_2. A positive value of β_1 indicates that a high degree of data connectedness (i.e., a high ρ_{ij}^{data}) is associated with more synchronized variations in firms' performances. Moreover, we expect β_2 to be negative, as ATT negatively impacts firms' data collection from operations, thereby weakening the mechanism behind performance comovements between data-connected firms. To facilitate the interpretation of the coefficients and comparison

[32]In Appendix D, we consider alternative time windows for calculating average data connectedness, such as the average of ρ_{ijt}^{data} in 2020 (the year immediately preceding ATT) and post-ATT averages of ρ_{ijt}^{data}. We also use pre- and post-ATT averages of ρ_{ijt}^{data}, respectively, to explain firm performance comovement in the corresponding periods. All results are consistent with our baseline findings based on the pre-ATT average of ρ_{ijt}^{data}.

TABLE 2: **Comovement in Customer Engagement (App Performance)**

	downloads		DAU	
	(1)	(2)	(3)	(4)
data connectedness	0.026***	0.024***	0.026***	0.023***
	(7.56)	(7.25)	(6.78)	(6.26)
ATT × data connectedness	-0.025***	-0.025***	-0.024***	-0.023***
	(-6.63)	(-6.74)	(-5.81)	(-5.75)
Other linkages	N	Y	N	Y
ATT × other linkages	N	Y	N	Y
Observations	1,401,082	1,401,082	1,399,426	1,399,426
R-sq	0.066	0.068	0.113	0.114

NOTE.—Table 2 shows the relationship between firm data connectedness and the comovement of app performance. Each observation represents a firm pair at a specific point in time. For each firm pair, the comovement of app performance is measured as the correlation between their quarterly log(downloads) and log(DAU), calculated separately for the periods before and after the introduction of ATT (2021Q2). The app performance data spans from 2014Q3 to 2023Q2. In even-numbered columns, we include controls for a comprehensive set of pairwise firm connections, as well as interaction terms between these connections and the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects (θ_{it} and ι_{jt}) and double-clustered standard errors by firm-i and firm-j. Time is defined relative to ATT. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

across different types of firm linkages, we normalize ρ_{ij}^{data} and elements in ρ_{ij}^{other} to have a zero mean and a standard deviation equal to one. The results are reported in Tables 2 to 4. In the Internet Appendix, Table A.3 to Table A.5 report the full results with estimates of β_3 and β_4.

Table 2 shows that higher data connectedness is associated with a significantly greater degree of app performance comovement (i.e., $\beta_1 > 0$). In Column 1 of Table 2, a one-standard-deviation increase in data connectedness leads to a 0.026-unit increase in the correlation of log(downloads) between two firms. Controlling for alternative types of firm linkages has little effect on the results in Column 2. We can compare the magnitude of these effects against the effects of other firm linkages. Based on Column 2 in Table A.3 in the Internet Appendix where we report the full results, the estimated coefficient of 0.024 for ρ_{ij}^{data} is 2.7 times larger than that for firms' product-market overlap, which is 0.009. We find similar results for the comovement in log(DAU).

This pattern is substantially weaker after ATT. Examining the ATT impact serves to validate that our measure of data connectedness indeed captures data-related linkages. If our measure were instead reflecting other economic relationships unrelated to data sharing, then ATT should have had a similar impact on other forms of firm linkages. The coefficients for the interaction terms between ATT and ρ_{ij}^{other}—the non-data linkages—are not significantly negative (see Table A.3).

In terms of magnitude, the coefficient of $ATT \times \rho_{ij}^{data}$ almost offsets the baseline effect. This does not imply that comovement driven by data sharing disappears. Our measure of data connectedness, ρ_{ij}^{data}, focuses on data sharing through SDKs in the iOS system where, before ATT,

TABLE 3: **Comovement in Financial Performance**

	earnings growth		sales/assets	
	(1)	(2)	(3)	(4)
data connectedness	0.002***	0.002**	0.005***	0.004***
	(2.82)	(2.33)	(3.64)	(2.79)
ATT × data connectedness	-0.003***	-0.003***	-0.003**	-0.003**
	(-3.16)	(-2.88)	(-2.08)	(-2.07)
Other linkages	N	Y	N	Y
ATT × other linkages	N	Y	N	Y
Observations	1,379,592	1,379,592	1,231,716	1,231,716
R-sq	0.005	0.005	0.184	0.186

NOTE.—Table 3 shows the relationship between firm data connectedness and the comovement of financial performance. Each observation represents a firm pair at a specific point in time. For each firm pair, the comovement of financial performance is measured as the correlation between their quarterly earnings growth and asset turnover (sales/assets), calculated separately for the periods before and after the introduction of ATT (2021Q2). The data on firms' financial performance spans from 2014Q3 to 2023Q2. In even-numbered columns, we include controls for a comprehensive set of pair-wise firm connections, as well as interaction terms between these connections and the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects (θ_{it} and ι_{jt}) and double cluster standard errors by firm-i and firm-j. Time is defined relative to ATT. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

data from different firms was merged by SDKs primarily based on IDFA (iOS app users' universal ID). ATT affected IDFA-based data merging. Other methods of data merging, for example, based on firms' proprietary customer IDs, persisted and may gain more adoption over time. In addition, ATT not only reduces the number of customers, whose data can be merged across firms based on IDFA, but also data representativeness. The iOS users who allowed IDFA-based tracking (between 4% and 18% in the first 12 months after ATT) may behave differently from the pre-ATT average iOS user. Such discontinuity in customer profiling may force firms to temporarily rely less on the SDK-transmitted data, thus weakening the performance comovement induced by data sharing.

Turning to the comovement in financial performance in Table 3, we find that a one-standard-deviation increase in data connectedness is associated with a 0.002 increase in the comovement of earnings growth before ATT. The smaller effect of data connectedness on earnings relative to that on app activities is intuitive, as firms have business activities and sources of profits unrelated to customers' app activities. This effect is again weakened by ATT. We obtain similar results with asset turnover as the measure of financial performance.

The economic magnitudes are meaningful. Because data connectedness enters the regressions standardized, each coefficient is the effect of a one-standard-deviation increase. For asset turnover, a one-standard-deviation increase in data connectedness raises the pairwise correlation by 0.005 (0.004 with the full set of other linkages and their ATT interactions), about 4–5% of the sample-mean correlation of 0.095. For earnings growth, the effect is 0.002. The same per-

standard-deviation metric renders the linkages comparable: the data-connectedness effect is about 25% of the product-market (horizontal) overlap effect for asset turnover and about 40% for earnings growth (Table A.4 in the Internet Appendix reports the full results).

To interpret the magnitudes in dollar terms, note that the best linear predictor of firm j's earnings growth given firm i's growth is $\mathbb{E}[g_j \mid g_i] = \rho_{ij} (\sigma_j/\sigma_i) g_i$, where ρ_{ij} denotes the correlation between firms i and j. Consider firms i and j with the same volatility, i.e., $\sigma_j/\sigma_i = 1$, in which case $\mathbb{E}[g_j \mid g_i] = \rho_{ij} g_i$. A move in firm i's quarterly earnings growth from the median to the 90th percentile, $\Delta g_i = 1.52$ (from 0.02 to 1.54; see Panel B of Table 1), changes the expected earnings growth of firm j by $1.52\rho_{ij}$. According to our regression coefficient, a one standard deviation increase of i and j's data connectedness increases ρ_{ij} by 0.002. This stronger comovement increases firm j's expected quarterly earnings growth by 1.52×0.002, i.e., 0.30 percentage points, or 1.2 percentage points at an annual rate.

Finally, we examine stock-return comovement. Similar to previous tables, Table 4 reports the results for raw returns (columns 1-2), abnormal returns based on CAPM (columns 3-4), and DGTW-adjusted returns (columns 5-6), with and without other types of firm linkages as control variables. The results are largely consistent across the three types of returns. For instance, based on the DGTW-adjusted returns in Column 6, a one-standard-deviation increase in data connectedness is associated with a 0.002 increase in return comovement. After ATT, this effect is reduced significantly. Notably, none of the alternative linkages experience a sharp drop in their effects on return comovement after ATT, except for mobile-user overlap (see Table A.5 in the Internet Appendix), which captures firms' app similarity and therefore can be correlated with data connectedness.

Comparing across the tables, the effects of data connectedness on the comovement in financial performance and stock returns are smaller than its effect on the comovement in app performance, relative to the effects of other firm linkages. This ordering is consistent with the data-sharing mechanism: shared data affects app engagement directly, while earnings, sales, and returns aggregate data-driven and non-data activities. Financial outcomes may also respond with a lag; consistent with this, the cyberattack spillovers in Section 3.2 appear immediately in peers' customer engagement but only at the two-quarter horizon in earnings growth. Finally, in the full results, the ATT interactions of some non-data linkages are also sizable; this is expected because these linkages partially overlap with data connectedness, so their interactions can absorb part of the same shock. The informative comparison is the horse race: data connectedness and its ATT interaction survive the inclusion of all other linkages and their ATT interactions, and no other linkage reproduces the joint pattern of a positive baseline effect and a negative ATT interaction across all performance dimensions.

TABLE 4: **Comovement in Stock Returns**

	raw return		return - CAPM		return - DGTW	
	(1)	(2)	(3)	(4)	(5)	(6)
data connectedness	0.004***	0.002***	0.005***	0.003***	0.003***	0.002***
	(6.52)	(3.64)	(6.71)	(4.35)	(6.25)	(3.76)
ATT × data connectedness	-0.002***	-0.002***	-0.003***	-0.003***	-0.002**	-0.002***
	(-3.00)	(-3.05)	(-3.86)	(-3.95)	(-2.53)	(-2.67)
Other linkages	N	Y	N	Y	N	Y
ATT × other linkages	N	Y	N	Y	N	Y
Observations	5,748,100	5,748,100	5,645,298	5,645,298	4,993,880	4,993,880
R-sq	0.442	0.449	0.097	0.110	0.022	0.028

NOTE.—Table 4 shows the relationship between firm data connectedness and return comovement. Each observation represents a firm pair at a specific point in time. For each firm pair, return comovement is calculated as the correlation between firms' monthly returns over non-overlapping 12-month windows, defined relative to the introduction of ATT in April 2021, for raw returns, CAPM abnormal returns, and DGTW-adjusted returns. In even-numbered columns, we include controls for a comprehensive set of firm-pair connections, along with interaction terms between these connections and the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects (θ_{it} and ι_{jt}) and double-clustered standard errors by firm-i and firm-j. Time is defined relative to ATT. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Robustness and placebo tests. In the Internet Appendix D, we consider an array of alternative measures of data connectedness and regression specifications. Our results remain robust. First, our baseline measure of firms' data connectedness relies on the top 50 SDKs. For robustness, we verify that the results hold when restricting to the top 20 SDKs. Second, we recompute data connectedness from binary SDK-presence vectors—an SDK either is or is not installed in a firm's apps, with no DAU-weighting (i.e., the pure extensive margin). Third, while the baseline measure treats all SDKs equally, our robustness tests allow for two sources of heterogeneity. We start by weighting SDKs by their market share, giving greater influence to widely used SDKs. We then scale s_{iakt} (used in ρ_{ijt}^{data} in Equation (1)) by a measure of data-collection intensity, reflecting how actively each SDK acts as a data conduit. This adjustment accounts for the fact that firms with a similar number of app users may transmit different volumes of data. Finally, we consider averaging ρ_{ijt}^{data} over different time periods (our main results are based on the pre-ATT average values). We find similar results as the data conduits (SDK overlap) remain stable over time.

Moreover, we explore two alternative samples. First, we exclude firms that are themselves SDK providers, as these firms may play confounding roles in the data economy. Second, we exclude firm pairs with strong product-market overlap and focus on firms' comovement induced by cross-industry data sharing. This addresses potential concerns about firms' willingness to share data with their competitors (see our discussion of Figure 3 in Section 2). In both cases, our estimates remain very similar to the baseline. We also show that the results remain highly significant when we cluster standard errors at alternative levels.

We also conduct a placebo test in Table A.6 of the Internet Appendix. The test speaks to two potential concerns: reverse causality—firms that are already connected through unobservable channels may align their app infrastructure and thereby adopt similar SDKs—and common developers—two firms may appear connected simply because they hire the same third-party app developers, who reuse their preferred toolkits. Using the same approach as in Section 2.2, we measure firms' SDK overlap based on the shared SDKs unrelated to data sharing, those enabling media display and in-app messaging. The channel of common developers or infrastructure alignment generates overlap in these non-data sharing SDKs as strongly as in the data-sharing SDKs. Yet neither this alternative SDK overlap nor its interaction with the ATT indicator explains firms' performance comovement. As a reminder, our main results have already validated that our data connectedness measure captures data sharing: its interaction with the post-ATT dummy shows a dampening effect from ATT, while this dampening effect of ATT is absent for other inter-firm linkages. The goal of these placebo tests is to provide further support for our methodology.[33]

Finally, we ask whether data connectedness reflects firms' common exposure to social-media platforms rather than data sharing. By construction, our measure is based only on data-sharing SDKs, so social-functionality components—social-media login and social-media sharing kits—do not enter it.[34] Nevertheless, several data-sharing SDKs are operated by social-media platforms. Firms that adopt these SDKs can be jointly exposed to shocks to the social-media platforms, which generates performance comovement unrelated to data sharing.[35] To address this concern, we construct a *social-SDK connectedness* measure, built in the same way as data connectedness in (1), and re-estimate Equation (2), controlling for social-SDK connectedness. The results in Tables D.1, D.2, and D.3 show that adding social-SDK connectedness leaves our estimates essentially unchanged.

Nonlinear network effects, higher-order connections, and clustering. Three further exercises characterize the structure of the network effects; the estimation results and the construction of the measures are reported in Internet Appendix D.3. The first exercise concerns nonlinearity. When data is used for prediction, diminishing returns on data (including other firms' data) are naturally expected (Farboodi and Veldkamp, 2025). We therefore add a squared data-connectedness term, and its ATT interaction, to Equation (2) for app downloads and DAU. Before ATT, the linear term

[33]Another potential placebo is to consider Android connectedness. However, because firms sometimes deploy common SDKs across the iOS and Android versions of their apps, Android connectedness is not a clean placebo for the iOS connectedness. We instead include Android connectedness as a control to isolate variation in iOS connectedness beyond firms' shared SDK infrastructure across iOS and Android systems. Although the iOS estimates attenuate somewhat, they remain largely statistically significant. The results are reported in Internet Appendix D.

[34]Social SDKs allow integration with social-media platforms. Embedding one lets an app offer social-media login ("Sign in with Facebook"), share content, or embed other platform features (friend lists, widgets, etc.)

[35]For example, if both firms' apps enable information sharing via Meta's social-media applications, a negative shock to Meta's user engagement may cause a decline in the use of these sharing functions at both firms' apps, which in turn may reduce customers' overall app engagement at both firms.

is positive and the squared term is negative, consistent with diminishing returns to data (Table D.4). The ATT interactions carry the opposite signs with comparable magnitudes.

The second exercise focuses on the higher-order propagation emphasized by our model for dynamic valuation of data assets in Section 4. In our model, the network adjacency matrix, Γ, summarizes the strength of bilateral connections with the ij-th element capturing the pairwise data connectedness in our empirical setting. We construct a second-order data connectedness measure from the square of Γ—that is, firms i and j are second-order connected if the ij-th element of Γ^2 is not zero; it captures the linkage between two firms through common intermediaries, and we include it alongside the direct linkage, each with its ATT interaction. Both terms enter positively and significantly: indirectly connected firms comove over and above their direct connectedness, while ATT attenuates both channels (Table D.5). The result is not specific to the second order: replacing the second-order term with the average of the second- through fifth-order connections yields the same pattern (Table D.6). The estimates thus indicate that the indirect component of network propagation carries independent explanatory power for comovement.

The third exercise examines the network topology. For each firm, we compute the local clustering coefficient—the share of a firm's neighbors in the data network that are themselves directly connected—and measure a pair's local density by the product of the two firms' clustering coefficients; with the firm-by-time fixed effects absorbing each firm's own clustering, this product captures the pair-level component of local density. Firm pairs embedded in dense local subnetworks comove more, and the effect of bilateral data connectedness is itself amplified by local density; ATT attenuates every layer of this structure—the direct effect of data connectedness, the amplification (the interaction between data connectedness and local clustering), and the own effect of local clustering (Table D.7). An alternative density measure based on the k-core number of the firm pair displays the same amplification pattern (Table D.8). Spillovers are thus strongest where the data-sharing network is locally dense, in line with the network as a comovement mechanism.

The advertising channel. The customer profiles that firms obtain from data-sharing SDKs are often used to target customers in marketing—the channel emphasized by our model in Section 4, in which data sharpens firms' knowledge of customer preferences and thereby helps stimulate customer engagement. If shared data is indeed the source of the comovement, its footprint should appear in firms' advertising: in how much firms spend, in how they allocate advertisements, and in what an advertising dollar achieves. In the Internet Appendix E, we examine each margin in turn.

We document advertising comovement using the gravity design of Equation (2): for each firm pair, the outcome is the correlation between firm i's and j's advertising variables, measured separately in the pre- and post-ATT periods. Nielsen Ad Intel measures *advertising spend* across media types—digital and non-digital. Apptopia's data covers advertisements hosted in apps in a

smaller sample, but they reveal *advertising strategy*: how firms allocate advertisements between apps and other channels. Finally, following Shapiro, Hitsch, and Tuchman (2021b), we estimate each firm's *advertising effectiveness*—the elasticity of its app performance with respect to its own advertising stock (a geometrically decaying sum of past advertising spend)—in rolling windows.

Table E.1 shows that an increase in data connectedness is associated with a significant increase in the correlation of two firms' digital advertising spend; for non-digital spend, the coefficient is close to zero and statistically insignificant. The non-digital spend serves as a placebo, ruling out comovement in advertising spend due to common shocks that move firms' overall advertisement budget together rather than data sharing.

Data connectedness also aligns firms' advertising strategies—the tilt of advertising toward the app channel, where the SDK-propagated consumer profiles are most directly valuable for customer targeting. For each firm, we measure the share of its advertising spend placed with app publishers—the app owners that sell advertising space inside their apps—and the share of its advertising impressions (individual displays of an advertisement to users) delivered inside apps. Both variables capture how strongly the firm's advertising leans toward apps relative to the web and other channels. Table E.2 shows that data connectedness significantly increases the comovement of both app-publisher spend shares and app-impression shares, while ATT reduces the effects significantly.

Finally, we examine advertising effectiveness in Table E.3: if shared data improves customer targeting, connected firms' returns on advertising should comove, and should decouple once data sharing is weakened. Data connectedness indeed explains comovement in the estimated advertising elasticities of downloads and monthly active users (MAU), and ATT reduces this comovement.[36]

3.2 Cyberattack ripple effects

Performance comovement emerges from shock propagation across firms—that is, a shock that originates from one firm can impact another firm through data connectedness, thus generating comovement. Next, we characterize such shock propagation. Specifically, we examine the propagation of cyberattacks, which is a rather salient type of shock in the data economy.

We collect major cyberattack events using data from Advisen that covers more than 90,000 cyber events between 2000 and 2023 from publicly verifiable sources. We identify 22 major cyber events involving at least 10 million exposed records (listed in Table B.1 in the Internet Appendix).

For a focal firm involved in a cyber event τ, we define a peer firm i's exposure as:

$$\text{Exposure}_{i\tau} = \frac{\sum_P \rho_{i\tau}^{\text{data},P} DAU_\tau^P}{\sum_P \sum_j \rho_{ij}^{\text{data},P} DAU_j^P}$$

[36]Note that the coefficient magnitudes can be reduced by measurement error as the dependent variable is an estimated elasticity of a firm's app performance with respect to its capitalized advertising spend.

TABLE 5: **Cyberattack Spillover Effects on App Performances**

	log(downloads)		log(DAU)	
	(1)	(2)	(3)	(4)
cyber event × high exposure	−0.116***	−0.107***	−0.148***	−0.137***
	(−6.60)	(−5.31)	(−7.28)	(−5.75)
cyber event × other linkages	N	Y	N	Y
Observations	81,373	81,373	81,373	81,373
R-sq	0.955	0.955	0.957	0.957

NOTE.—Table 5 presents the cross-firm spillover effects of major cyber events using a stacked difference-in-differences (DiD) specification in 16-quarter event windows. Major cyber events are defined as those resulting in the exposure of over 10 million records. A firm k is considered an important peer if Exposure$_{ik} \geq 0.01$, corresponding to the 75th percentile of the exposure distribution. Firms with Exposure$_{ik} < 0.01$ are considered as control firms. Each regression includes the following firm-level controls: firm size (log of assets), long-term debt, tangible assets, cash, R&D expenditure, and capital expenditures, with the latter five normalized by total assets. Additionally, we control for firm×event fixed effects and event-specific relative quarter fixed effects. Standard errors are double-clustered by event and firm. t-statistics are provided in parentheses. Statistical significance at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively.

where j represents a firm connected to firm i via data sharing, and P represents platforms, $P \in \{\text{iOS, Android}\}$, and $\rho_{i\tau}^{\text{data},P}$ is data connectedness between firm i and the focal firm involved in event τ, as defined in Equation (1).[37] A higher value of Exposure$_{i\tau}$ implies that the impacted firm's data is important for firm i relative to all the other connected firms j. A firm i is considered to be highly exposed to the event if Exposure$_{i\tau} \geqslant 0.01$, corresponding to the 75th percentile of the distribution. Firms with Exposure$_{i\tau} < 0.01$ are considered control firms. We assign an indicator variable, *high exposure*$_{i\tau}$, which takes the value of 1 for highly exposed firms and 0 otherwise.

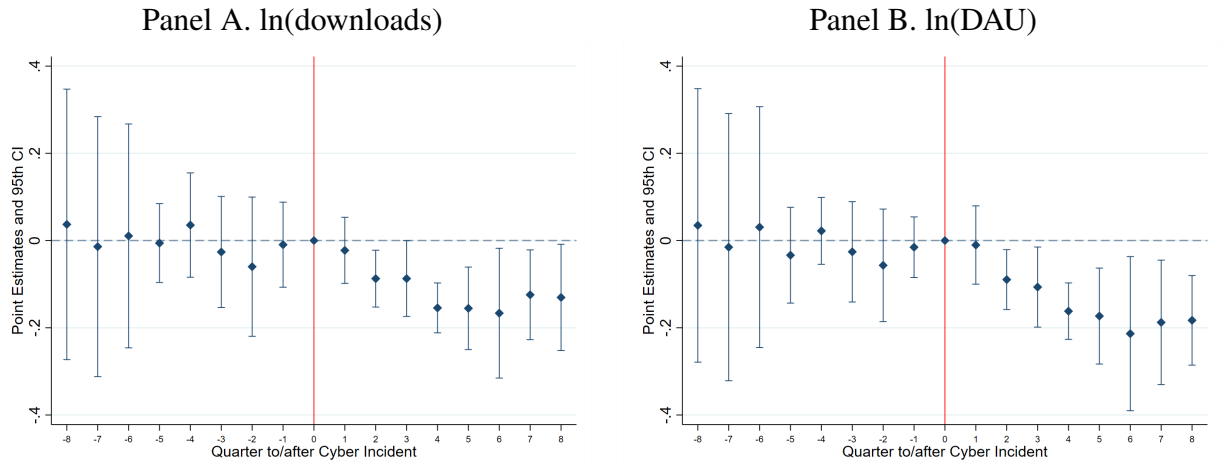
We estimate the following equation for the 22 cyber events using a stacked difference-in-differences approach:

$$Y_{i\tau t} = \alpha + \beta_1 cyber_\tau \times high\ exposure_{i\tau} + \beta_2 cyber_\tau \times \rho_{i\tau}^{other} + \beta_3' \mathbf{C_{it}} + \theta_{i\tau} + \iota_{\tau t} + \varepsilon_{i\tau t}, \quad (3)$$

where τ indexes events, i indexes firms, and t indexes the quarter relative to each event. We include 8 quarters before and 8 quarters after the event. We consider two outcomes for Y_{it}: app downloads and DAU (daily active users) at firm-quarter level, both in logarithm. *cyber*$_\tau$ is an indicator variable that takes the value of 1 after the cyberattack. $\rho_{i\tau}^{other}$ includes non-data linkages between i and the focal firm in the event τ. In all specifications, we include the following firm-level controls, $\mathbf{C_{it}}$: firm size (log of assets), long-term debt, tangible assets, cash, R&D expenditure, and

[37] Note that in the regression analysis of firms' performance comovement, our data connectedness measure is based on ρ_{ij}^{iOS}, as ATT affects data collection and sharing only within the iOS ecosystem. In contrast, for the analysis of cyberattack impacts, we consider data connections between firms across both iOS and Android systems, since unlike ATT, which is an iOS-specific shock, cyberattacks affect firms regardless of their customers' operating systems and can propagate through both the iOS and Android data networks.

FIGURE 4: **Cyberattack Spillover Effects on App Performances**



NOTE.—Figure 4 shows the cross-firm spillover effects of 22 major cyber events. The dynamic DiD coefficients are obtained from estimating the dynamic version of Equation (3). A firm k is considered an important peer if $\text{Exposure}_{ik} \geq 0.01$, corresponding to the 75th percentile of the exposure distribution. Firms with $\text{Exposure}_{ik} < 0.01$ are considered as control firms. We include the following firm-level controls: firm size (log of assets), long-term debt, tangible assets, cash, R&D expenditure, and capital expenditures, with the latter five normalized by total assets. . Additionally, we control for firm$\times$event fixed effects and event-specific relative quarter fixed effects. Standard errors are double-clustered by event and firm.

capital expenditures, with the latter five normalized by total assets. Additionally, we control for firm-by-event fixed effects ($\theta_{i\tau}$) to isolate event-specific effects, as a given firm may be exposed to multiple events. We also include event-specific quarter fixed effects ($\iota_{\tau t}$) to absorb common shocks affecting all firms around the event. Standard errors are double-clustered by event and firm.

We report the estimation results in Table 5, with the first two columns presenting the results on ln(downloads) and the last two columns on ln(DAU). In all columns, β_1 is negative and statistically significant at the 1% level. The magnitude of the cross-firm spillover effect of these major cyberattacks is economically large. Specifically, relative to the control firms, the highly exposed firms experience an 11.6% drop in quarterly downloads (Column 1) and a 14.8% drop in DAU (Column 3). Controlling for the interaction of $cyber_{\tau}$ with other firm linkages, these point estimates drop slightly to 10.7% and 13.7%, as reported in Column 2 and Column 4, respectively.

Additionally, when estimating the dynamic version of Equation (3), we show in Figure 4 that our results are not driven by pre-trends between control and treated firms. The treatment effect begins to emerge immediately after the cyber event occurs, continuing through the fifth quarter post-event. Finally, it stabilizes after six quarters with a slight upward reversal.

Furthermore, the spillover effect is not driven by cybersecurity vulnerability in firms' shared software design. In Table A.8 in the Internet Appendix, we show that the spillover effect disappears when we replace the data-sharing SDK overlap with overlap in SDKs that are unrelated to data

sharing but reflect firms' overall app design, such as those that enable media display and within-app messaging. This placebo test confirms that the spillover effects emerge from data sharing.

The spillover effects extend beyond customer engagement to financial performance. Because cyber events are often small relative to firms' overall operations, and the transmission from this specific operational side to financial metrics occurs with a delay, we estimate Equation (3) with two-quarter-ahead earnings growth and asset turnover as outcomes in Table A.9 in the Internet Appendix. Relative to the control group, the exposed peers experience a decline of earnings growth and asset turnover. Controlling for the interactions of the cyber-event indicator with the full set of non-data linkages leaves the estimates essentially unchanged (Columns 2 and 4), and none of these other interactions shows a significant negative effect. The timing is intuitive: customer engagement deteriorates immediately after the event, whereas accounting outcomes deteriorate with a lag—consistent with the shock propagating from customer engagement into financial metrics.

Our findings provide the first evidence of how shocks to firms spill over to their peers through the data-sharing network. Such shock propagation applies to firms' operational, financial, and stock-market performances. In the Internet Appendix, we conduct event studies on data-connected firms' stock performances surrounding cyberattacks on the focal firm.[38] Firms that are data-connected to the focal firms experience a 2–3% decline in cumulative (CAPM-based) abnormal returns in the month. Using other benchmark models for computing abnormal returns, such as the Fama–French three-factor model and DGTW, yields qualitatively similar results.

4 A Network Model of Data Economy

We develop a model of data economy where firms are interconnected through data sharing. Data improves firms' understanding of customer preferences and thereby stimulates customer engagement and contributes to revenue generation. The model captures our empirical findings on firms' performance comovements and cross-firm shock propagation in Section 3. Importantly, the model serves as a measurement tool. It embeds the empirically measured network of data conduits, Γ, and firm-level data stocks, proxied by DAU, into a dynamic setting with direct and indirect data spillovers across firms. It converts the resulting cash flows into present values, delivering a valuation of data as a networked asset and permitting counterfactual comparisons before and after ATT (Section 5). As we introduce each ingredient below, we note the measurement task it supports.

[38] In Figure A.4, Panel A and Panel B use the incident dates and notice dates, respectively, as the event dates. Market participants were likely to be made aware of those events between the incident dates when cyberattacks happened and notice dates when firms made announcements typically following preliminary investigations.

4.1 The setup

Data dynamics. The economy has N firms. We consider firm i's problem, $i \in \{1, ..., N\}$. Let $\delta_{i,t}$ denote the firm's stock of raw data on its customers. The firm collects data from interaction with customers ("customer engagement"), denoted by $y_{i,t}$, and $\delta_{i,t}$ evolves as

$$d\delta_{i,t} = \theta y_{i,t} dt + \delta_{i,t} \left(\mu_\delta dt + \sigma_{i,\delta} dz_{i,t} \right), \tag{4}$$

where the terms in the bracket are a stochastic growth rate, and $z_{i,t}$ is a standard Brownian motion that is independent across firms.[39] Data depreciates when the last term is negative. Faster customer turnover is associated with faster data depreciation, as data is typically used for customer profiling.

The first term on the right side of (4), $\theta y_{i,t} dt$, captures the idea that data is generated by customer activities, where θ (> 0) represents data collection efficiency. An e-commerce platform collects data on consumers by observing their transactions and search activities. For a retail company, promotions of products generate interactions with customers, and the feedback from customers is informative of their preferences. Software companies learn about users' preferences through their usage patterns. Our empirical setting covers many industries. Accordingly, we set up our model to be sufficiently generic to highlight the commonality among these firms.

In the law of motion (4), customer engagement, $y_{i,t}$, generates new data. Data in turn helps firms stimulate customer activities by making firms more informed—that is, data contributes to $y_{i,t}$.[40] This force leads to the self-perpetuating growth of data in the spirit of Farboodi and Veldkamp (2025). Next, we discuss how $y_{i,t}$ is determined and its dependence on data.

In our model, data has two forms, the raw data collected at the firm level, $\delta_{i,t}$, and the processed and aggregated data, $D_{i,t}$, that corresponds to the comprehensive customer profiles that firms obtain from the SDKs in our empirical setting:

$$D_{i,t} = \sum_{j=1}^{N} \gamma_{ij} \delta_{j,t} = \{\Gamma \bar{\delta}_t\}_i, \tag{5}$$

where γ_{ij} represents the data conduit from firm j to i facilitated by the shared SDKs. Note that consistent with our empirical findings, the data conduits are stable over time, so γ_{ij} does not have a time subscript. As discussed in Section 2, the universal IDs allow data from different firms to be merged into customer profiles. Let $\bar{\delta}_t$ denote the column vector of all firms' raw data:

[39] A negative shock may reflect a direct loss of data, for example, due to cyberattack. Florackis, Louca, Michaely, and Weber (2022) measure cybersecurity risk from corporate disclosure. In contrast, following a positive shock, the firm gains more information on its customers, for example, through an increase in product reviews.

[40] For example, a retail manufacturer can interact with its customers and stimulate customer activities only if it knows where the customers shop, post reviews, and view advertisements in physical locations and on the internet. For software companies, knowing customers' preferences and habits is critical for increasing time spent on the product.

$\bar{\delta}_t = [\delta_{1,t},\ ...,\ \delta_{N,t}]^\top$. We introduce the notation Γ where the ij-th element is γ_{ij} and the row sums are equal to one. Multiplying Γ on $\bar{\delta}_t$ we obtain a vector of SDK-transmitted data, and the i-th element of $\Gamma\bar{\delta}_t$, denoted by $\{\Gamma\bar{\delta}_t\}_i$, is firm i's information on its customers, $D_{i,t}$.[41]

Firm i faces a unit mass of customers and uses its information to forecast their preferences. A customer's preference at time t, denoted by $\widetilde{\eta}_{i,t}$, is drawn independently from a Gaussian distribution, $\mathcal{N}(0,\ \sigma_\eta^2)$, and firm i receives a Gaussian signal, denoted by $h_{i,t}$. Firm i forms a forecast, $a(h_{i,t})$, that minimizes the mean-squared error:

$$\lambda_{i,t}^2 := \left(\min_{a(h_{i,t})} \mathbb{E}\left[(a(h_{i,t}) - \widetilde{\eta}_{i,t})^2\right]\right)^{-1}, \tag{6}$$

where $\lambda_{i,t}^2$ is the precision. The informativeness of $h_{i,t}$ is constrained by $\mathcal{H}\left(\widetilde{\eta}_{i,t}\right) - \mathcal{H}\left(\widetilde{\eta}_{i,t}|h_{i,t}\right) \leq \ln\left(\kappa D_{i,t}\right)$, where $\mathcal{H}\left(\widetilde{\eta}_{i,t}\right)$ and $\mathcal{H}\left(\widetilde{\eta}_{i,t}|h_{i,t}\right)$ represent the unconditional and h-conditional entropy of $\widetilde{\eta}_{i,t}$, respectively (Sims, 2003). We assume $\kappa D_{i,t} \geq 1$ so that the information constraint is well defined, where κ scales $D_{i,t}$ into information units in logarithm. This constraint restricts the entropy reduction brought by the signal, $h_{i,t}$, and can be equivalently written as

$$\lambda_{i,t} \leq \kappa D_{i,t}/\sigma_\eta. \tag{7}$$

Under the classic solution, $a(h_{i,t}) = \mathbb{E}\left[\widetilde{\eta}_{i,t}|h_{i,t}\right]$, the information constraint binds: $\lambda_{i,t} = \kappa D_{i,t}/\sigma_\eta$.

Note that, a customer's $\widetilde{\eta}_{i,t}$ is drawn independently. Firm i's signal on $\tilde{\eta}_{i,t}$ is $h_{i,t}$, and facing a continuum of customers, the firm has a continuum of signals. Therefore, what matters for firm i is not the forecasting error for a single customer's preferences but the overall forecasting precision, which improves through an increase in $D_{i,t}$, due to either a higher $\delta_{i,t}$ (firm i's own data) or a higher $\delta_{j,t}$ (a connected peer's data). In summary, a firm's information set includes a continuum of customer-level signals, which enters the rest of the model ingredients only through the precision index $\lambda_{i,t}$, and the data vector $\bar{\delta}_t = [\delta_{1,t},\ ...,\ \delta_{N,t}]^\top$, which, through $D_{i,t}$, determines $\lambda_{i,t}$.

We specify customer engagement, $y_{i,t}$, as follows:

$$y_{i,t} = \alpha\lambda_{i,t} + x_{i,t}, \tag{8}$$

where $\alpha > 0$, implying the self-perpetuating growth of data: firm i's raw data $\delta_{i,t}$ enters $D_{i,t}$ in (5), which in turn contributes to $y_{i,t}$ through the forecasting precision, $\lambda_{i,t}$, in (8), and $y_{i,t}$ generates more data $d\delta_{i,t}$ in (4). Different from Farboodi and Veldkamp (2025), our model features data

[41] Under this specification, smaller firms rely more on external data, which is in line with the evidence: Aridor et al. (2026) show that ATT reduced the click-through rate of advertisements on Meta by 37%, with the exposed e-commerce firms' revenue losses concentrated among smaller firms; and Wernerfelt et al. (2024) find experimentally that removing access to offsite data disproportionately reduces advertising effectiveness for smaller advertisers on Meta.

spillover and interconnected growth, as other firms' data affects firm i's data growth through the forecasting precision, $\lambda_{i,t}$, and then $y_{i,t}$, and vice versa.

Beyond a data-enabled level of customer engagement in the first term in (8), the firm can enhance customer engagement by an extra amount, $x_{i,t}$, through a product-design choice: a higher $x_{i,t}$ makes its product more suitable for generating customer engagement rather than cash flows (monetization). Next, we introduce the trade-off in firm i's choice of $x_{i,t}$.

Cash flows. Customer engagement does not equate to cash-flow generation but they are connected in two ways. First, as shown in (4), $\delta_{i,t}$ represents firm i's raw data stock collected from the history of customer engagement, $y_{i,t}$, that has accumulated over time subject to stochastic growth or depreciation due to customer turnover. This history of interaction with customers forms firm i's customer capital that generates cash flow $\zeta\delta_{i,t}$ at time t.[42] Note that customer capital arises from the *cumulative* customer engagement, $\delta_{i,t}$, rather than the current level of engagement, $y_{i,t}$.

The second connection (or more precisely, tension) between customer engagement and cash flows is that when firm i prioritizes engagement in product design (i.e., it increases $x_{i,t}$ in (8)), such choice reduces cash flows. For example, an app design focused on engagement may offer rich free features to maximize customers' time spent in the app. In contrast, a design that prioritizes monetization can increase revenues by moving more features behind the paywall at the expense of customer engagement. The negative impact of $x_{i,t}$ on firm i's cash flow is captured by a function $g\left(x_{i,t}, \lambda_{i,t}\right)$ with $g_x\left(x_{i,t}, \lambda_{i,t}\right) < 0$. Intuitively, we also assume that when firm i is more informed about its customers, the negative cash-flow impact of prioritizing customer engagement over monetization is mitigated, i.e., $g_{x\lambda}\left(x_{i,t}, \lambda_{i,t}\right) > 0$. For example, if a software company knows its customers better, it can prioritize engagement by offering many free features without reducing cash flows, as it can more effectively target high-purchase-intent users with its premium services.

Introducing this product-design choice serves our measurement goal rather than simply adding realism. Valuing data as a networked asset must account for firms' endogenous response to data availability and incentives to accumulate data. Because customer engagement generates data, firms face an intertemporal trade-off: product designs that sacrifice current monetization to increase engagement build data that improve monetization in the future. In addition, introducing this product-design choice allows us to examine our model's prediction out of sample. When calibrating our model, we intentionally leave out firms' product-design choices, but we will show that the calibrated model generates the empirical patterns.

[42] Following Gourio and Rudanko (2014a,b), we have cash flows linear in customer capital, consistent with evidence in Einav, Klenow, Levin, and Murciano-Goroff (2021) that firms' revenues largely scale with the number of customers.

In summary, firm i's cash flows are given by

$$F_{i,t} = \zeta \delta_{i,t} + g\left(x_{i,t}, \lambda_{i,t}\right). \tag{9}$$

Let ρ denote the discount rate. Firm i's valuation (i.e., its value function at $t = 0$) is given by

$$V^i\left(\delta_{i,0}, \{\delta_{j,0}\}_{j \neq i}\right) = \max_{\{x_{i,t}\}_{t=0}^{\infty}} \mathbb{E} \int_{t=0}^{+\infty} e^{-\rho t} F_{i,t} dt. \tag{10}$$

In summary, $D_{i,t}$ relaxes the constraint (7) on forecasting customers' preferences and thereby contributes to cash-flow generation in two ways. First, a better knowledge of customers (i.e., a higher $\lambda_{i,t}$) increases customer engagement (see (8)), and the accumulated interactions with customers ("customer capital") generate the first component of cash flow. Second, $\lambda_{i,t}$ affects the trade-off between customer engagement and monetization in product design.[43]

We follow the literature that models data as a by-product of firms' productive activities and as a key input in production (e.g., Bergemann et al., 2022; Farboodi and Veldkamp, 2025).[44] Our model extends the framework in three aspects. First, we separate productive activities in two related areas, customer engagement and cash-flow generation. The former generates data and, over time, allows customer capital to accumulate, indirectly benefiting the latter in the long run. Second, we introduce a contemporaneous trade-off between these two areas of productive activities. Third, by improving firms' forecasting precision (i.e., knowledge on customers' random preferences), data contributes positively to both areas through $\lambda_{i,t}$.[45] Separating the two areas of productive activities brings out a meaningful intertemporal trade-off: increasing $x_{i,t}$ reduces current profits but, by enhancing customer engagement, $y_{i,t}$, it increases $\delta_{i,t}$ that in turn boosts future profits.

Therefore, the choice of $x_{i,t}$ is akin to a (data-) investment decision. Our model differs from the classic investment theories (e.g., Hayashi, 1982; Abel and Eberly, 1994) in two key aspects. First, data investment has a positive externality.[46] Second, firms' investment decisions are interconnected through data sharing. In the next subsection, we show that a firm's data marginal q (the derivative of value function with respect to data stock) drives the optimal $x_{i,t}$ and incorporates the expected trajectories of data inflows from other firms. Figure 5 illustrates the three blocks of our model, data dynamics, product-design choice, and firm valuation that we solve in Section 4.2.

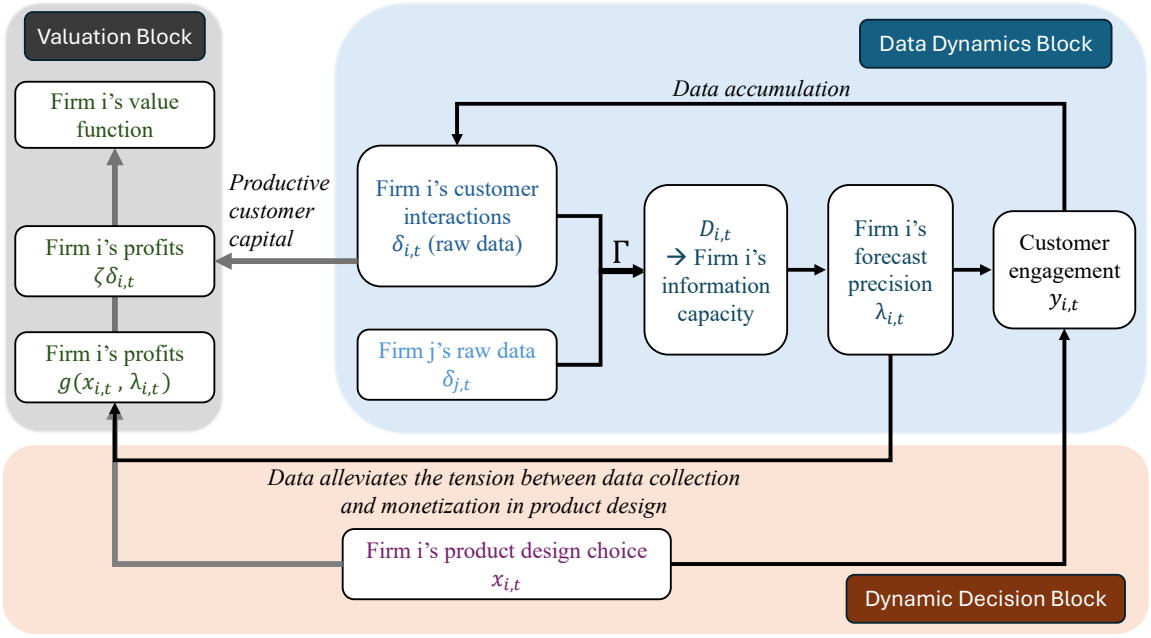
[43] Firms may use data to improve customers' willingness to pay (Ichihashi, 2020). The potential harmful impact of price discrimination is beyond the scope of this paper, as our focus is on firm dynamics rather than consumer welfare.

[44] The notion that data is a by-product of economic activity was well established in the information economics literature (e.g., Veldkamp, 2005; Ordoñez, 2013; Fajgelbaum et al., 2017).

[45] In models that do not separate productive activities in two areas, data, often modeled through forecasting precision, typically contributes directly to firms' productivity (e.g., Farboodi and Veldkamp, 2025).

[46] This stands in contrast to the traditional investment dynamics where one firm's investment often crowds out others' investment. For example, investment by one firm may increase the cost of investment inputs and financing or intensify product-market competition that other firms face (e.g., Asriyan et al., 2024).

FIGURE 5: **A Summary of Model Structure**



Discussion: the characteristics of data assets. Our model captures the three features of data as a productive asset. First, as previously discussed, data is a by-product of firms' operations and in turn contributes to firms' operations. Second, data is non-rival: sharing data with other firms does not prevent a firm from using its data or cause it to lose data (e.g., Jones and Tonetti, 2020). Third, data has externality: data on one firm's customers can be informative about other firms' customers (e.g., Ichihashi, 2020). The second and third features explain why firms are willing to share data.

Firm-level data externality deviates from externality at the individual level—one person sharing data reveals other people's attributes. Choi et al. (2019) and Acemoglu et al. (2022) point out that data externality leads to excessive data sharing and collection. We will show that in our model, firms under-invest in data accumulation under positive data externalities (and thus monetize excessively rather than prioritizing customer engagement). The difference lies in the fact that in Choi et al. (2019) and Acemoglu et al. (2022), it is the consumers who decide on sharing data, while in our model, firms set the speed of data accumulation via product-design choices.

The combination of these features distinguishes data from other intangible assets. One example is R&D, which also generates knowledge that is a non-rival asset and has positive spillover effects (i.e., the second and third features of data). However, R&D is typically not a by-product of firms' interactions with customers.[47] In contrast, data is generated as a by-product of business operations. While firms may incur costs to stimulate customer activities and generate more data, the

[47]R&D is not a by-product of customer engagement. It often requires deliberate, resource-intensive investment, separate from the normal production process (Corrado, Hulten, and Sichel, 2005; Hall, Mairesse, and Mohnen, 2010).

baseline level of data generation is free, and therefore, through data sharing, there exists a baseline positive externality of one firm's production and the associated data generation on other firms.

4.2 Equilibrium

Firm i's state variables include its own raw data stock, $\delta_{i,t}$, and other firms' $\delta_{j,t}$ (i.e., the full vector, $\bar{\delta}_t = \{\delta_{j,t}\}_{j=1}^N$). The Hamilton-Jacobi-Bellman (HJB) equation for the value function of firm i at time t, denoted by $V^i(\bar{\delta}_t) = V^i\left(\delta_{i,t}, \{\delta_{j,t}\}_{j\neq i}\right)$, is given by

$$\rho V^i\left(\delta_{i,t}, \{\delta_{j,t}\}_{j\neq i}\right) = \max_{x_{i,t}} \zeta\delta_{i,t} + g\left(x_{i,t}, \lambda_{i,t}\right) + V^i_{\delta_{i,t}}\left[\theta(\alpha\lambda_{i,t} + x_{i,t}) + \mu_\delta\delta_{i,t}\right] + \frac{1}{2}V^i_{\delta_{i,t}\delta_{i,t}}\delta^2_{i,t}\sigma^2_{i,\delta}$$
$$+ \sum_{j\neq i}\left[V^i_{\delta_{j,t}}\left[\theta(\alpha\lambda_{j,t} + x_{j,t}) + \mu_\delta\delta_{j,t}\right] + \frac{1}{2}V^i_{\delta_{j,t}\delta_{j,t}}\delta^2_{j,t}\sigma^2_{j,\delta}\right]. \tag{11}$$

To highlight the intertemporal linkage between $x_{i,t}$ and data accumulation, we substitute out $y_{i,t}$ in the drift of $\delta_{i,t}$ using (8), i.e., $y_{i,t} = \alpha\lambda_{i,t} + x_{i,t}$, where, as previously discussed, $\lambda_{i,t} = \kappa D_{i,t}/\sigma_\eta$.[48]

We characterize a feedback (Markovian) Nash equilibrium under complete information (Başar and Olsder, 1999)—that is, a Markov perfect equilibrium in which firms' strategies condition only on the payoff-relevant state vector, $\bar{\delta}_t$. The equilibrium objects include, for any i ($\in \{1, ..., N\}$), firm i's strategy that is a mapping from its state variables $\{\delta_{j,t}\}_{j=1}^N$ to the optimal choice of $x_{i,t}$, firm i's value function, and the law of motion of the state variables. Formally, let $x_i(\bar{\delta})$ denote firm i's policy function. An equilibrium is a collection of policy functions and value functions, $\{x_i(\bar{\delta}),\ V^i(\bar{\delta})\}_{i=1}^N$, such that, for each firm i, given the other firms' policy functions, $x_i(\bar{\delta})$ attains the maximum in the HJB equation (11), and $V^i(\bar{\delta})$ is the value function under the implied law of motion of $\bar{\delta}_t$; the feedback structure implies that the equilibrium is subgame perfect.[49] All firms observe the state variables $\{\delta_{j,t}\}_{j=1}^N$ under complete information.

Note that even though at time t, firm i's information set includes both the state variables $\{\delta_{j,t}\}_{j=1}^N$ and the customer-level signals for forecasting customer preferences, what matters for the equilibrium dynamics are $\{\delta_{j,t}\}_{j=1}^N$. As previously discussed, the forecasting problem is reduced to the determination of precision, $\lambda_{i,t} = \kappa D_{i,t}/\sigma_\eta$, which only depends on $D_{i,t} = \sum_{j=1}^N \gamma_{ij}\delta_{j,t}$. It is the precision of signal (rather than the signal itself) that enters the rest of the model ingredients after the firm solves the optimal forecasting problem—that is, the customer-level signals are not state variables of the dynamic problem, as their effect on the intertemporal problem operates only

[48] Note that given the state variables $\{\delta_{j,t}\}_{j=1}^N$, firm i knows $\lambda_{j,t}$ as it is a linear combination of $\{\delta_{j,t}\}_{j=1}^N$ through $D_{j,t}$. Therefore, given $\{\delta_{j,t}\}_{j=1}^N$, it does not need to track $\lambda_{j,t}$. In addition, as in the standard Nash framework, all firms' mappings from the state variables to optimal choices $x_{j,t}$ (i.e., firms' optimal strategies in equilibrium) are assumed to be known by all firms, so given $\{\delta_{j,t}\}_{j=1}^N$, firm i does not need to track $x_{j,t}$ ($j \neq i$).

[49] Admissibility requires feasible state and control paths almost surely and finite expected discounted payoffs.

through $\lambda_{i,t}$, which is a function of the state vector, $\bar{\delta}_t = \{\delta_{j,t}\}_{j=1}^N$. In summary, given $\bar{\delta}_t$, $\lambda_{i,t}$ and $x_{i,t}$ are determined, which feeds into the law of motion of state variables through (4) and (8).

A related question is whether firm i could use less information—its own data stock, $\delta_{i,t}$, together with $D_{i,t}$—rather than on the full vector $\bar{\delta}_t$. It cannot: $D_{i,t}$ is not a sufficient statistic for firm i's problem. Different values of the vector $\bar{\delta}_t$ can generate the same $D_{i,t}$ for firm i while implying different $D_{j,t}$ for the other firms and thereby—through the other firms' decisions—different paths of $\bar{\delta}_t$ and different values for firm i. The closed-form value function in Proposition 2 below reflects this: firm i's value is a linear function of the entire vector $\bar{\delta}_t$ that is not proportional to $D_{i,t}$.[50]

The following proposition characterizes the optimal $x_{i,t}$ through a Q-theory of data investment. The intertemporal trade-off is between the impact on current profits under $g_x(x_{i,t}, \lambda_{i,t}) < 0$, and data accumulation evaluated at the marginal value of data (or "*data marginal q*"), $V_{\delta_{i,t}}$.

Proposition 1 (Q-theory of Data Accumulation) *The first-order condition for $x_{i,t}$ is given by*

$$-g_x(x_{i,t}, \lambda_{i,t}) = V^i_{\delta_{i,t}} \theta \,. \tag{12}$$

The marginal cost of stimulating customer activities at the expense of profit generation is equal to the marginal benefit (marginal q) of data, $V_{\delta_{i,t}}$, multiplied by data generation efficiency, θ.

Data functions as a form of productive capital, with a firm's product-design choices mirroring investment decisions.[51] The marginal q of a firm's data drives the choice of $x_{i,t}$, i.e., whether to prioritize customer engagement and data collection over monetization in product design.[52] A key distinction from the traditional investment models is that firms' choices of $x_{i,t}$ are interconnected through data sharing. Other firms' data, $\delta_{j,t}$, enters firm i's optimality condition (12) via firm i's information on its customers, $D_{i,t}$ and the precision of its forecast of customer preferences, $\lambda_{i,t}$.

To sharpen the intuition, we adopt the following functional form $g(x_{i,t}, \lambda_{i,t}) = \zeta \log(\phi_\lambda \lambda_{i,t} - \phi_1 x_{i,t})$ that satisfies the properties of $g(x_{i,t}, \lambda_{i,t})$ discussed in Section 4.1 (i.e., $g_x(x_{i,t}, \lambda_{i,t}) < 0$ and

[50] Decomposing the state-dependent component of the valuation in Proposition 2, $\zeta\{(\hat{\rho}\mathbf{I} - \beta\Gamma)^{-1}\bar{\delta}_t\}_i = (\zeta/\hat{\rho})\,\delta_{i,t} + (\beta\zeta/\hat{\rho}^2)\{\sum_{k=0}^{\infty}(\beta\Gamma/\hat{\rho})^k D_t\}_i$, where D_t is the column vector whose j-th element is $D_{j,t}$. To compute its value—one of the equilibrium objects—firm i must track the composite signals of all firms, each of which is a different linear combination of $\bar{\delta}_t$. Therefore, firm i has to essentially track the full vector $\bar{\delta}_t$ as its state variables in equilibrium.

[51] Increasing $x_{i,t}$ to stimulate customer engagement and data accumulation at the expense of monetization is a form of "active experimentation" in line with Farboodi et al. (2019). In Farboodi et al. (2019), the impact of experimentation on profits can be positive or negative (rather than always negative as in our model), but the impact is negative at the optimum: the optimal scale of experimentation (the choice of production scale in their model) is sufficiently large such that the marginal impact on current profits is negative. The firm accepts the negative impact on profits because the marginal value of data acquired through experimentation is positive. We explicitly focus on the region where such an explicit trade-off between current profits and data acquisition emerges.

[52] In our model, firms adjust product design while facing an intertemporal trade-off between current cash flows and future cash flows that benefit from data accumulation. In Chen (2026), firms face a different trade-off: improving product quality is costly but earns customers' attention that can be monetized through selling advertisements.

$g_{xx}(x_{i,t}, \lambda_{i,t}) < 0$). We define $\phi_0 = \phi_\lambda \kappa / \sigma_\eta$ so that given $\lambda_{i,t} = \kappa D_{i,t}/\sigma_\eta$, we have $g(x_{i,t}, \lambda_{i,t}) = \zeta \log(\phi_0 D_{i,t} - \phi_1 x_{i,t})$, which directly depends on $D_{i,t}$. Under this functional form, we obtain closed-form solutions of the value function. Below we define the *user cost of data*:

$$\hat{\rho} = \rho - \mu_\delta. \tag{13}$$

In (4), a negative μ_δ represents data depreciation—stale information on customer behavior is less informative due to customer turnover. Therefore, $\hat{\rho}$ resembles the *user cost of capital* in investment theory, i.e., the sum of required rate of return on capital (discount rate) and depreciation rate. The next proposition summarizes the solution of value function. Appendix F provides the proof.

Proposition 2 (Network-Augmented Gordon Growth Formula) *In the admissible interior Markov equilibrium, under the parameter condition,*

$$\beta = \theta \left(\frac{\alpha \kappa}{\sigma_\eta} + \frac{\phi_0}{\phi_1} \right) < \hat{\rho}, \tag{14}$$

firm i's value at time t is given by

$$V_{i,t} \equiv V^i(\bar{\delta}_t) = \zeta \left\{ (\hat{\rho}\mathbf{I} - \beta\Gamma)^{-1} \bar{\delta}_t \right\}_i + v_{i,0}, \tag{15}$$

where $v_{i,0}$ is constant and the operator, $\{\cdot\}_i$, picks the i-th element of a vector.

In the absence of data sharing—that is, when $\Gamma = \mathbf{I}$ ($\gamma_{ii} = 1$ and $\gamma_{ij} = 0$ for $j \neq i$)—each firm relies on its own data, and firm i's valuation reduces to the standard Gordon growth formula:

$$V_{i,t} = \left(\frac{\zeta}{\hat{\rho} - \beta} \right) \delta_{i,t} + v_{i,0},$$

where ζ is the cash flow per unit of $\delta_{i,t}$, and $\hat{\rho}$ and β play the roles of discount rate and growth rate, respectively.[53] Under data sharing, the firm-level growth is replaced with "communal growth" of the whole data economy, which in turn depends on Γ, the data-conduit matrix.

The composite parameter for growth, β, reflects a sequence of events that lead to firm growth. Below we explain each step. All firms' data is recorded in the vector $\bar{\delta}_t$. Firms' information on customers is given by $D_t = \Gamma \bar{\delta}_t$, where the data conduits, Γ, allow raw data to be transferred and merged across firms. When forecasting customer preferences, firm i's precision, $\lambda_{i,t}$, depends on its information on customers, $D_{i,t}$ (the i-th element of D_t), through κ/σ_η, and then, $\lambda_{i,t}$ drives customer engagement, $y_{i,t}$, via $\alpha\lambda_{i,t}$ (see (8)). This channel of customer engagement explains

[53] The last term, as in Abel (1985), reflects the fact that the firm value, $V_{i,t}$, is achieved under the optimized investment whose infra-marginal units, with a marginal benefit greater than the marginal cost, generate economic profits.

$\alpha\kappa/\sigma_\eta$ in β. Next, customer engagement, $y_{i,t}$, also depends on $x_{i,t}$, but increasing $x_{i,t}$ (prioritizing engagement over monetization) reduces cash flows. When ϕ_0 is higher, firm i's information on customers mitigates this negative impact, and a lower ϕ_1 also reduces this negative impact—that is, overall, a high ratio of ϕ_0/ϕ_1 allows firm i to design products that stimulate customer engagement without sacrificing current profits too much. Therefore, through this second channel of customer engagement, ϕ_0/ϕ_1 enters β. Finally, firm i grows by accumulating raw data and customer capital, and $d\delta_{i,t}$ depends on customer engagement $y_{i,t}$ through the parameter θ (see (4)). So, θ multiplies the two channels of customer engagement, $\alpha\kappa/\sigma_\eta$ and ϕ_0/ϕ_1, generating β.

The data-conduit matrix, Γ, appears in $\zeta\left\{(\hat{\rho}\mathbf{I}-\beta\Gamma)^{-1}\bar{\delta}_t\right\}_i$. This term can be decomposed as follows to show the rounds of data propagation through the data-conduit network:

$$\zeta\left\{(\hat{\rho}\mathbf{I}-\beta\Gamma)^{-1}\bar{\delta}_t\right\}_i=\frac{\zeta}{\hat{\rho}}\left\{\left(\mathbf{I}-\frac{\beta}{\hat{\rho}}\Gamma\right)^{-1}\bar{\delta}_t\right\}_i=\frac{\zeta}{\hat{\rho}}\left\{\sum_{k=0}^{\infty}\left(\frac{\beta}{\hat{\rho}}\Gamma\right)^k\bar{\delta}_t\right\}_i. \tag{16}$$

The first component, $\zeta/\hat{\rho}$, is the value of a cash-flow perpetuity without growth. The growth operator, $\sum_{k=0}^{\infty}\left(\frac{\beta}{\hat{\rho}}\Gamma\right)^k$, is applied to the data vector $\bar{\delta}_t$, incorporating direct and indirect network effects. Firm i is affected by a connected firm's data through its precision in forecasting customer preferences, which in turn influences its product-design decision, customer engagement, data collection, and growth. The variation in firm i's data then affects other firms, resulting in second-order spillover effects. The infinite sum accounts for all the direct and indirect channels of network propagation over time with $\beta/\hat{\rho}$ being the attenuation factor. The convergence of the geometric sequence is guaranteed by $\beta/\hat{\rho}<1$ under the condition (14).[54] The spillover effects weaken as the distance increases: k-degree network propagation, Γ^k, is "discounted" by $(\beta/\hat{\rho})^k$. The strength of network propagation depends on the aforementioned channels of data-driven growth, with each step captured by a component of β as previously discussed.

Our valuation metrics naturally encompass the impact of data variation at one firm on all other firms across multiple time horizons and network spillover effects of first, second, and all the higher-order degrees. This stands in contrast to the traditional cost-based method of valuing intangible capital. Broadly speaking, intangible capital exhibits positive network externalities through knowledge spillover across firms and the scope of usage across firms and industries. Data is one salient example. We provide a direct measure of spillover effects, i.e., the data-sharing network Γ, and incorporate it into a valuation framework to evaluate its economic significance.

Motivated by our findings in Section 3, the model setup directly captures the comovement in firms' app performances (customer engagement, $y_{i,t}$) and cash flows due to data sharing. In addition, after solving firms' valuations, we compute the stock return, $dR_{i,t}=dV_{i,t}/V_{i,t}$.[55] The

[54] As a reminder, the data-conduit matrix, Γ, is right-stochastic so its largest eigenvalue is one.

[55] Our valuation exercise and return calculation take the perspective of an investor with perfect information. We

next corollary shows that data sharing induces return comovement. Appendix F provides the proof.

Corollary 1 (Stock Return Comovement) *The stock return of firm i at time t, $dR_{i,t} = dV_{i,t}/V_{i,t}$, has a correlation, $\rho_{r,i,j} = corr(dR_{i,t}, dR_{j,t})$, with firm j's stock return that is increasing in the dependence of firm i on firm j's data, γ_{ij}, i.e., $\partial\rho_{r,i,j}/\partial\gamma_{ij} > 0$.*

4.3 Calibration and model-implied comovements

Next, we calibrate our model to match the empirical patterns of comovements in firm performances induced by data sharing in the pre-ATT period.[56] The data-sharing connection between firms i and j, γ_{ij} (the ij-th element of Γ), is computed in Section 2, and we normalize Γ so that each row sums up to one as in our model, effectively turning Γ into a Markov transition matrix that reflects how shocks are propagated from one firm to another with different strengths. Therefore, the model takes the entire data-sharing network as an input.

One unit of time is set to one quarter. The parameter that converts a unit of data (proxied by DAU) into cash flow, ζ, is estimated from firms' data-driven cash flows. Specifically, using information from Compustat business segments and our data-reliance score (see Section 2), we construct each firm's annual cash flow generated by data-reliant segments—revenue net of the cost of goods sold, operating expenses, and capital expenditures attributable to those segments—and regress it on the firm's DAU, controlling for firm characteristics and year fixed effects, as described in Appendix C.2. The estimate implies that one million DAU is associated with \$28.983 million of annual cash flow; at the model's quarterly frequency, $\zeta = 28.983/4 = 7.246$, that is, \$7.246 of quarterly cash flow per daily active user.[57]

For each firm on the network, we compute the volatility of quarterly growth rate of DAU (daily active users), our proxy for $\delta_{i,t}$, to pin down $\sigma_{i,\delta}$, capping the quarterly volatility at 40%. Firms with volatility less than 40% account for more than 98% of DAU in our sample.

We pin down a subset of parameters with estimates from prior studies. Discount rate ρ has two components, interest rate and a Poisson intensity of firm exit. The former is set to 1%, and the latter set to 2% per quarter, consistent with the 8% annual exit rate in Jones and Kim (2018). We set a quarterly depreciation rate of 7.5% ($\mu_\delta = -0.075$), consistent with the 30% annual amortization

emphasize return comovement due to firms' interconnected fundamentals (networked data asset). This is different from Veldkamp (2006) that emphasizes comovements in asset prices due to market participants' information choices.

[56] In Section 5, we introduce a new parameter that captures the impact of ATT and calibrate it to generate the ATT impact that we estimate in Section 3.

[57] Our estimate of \$28.983 annual cash flow per daily active user is in line with numbers reported by major companies. For example, Meta's 2021 average revenue per user (ARPU) was approximately \$41 worldwide and \$214 in the United States and Canada. Applying its free-cash-flow margin of roughly one third implies free cash flow of about \$13 and \$70 per user, respectively. Our estimate of roughly \$29 lies between these benchmarks.

TABLE 6: **Model Calibration and Model-Implied Comovements**

A. Moments			
		Data	Model
Regression coefficient of			
– App performance comovement on data connectedness		0.023	0.016
– Financial performance comovement on data connectedness		0.004	0.002
– Return comovement on data connectedness		0.003	0.004
B. Parameters			
Description	Source	Notation	Parameter Value
Data economy quarterly growth rate	Moment Matching	β	0.083
Quarterly data depreciation rate	External	μ_δ	-0.075
Quarterly discount rate	External	ρ	0.030
Quarterly data-driven free cash flow per DAU ($)	Data	ζ	7.246
Volatility of customer activities	Data	$\{\sigma_{i,\delta}\}_{i=1}^{N}$	In-text discussion
Data network matrix	Data	$\{\gamma_{ij}\}_{i,j=1}^{N}$	In-text discussion

NOTE.—Table 6 presents the calibration of the model. Panel A describes the set of moments that we target, and Panel B presents the other calibrated parameters. The target set of moments, shown in the upper panel, includes the regression coefficients of comovement between app performance, financial performance and stock returns on data connectedness.

schedule applied to data assets under the current accounting standards.[58]

The last parameter, β, is calibrated to match three moments from our empirical findings in Section 3, as discussed below. Note that β, defined in Equation (14), is a composite parameter. We calibrate β directly rather than calibrating its underlying parameters separately.[59] We set β to 0.083. We summarize the calibrated parameters in Panel B of Table 6.

Under these parameter values, we simulate our model with the number of firms equal to that of our sample and firms interconnected via Γ. The simulation is done one hundred times, and each is run for 20 quarters. Using the model-simulated data, we replicate the regressions reported in Column (4) of Table 2, Table 3, and Table 4 and report the median estimates of the coefficients of data connectedness in the top panel of Table 6. These regressions target the impact of data connectedness on comovement in firms' operational, financial, and stock-market performances, in the pre-ATT period, respectively. Note that there are three regression coefficients but only one parameter, β, that we adjust to match these coefficients. The comparison in Table 6 shows that our

[58]Farboodi and Veldkamp (2025) point out that accounting rules depreciate data like software at 30% annually, but the actual depreciation rate of data may depend on whether the forecasting targets are static (e.g., consumers' tastes) or something more ephemeral like stock trading order flows. Abis and Veldkamp (2023) use a 3% monthly depreciation rate. The intangible capital literature reports similar figures. For example, Ewens et al. (2024) estimate 33% for knowledge capital and 28% for organizational capital. Customer churn rates, estimated to be around 30–35% annually, provide another justification, see He, Mostrom, and Sufi (2024) and Baker, Baugh, and Sammon (2023).

[59]When applying the model, our main focus is on the value of data. The last term of (15) depends on the deep parameters (e.g., ϕ_0 and ϕ_1) but does not load on data stocks, $\delta_{i,t}$ or $\delta_{j,t}$, so it can be omitted in data valuation.

model captures the comovements in firms' performances reasonably well.[60]

4.4 Shock propagation

The calibrated model allows us to trace how a shock to one firm's data propagates through the entire data-sharing network. We conduct an impulse-response exercise using 1,000 simulations of our model. We choose Alphabet as the focal firm and initialize firms' data stocks at their mean DAU in our sample period. At quarter 1, we impose a one-time shock that reduces the focal firm's $\delta_{i,t}$ by 20%, and we compare the shocked path with a no-shock baseline path that loads on the same subsequent shocks. Responses are reported as percentage deviations from the baseline.

Figure 6 plots the focal firm's responses. The shaded bands report the 5th–95th percentile range. The data-loss shock generates a sharp and persistent decline in the firm's raw data stock. The persistence is economically meaningful: the loss of data compromises the firm's ability to forecast customers' preferences, leading to a reduction of customer engagement and thus a slowdown of data accumulation (Panels A and B). This channel reflects the self-perpetuating loop of data growth that generates persistent shock impact on both the upside and downside. In addition, the shock impact also persists via the active data acquisition margin—i.e., the firm's product-design choice. The loss of data also elevates the cost of engagement-oriented product design, which in turn causes the firm to tilt its product design away from customer engagement, resulting in another slowdown of data accumulation (Panel C). Cash flow and firm value fall because the shock reduces both current payoffs and the continuation value that relies on data-driven growth.

We then trace the propagation of the shock to the focal firm's data-connected peers. Direct exposure—the element γ_{ji} of the data-adjacency matrix Γ—is not sufficient for this exercise, because propagation operates through repeated rounds of network feedback: the shock to $\delta_{i,t}$ affects other firms' composite signals, product-design choices, and data accumulation, which feed into later rounds of data flows. We therefore rank peers by the full network exposure implied by the model solution:

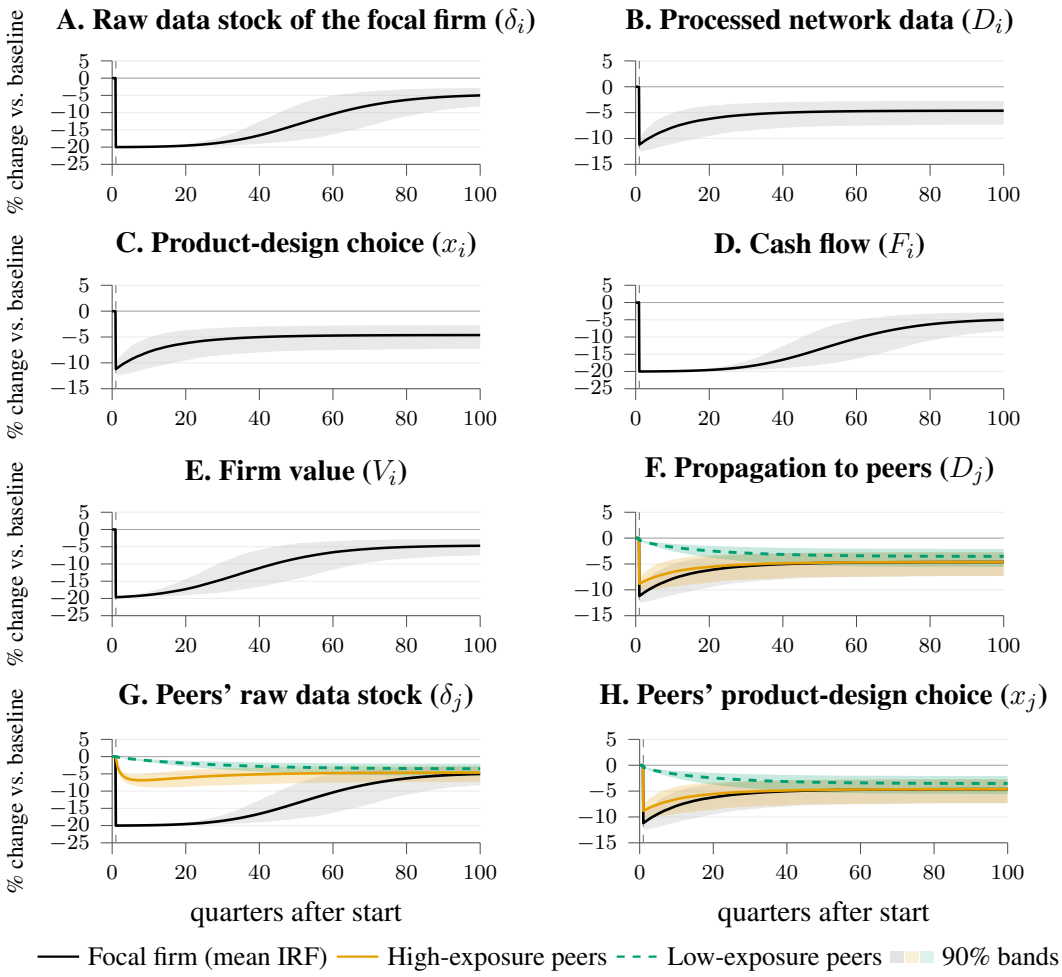
$$s_j = \left[\left(\mathbf{I} - \frac{\beta}{\hat{\rho}} \Gamma \right)^{-1} \frac{\beta}{\hat{\rho}} \Gamma \mathbf{e}_i \right]_j = \sum_{k=1}^{\infty} \left(\frac{\beta}{\hat{\rho}} \right)^k \left[\Gamma^k \right]_{ji}$$

where $\mathbf{e}_i$ is the unit vector of the focal firm. The measure cumulates the direct and higher-order paths from the focal firm to peer j, with paths of length k discounted by $(\beta/\hat{\rho})^k$; the focal firm is excluded from the ranking. High- and low-exposure peers are in the top and bottom quintiles.

Panel F of Figure 6 plots peers' D_j, which incorporates the focal firm's data and determines their forecasting precision—i.e., the entry point of the focal firm's shock. Panel G shows the peer firms' data stock, whose evolution reflects the self-perpetuating data loop, and Panel H shows the

[60] Market participants may not be informed about firms' data connectedness, so there can be a gap between our valuations and stock-market valuations. This partly explains the discrepancy between model-implied and data moments.

FIGURE 6: **Impulse Responses to a Data Shock**



NOTE.–Figure 6 plots impulse responses to a one-firm data shock in the calibrated pre-ATT network. The focal firm is Alphabet. In quarter 1, its raw data stock is reduced once by 20 percent. Each simulation compares the shocked path with a no-shock path that uses the same subsequent innovations. Responses are percentage deviations from the baseline path, $100 \times (X_t^{\text{shock}}/X_t^{\text{base}} - 1)$. Panels A–E report the focal firm's raw data stock δ_i, processed network data D_i, product-design choice x_i, cash flow F_i, and firm value V_i. Panels F–H report the responses of the focal firm and of the high- and low-exposure peers for processed network data D, raw data stock δ, and product-design choice x, respectively. Peers are ranked by $s_j = [(I - r\Gamma)^{-1} r\Gamma e_i]_j$, where $r = \beta/\hat{\rho}$ and $\hat{\rho} = \rho - \mu_\delta$; the focal firm is excluded. High- and low-exposure peers are the top and bottom quintiles. The exercise uses 1,000 simulations. For the focal firm, the lines are mean responses across simulations and the shaded areas are their pointwise 5th–95th percentile ranges. For peers in Panels F–H, firm-level responses are first averaged across firms within the high- and low-exposure quintiles in each simulation; the lines are the means of these quintile averages across the 1,000 simulations, and the shaded areas are the 5th–95th percentile ranges.

peer firms' product-design choice—the active data acquisition margin—where a decline means a tilt away from customer engagement and data acquisition. High-exposure peers, defined as firms in the top quintile, experience larger and more immediate declines across both margins of data accumulation than low-exposure peers in the bottom quintile. As time goes, the propagated impact on low- and high-exposure peers eventually becomes close. A data loss is thus not confined to

the firm that suffers it: through the data-sharing network, it alters the focal firm's long-term data growth and valuation and propagates to other firms through direct and higher-order links, echoing the cyberattack spillovers documented in Section 3.2.

The reduced-form findings on firms' performance comovement in Section 3 are based on firms' direct connections through data sharing. These impulse responses reflect higher-order spillover effects. A firm's data has a persistent effect over time on both itself and its connected peers by affecting the trajectory of data growth. Next, we present our results on firms' data valuation and use valuation metrics to identify systemically important firms. In our model, a firm's data value is the present value of data-generated cash flows. Therefore, the valuation metrics summarize the network externalities of higher orders and over multiple time horizons.

4.5 Valuing data as a networked asset

A central purpose of the model is measurement: it embeds the empirically measured data-sharing network into a data valuation framework. The value of data cannot be measured solely by the cash flows it generates at the data-originating firm. Through data sharing, one firm's data improves other firms' knowledge of their customers, raises their customer engagement and cash flows, and accelerates their own data accumulation, which in turn benefits their data-connected peers. Valuing data as a networked asset therefore requires computing the present value of the cash flows that a firm's data generates at all firms, through all rounds of network propagation, and over all horizons. The results in Proposition 2 make this calculation feasible.

Proposition 2 expresses the present value of data-generated cash flows for a single firm. Aggregating across firms, the total value of data-generated cash flows in the economy is

$$\zeta \mathbf{1}^\top \left(\hat{\rho}\mathbf{I} - \beta\Gamma\right)^{-1} \bar{\delta}_t. \tag{17}$$

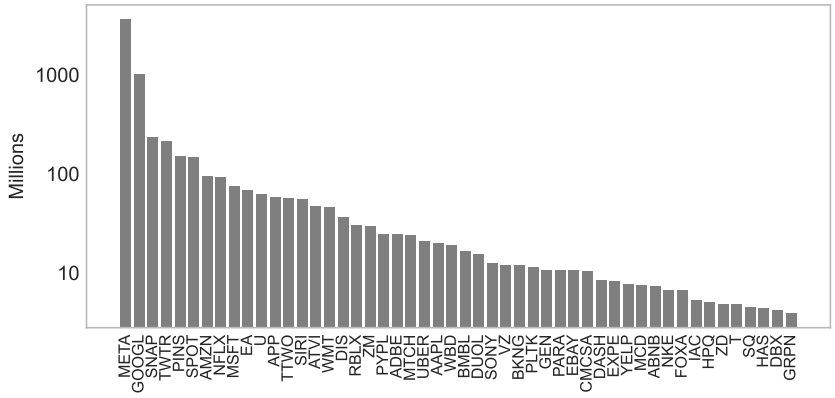
Because (17) is linear in the vector of firm-level data stocks δ_t, it admits an exact attribution by the origin of data: the contribution of firm i's data to the aggregate value is

$$C_i := \zeta \{\mathbf{1}^\top \left(\hat{\rho}\mathbf{I} - \beta\Gamma\right)^{-1}\}_{.i} \delta_{i,t}, \tag{18}$$

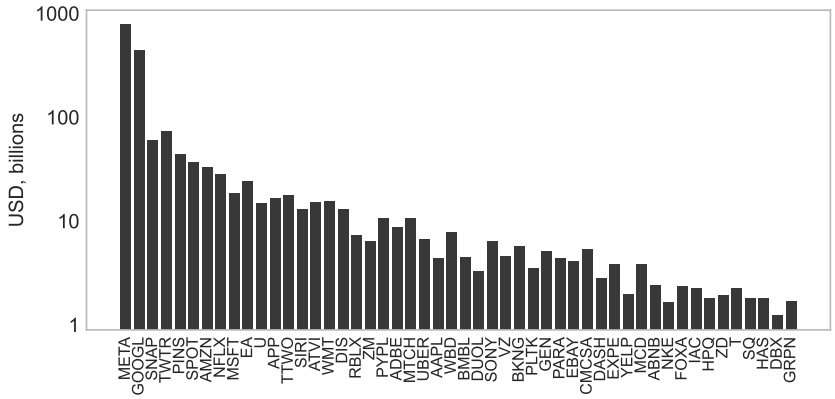
where $\{\cdot\}_{.i}$ picks the i-th column of a matrix. Equation (18) is our formula for valuing data as a networked asset, and the data-sharing network, Γ, features prominently in it. The contribution of firm i's data is the product of two components: the firm's own raw data stock, $\delta_{i,t}$, and a firm-specific network multiplier, $\{\mathbf{1}^\top(\hat{\rho}\mathbf{I} - \beta\Gamma)^{-1}\}_{.i}$, which encodes every route through which firm i's data reach other firms. Expanding the multiplier as in (16) decomposes the value into rounds of propagation: the zeroth-order term is the value that the data would create at firm i before accounting for

FIGURE 7: **Data Valuation**

A. Top DAU Firms



B. Data Valuation



NOTE.—Figure 7 displays the top 50 firms, whose main customer base is in the United States, ranked by their average daily active users (DAU) within the sample period. Firms are ordered by average DAU from left to right. Panel A reports average DAU in millions of active users. Panel B reports each firm's contribution to aggregate data-network valuation in USD billions, using the same firm ordering. Both panels use a log scale.

data-driven feedback, the first-order term captures the value created at firm i's direct data-sharing partners, and higher-order terms capture the value created through chains of intermediaries, with round k discounted by $(\beta/\hat{\rho})^k$—the strength of data-driven growth relative to the effective discount rate. The formula thus traces the value creation of firm i's data through the whole economy.

Figure 7 reports the valuation for the top 50 firms ranked by daily active users (DAU). Panel A plots firms' DAU, our proxy for the stock of customer data that each firm generates on its own: the more active users a firm has on its apps, the more data it collects about its customers. The bars decline monotonically by construction, as the horizontal axis orders firms by DAU. Meta and Alphabet occupy the top positions, and the steep decay of the bars shows that own-data generation is highly concentrated: the largest platforms already collect vast amounts of consumer data by themselves, even before accessing other firms' data through the network.

In fact, Panel A of Figure 7 serves as a benchmark: if data creates value only where it originates—no spillovers due to data sharing—the value created by each firm's data would follow the DAU ranking. Panel B shows that it does not. Panel B reports the contribution of each firm's data to the whole economy, computed from (18). The magnitudes are economically large: Meta's pre-ATT contribution is about $780 billion, and Alphabet's is about $442 billion. More importantly, the ranking of contributions departs visibly from the ranking of DAU: firms with moderate user bases can create outsized value when their data conduits place them centrally in the network, while some large firms contribute less than their size suggests because their data reach fewer peers. Valuing data as an asset without accounting for the network structure of data flows would misstate both the level and the cross-firm distribution of data value.

The aggregate magnitude is also economically significant. Summing the data values across all firms, the pre-ATT value is approximately $1.984 trillion, which is around 8.4% of US nominal GDP of $23.7 trillion in 2021. Our estimate complements the measurement in Bhutani et al. (2026), who estimate the missing value of data at about 1.5% of GDP annually. Their economy-wide measure captures the capitalized value of the data newly acquired in a given year, whereas ours capitalizes all the future cash flows of the data stock.

Our analysis is confined to publicly listed firms. The establishment-level records from the Infogroup Historical Business Database show that private firms account for more than 80% of establishments and roughly 60–70% of employment and sales in data-reliant sectors over 2016–2023 (Figure C.3 in the Internet Appendix), where data-reliant sectors are identified using the same data-reliance scores that underlie our business segment classification (Appendix C.1). This suggests that our valuation of data as a networked asset understates the aggregate importance of data, as we do not cover data from private companies and the benefit that private companies enjoy from access to public companies' data via the data-sharing SDKs embedded in mobile apps.

5 Policy Implications: The ATT Impact

This section consolidates the model-based analysis of ATT. As discussed in Section 2, ATT—App Tracking Transparency—was introduced in 2021 by Apple to enable iOS users to hide their IDFA, which has been a key input for merging data from different apps (the de facto merge key at individual consumer level in the SDK ecosystem). Section 3 established the reduced-form evidence: ATT weakened the performance comovement of data-connected firms across different metrics. Here we introduce the policy into the model of Section 4, calibrate its severity, and quantify its consequences for the valuation of data as a networked asset.

ATT in the model. ATT allows iOS users to sever the link between their data and universal identifiers on which the SDKs rely to merge data across firms (Section 2). We capture the policy by assuming that, post-policy, a fraction $\xi \in [0, 1]$ of data is linked to universal IDs and can thus be merged across firms: firm i's information on its customers, defined in (5), generalizes to

$$D_{i,t} = \sum_{j=1}^{N} \gamma_{ij} \xi \delta_{j,t} = \{\xi \Gamma \bar{\delta}_t\}_i, \tag{19}$$

Before ATT, $\xi = 1$, and the model reduces to that of Section 4; after ATT, $\xi < 1$. Because ξ scales every data conduit, it enters the composite growth parameter β defined in (14):

$$\beta = \theta \left(\frac{\alpha \kappa}{\sigma_\eta} + \frac{\phi_0}{\phi_1} \right) \xi \,; \tag{20}$$

the pre-ATT benchmark of Section 4 corresponds to $\xi = 1$.

The next corollary summarizes how ATT, by reducing ξ, weakens the growth of the data economy and the network effects of data sharing; this follows directly from Equation (16).

Corollary 2 (ATT Impact) *ATT reduces ξ, the fraction of data linked through universal IDs, and thereby reduces β, the composite growth rate of the data economy, and weakens the network effects of data sharing. Holding Γ fixed, a reduction in ξ attenuates every network-propagation term $(\beta/\hat{\rho})^k \Gamma^k$, $k \geq 1$, in the valuation operator in* (16)*.*

One clarification on interpretation is in order. ATT primarily restricts the merging of data *across* firms whereas a firm's collection of data within its own apps is less affected. Our specification applies ξ uniformly to all data conduits, including the own-firm element γ_{ii}, and thus does not capture this asymmetry. This uniform treatment reflects the focus of our model on data as a networked asset: the ATT impact on $D_{i,t}$ applies to the firm's own data as well as other firms' data, under the assumption that, without proper merging enabled by the universal identifiers, the firm's own data also becomes less useful; in reality, the use of a firm's own data is likely to be less affected—we acknowledge this, and extending the model by allowing such heterogeneous impact of ATT would lead to a more comprehensive quantification of the ATT impact on data value.[61] We adopt the uniform specification for parsimony: it introduces a single policy parameter.

Calibrating the ATT shock. We calibrate the post-ATT value of ξ to match the three ATT difference-in-differences coefficients estimated in Section 3.1—the ATT interactions with data

[61] The distinct roles of own and cross-firm linkages in the transmission of shocks through networks are emphasized in the production-network literature; see Huo et al. (2025).

TABLE 7: **Post-ATT Calibration and Model-Implied Comovements**

A. Moments			
		Data	Model
ATT DiD coefficient of			
– App performance comovement on data connectedness		-0.023	-0.014
– Financial performance comovement on data connectedness		-0.003	-0.001
– Return comovement on data connectedness		-0.003	-0.003
B. Parameter			
Description	Source	Notation	Parameter Value
Post-ATT fraction of data linked to universal IDs	Moment Matching	ξ^{post}	0.247

NOTE.—Panel A reports the three ATT interaction coefficients used to calibrate ξ^{post}. The pre-ATT calibration is held fixed, while the post-ATT composite growth parameter is given by $\beta^{\text{post}} = \xi^{\text{post}}\beta^{\text{pre}}$, with ξ^{pre} normalized to one. The model-implied moments are computed from 100 paths of simulation.

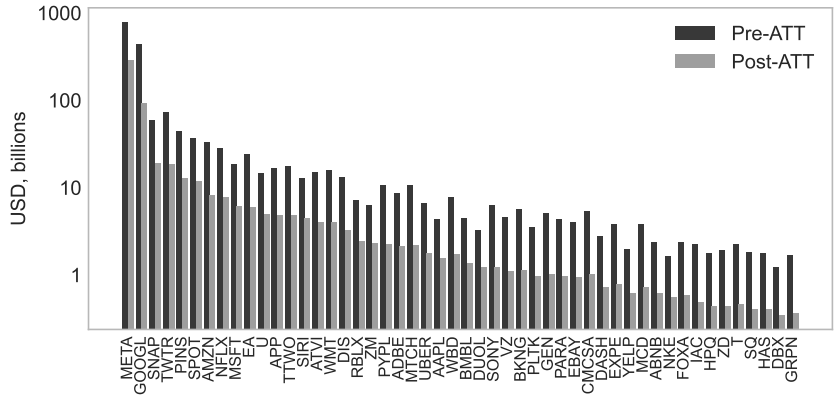
connectedness in the app-performance, financial-performance, and stock-return comovement regressions. To obtain the model-implied counterparts, we introduce a reduction of ξ at the ATT date in the simulations, holding all other parameters fixed, and estimate Equation (2) on the simulated data with the interaction between the ATT indicator and data connectedness. Matching the three coefficients yields $\xi^{post} = 0.247$: about a quarter of the data previously merged through universal identifiers remains linkable across firms after ATT.[62] As shown in (20), the 75% reduction of ξ translates into an equal proportional decline of the composite parameter β. Table 7 compares the model-implied coefficients with their empirical counterparts: with this single parameter, the model reproduces the sign of the ATT-induced decline in comovement across all three outcome variables, matches the return-comovement coefficient exactly, and generates somewhat smaller declines in app- and financial-performance comovement than in the data.

ATT and data valuation. The impact of ATT on the value of firm i's data is given by $\Delta C_i^{\text{ATT}} = C_i(\beta^{\text{pre}}) - C_i(\beta^{\text{post}})$, where $C_i(\beta)$ denotes the contribution of firm i's data to the whole data economy (defined in equation (18)) evaluated at the corresponding parameter β. Figure 8 reports the pre- and post-ATT values for the top 50 firms ranked by DAU. The magnitudes are large: Meta's contribution to the aggregate value of data-enabled cash flows falls from \$780 billion to \$288 billion, a decline of 63%, and Alphabet's falls from \$442 billion to \$92 billion. Aggregating across all firms, the total value of the data network falls from \$1.984 trillion to \$0.586 trillion.

We make two remarks on the interpretation. First, these dollar values are contributions to the

[62] Although around 82% of iOS users opted out of IDFA tracking as of April 2022, ATT's impact is likely smaller than the opt-out rate suggests. The SDKs adopted other methods of linking data from different firms; while such methods are imperfect substitutes for the IDFA, they partially restored cross-app data linkage.

FIGURE 8: **ATT and Data Valuation**



NOTE.—Figure 8 reports each firm's contribution to aggregate data-network valuation before and after the App Tracking Transparency (ATT) policy shock. Values are measured in USD billions. We display the top 50 firms ranked by their average daily active users (DAU) within the sample and use the same firm ordering for the pre-ATT and post-ATT bars. The y-axis uses a log scale.

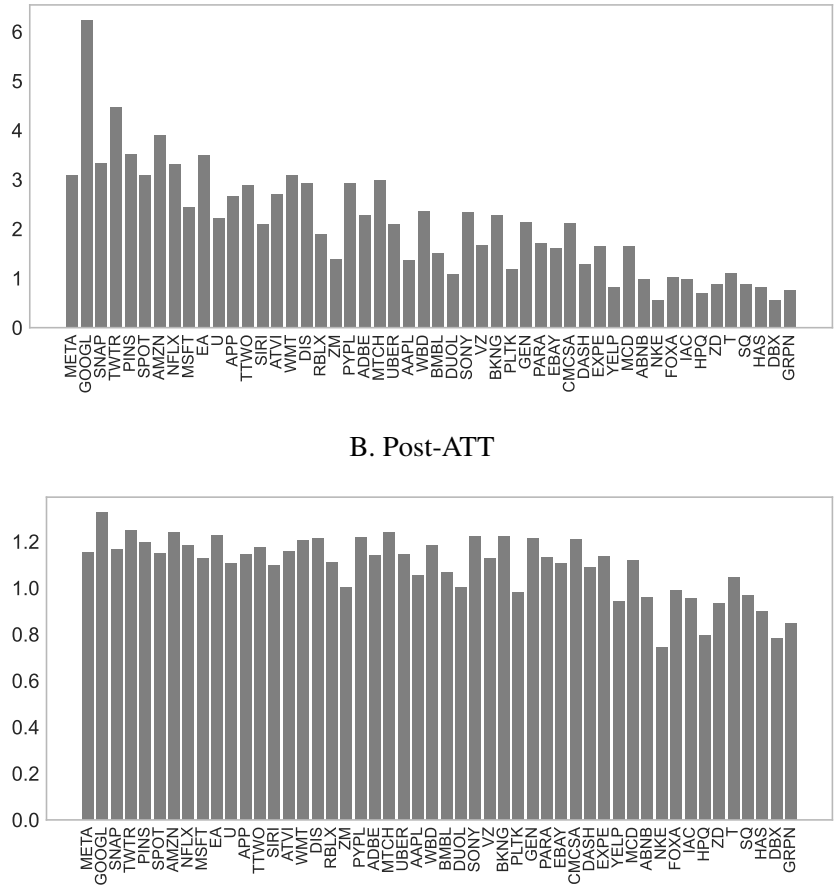
aggregate data-driven cash flows in present-value terms, not a welfare calculation. From the firms' perspective, ATT reduced the collective cash-flow value generated by data sharing; from a privacy perspective, consumers' welfare may increase. Second, these magnitudes are in line with market assessment. While Meta itself disclosed a deliberately conservative revenue headwind of about $10 billion for 2022, roughly 8.5% of its prior-year revenue (see Table C.4 in the Internet Appendix), the market capitalized the shock into a far larger revaluation: Meta's market capitalization fell from about $1 trillion to roughly $250 billion between late 2021 and late 2022, including a record single-day loss of $232 billion on February 3, 2022, when Meta first quantified the ATT impact.[63] Although other forces also weighed on Meta's valuation over this period, the comparison indicates that the market repriced the firm in present-value terms by far more than the disclosed cash-flow effect, and such repricing aligned in magnitude with our model-implied values of ATT impact. Note that such market repricing only accounts for the loss of data value for one firm, while our model-implied values reveal the loss of data value for the entire networked data economy.

ATT and network externality. Figure 9 quantifies the impact of ATT on the network externality from data sharing. A firm's contribution to the value of aggregate cash flows differs from its own valuation; Panels A and B sort firms by their DAU and report the ratio of the two,

$$\frac{\zeta\{\mathbf{1}^\top(\hat{\rho}\mathbf{I}-\beta\Gamma)^{-1}\}_{.i}\delta_{i,t}}{\zeta\mathbf{e}_i^\top(\hat{\rho}\mathbf{I}-\beta\Gamma)^{-1}\bar{\delta}_t}, \tag{21}$$

[63] See `https://companiesmarketcap.com/meta-platforms/marketcap/` and "Facebook's $232 Billion Drop in Value Sets All-Time Record," CNBC, February 3, 2022.

FIGURE 9: **ATT and Network Externality**



NOTE.—Figure 9 reports data-network valuation measures before and after the App Tracking Transparency (ATT) policy shock. We display the top 50 firms ranked by their average daily active users (DAU) within the sample and use the same firm ranking in both panels. Panels A and B report the ratio of each firm's contribution to aggregate data-network valuation relative to its own data-enabled valuation, before and after ATT, respectively.

computed under the pre-ATT value of ξ $(= 1)$ and the post-ATT value ($\xi^{post} = 0.247$), where $\mathbf{e}_i$ picks the i-th element of a vector. A ratio far above one indicates that most of the value created by a firm's data accrues at *other* firms through the sharing network.

Before ATT, the contribution of Meta (formerly Facebook) to the value of aggregate cash flows in the data economy is about 3 times its own data-enabled valuation, and Alphabet's ratio is about 6; after ATT curtails the cross-firm data flows, the ratios decline significantly to 1.15 and 1.33, indicating a sharp reduction of the network spillover effect of data sharing.

The contribution metric also identifies the firms whose data underpin the value of the entire data economy. Following Ballester et al. (2006) and Denbee et al. (2021), we define the valuation key player as the firm with the largest relative contribution in (18).[64] These systemically important

[64]Our focus on the scope of data usage across firms echoes Crouzet et al. (2024), who emphasize the scope of

firms are natural focal points for assessing shocks to the data economy: precisely because their data travels widely through the network, disruptions to their data—or to the conduits that carry it—have outsized aggregate consequences. The ATT experiment above illustrates this mechanism at work: the policy destroyed no firm's data stock, yet by weakening the conduits that carry data across firms, it eliminated most of the cross-firm layer of data value.

6 Product-Design Dynamics

In Section 4, we develop a theoretical framework to capture the salient features of data economy that emerge from data sharing as documented in Section 3. Our model highlights that when designing products, firms face an intertemporal trade-off between monetization and data accumulation. The optimal product-design decision is characterized by a Q-theory of data investment in Proposition 1. Next, we show that under data sharing, firms rationally mimic the product-design choices of one another. Such herding behavior in investment decisions is unique to data economy.

This section complements the main analysis. Its purpose is twofold. First, it provides an out-of-sample validation of the model: the herding in product design is a distinct prediction of the mechanism that generates the comovement results—data sharing interconnects not only firms' performances but also their incentives to accumulate data—and the evidence below confirms both the prediction and its weakening after ATT, even though these moments are not targeted in the calibration. Second, it completes the account of ATT's impact: beyond reducing the value of data-enabled cash flows and dampening the network externality (Section 5), the policy weakened the strategic complementarity in firms' data investment.

Herding in product design. The optimality condition for firm i's product-design decisions $x_{i,t}$ in Proposition 1 and the value function in Proposition 2 show that other firms' data, $\delta_{j,t}$, enters into firm i's choice of $x_{i,t}$ via firm i's information on its customers, $D_{i,t}$. Next, we show that $x_{i,t}$ is increasing in $D_{i,t}$, and herding behavior in firms' product-design decisions emerges due to data sharing. ATT weakens such behavior. Appendix F provides the proof.

Proposition 3 (Optimal Product Design) *Firm i prioritizes customer engagement over monetization (increases $x_{i,t}$) when $D_{i,t}$ is higher. The expected change of $x_{i,t}$, $\mathbb{E}_t[dx_{i,t}]$, is increasing in $x_{j,t}$ for $j \neq i$ with $\gamma_{ij} > 0$, and the sensitivity is increasing in ξ, i.e., $\frac{\partial^2 \mathbb{E}_t[dx_{i,t}]}{\partial x_{j,t} \partial \xi} > 0$.*

Consider an increase in other firms' current product-design choice of $x_{j,t}$, which leads to an increase in their data stock, $\delta_{j,t}$, and an increase from t to $t + dt$ in firm i's information on its

intangible capital usage across divisions within firms and by competing imitators.

customers, $D_{i,t}$. Being more informed allows firm i to raise customer engagement via a higher $x_{i,t}$ without sacrificing monetization as much—that is, data alleviates the tension in product design between customer engagement and monetization as we have discussed in Section 4.2. Therefore, from t to $t + dt$, firm i's choice of $x_{i,t}$ increases, resulting in a "herding" or cross-firm momentum in prioritizing data accumulation over monetization in product design. Thus, a positive data investment externality exists, which is in sharp contrast with the investment dynamics of traditional firms whose investment is likely to crowd out other firms' investment (for example, by raising the prices of investment inputs and financing costs). Below, we provide empirical evidence.

The product-design dynamics also imply that waves of data collection or monetization emerge in the data economy: because data sharing interconnects firms' marginal costs of data investment, a positive shock to one firm leads its connected peers to prioritize customer engagement and data collection, whereas a negative shock propagates through the network and pushes the connected peers toward monetization at the expense of engagement. Moreover, because one firm's data accumulation lowers other firms' cost of data investment—a positive spillover that individual firms do not internalize—the equilibrium features under-investment in data accumulation.

Proposition 3 shows that by reducing ξ (the fraction of data linked to universal IDs), ATT weakens the cross-firm momentum in product-design decisions, i.e., $\frac{\partial^2 \mathbb{E}_t[dx_{i,t}]}{\partial x_{j,t} \partial \xi} > 0$. This is an unintended consequence of ATT that has not been studied before. Beyond the impact on herding in product design, reducing ξ directly reduces firms' incentive to acquire data (i.e., decreases data marginal q) by weakening the data-driven firm growth, as ξ is part of the composite growth parameter β introduced in (20). As a result, ATT encourages firms to prioritize monetization. This prediction is in line with the findings in Kesler (2023). The next proposition summarizes this result.

Proposition 4 (ATT and Product Design) *Reducing ξ reduces the data marginal q, $\frac{\partial V_{i,t}}{\partial \delta_{i,t}}$, thereby causing all firms to prioritize monetization over customer engagement (i.e., $x_{i,t}$ decreases), and reduces $\frac{\partial \mathbb{E}_t[dx_{i,t}]}{\partial x_{j,t}}$, weakening the cross-firm momentum in product-design decisions.*

Evidence and model simulation. We test the herding behavior in firms' product-design decisions in Proposition 3 and the ATT impact. To examine firms' product-design decisions (prioritizing monetization vs. customer engagement) and how they are influenced by their peers, we design the following empirical strategy. We use the functionality-SDK measures introduced in Section 2: payment SDKs are directly related to monetization, whereas SDKs that support data security, customer service, and user reviews are associated with customer engagement (examples are provided in Internet Appendix B.2). For each category, $X_{i,t}$ denotes the number of unique SDKs used by

TABLE 8: **Product-Design Decisions**

	Monetization	User Engagement			
Feature:	Payment	Security	Support	Review	Model
	(1)	(2)	(3)	(4)	(5)
L1.product feature (peers)	0.007***	0.003***	0.001***	0.001***	0.005
	(5.22)	(4.86)	(4.17)	(3.96)	
ATT × L1.product feature (peers)	−0.011***	−0.002	−0.002***	−0.002***	−0.008
	(−3.92)	(−1.21)	(−2.82)	(−3.43)	
Industry#Quarter FE	Y	Y	Y	Y	
Firm FE	Y	Y	Y	Y	
Firm controls	Y	Y	Y	Y	
Observations	18,596	18,596	18,596	18,596	
R-sq	0.068	0.086	0.070	0.066	

NOTE.—Table 8 presents evidence of cross-firm momentum in data accumulation investments. The outcome variable is the change in a firm's product design, measured by the number of unique functionality SDKs, as defined in Section 6. Column 1 reports the results on payment SDKs, and Columns 2–4 report the results on SDKs that are likely to improve user engagement. Standard errors are clustered by firm. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. The last column reports the model-implied coefficients, based on median estimates from 100 simulated panels using the calibrations in Tables 6 and 7.

firm i, $\Delta X_{i,t}$ denotes its change, and

$$X_{-i,t-1} = \sum_{j \neq i} \rho_{i,j}^{\text{data}} X_{j,t-1}$$

is the data-connectedness-weighted stock of SDKs used by the focal firm's peers, which captures the product-design choices of the firm's data-connected peers.

We then estimate the following equation separately for different SDK categories:

$$\Delta X_{i,t} = \alpha + \beta_1 X_{-i,t-1} + \beta_2 \text{ATT} \times X_{-i,t-1} + \beta_3 X_{i,t-1} + \beta_4' \mathbf{C_{it}} + \iota_{st} + \varphi_i + \varepsilon_{i,t}, \tag{22}$$

where $\mathbf{C_{it}}$ includes the same set of firm-level controls as in Equation (3). We also include industry-quarter fixed effects (ι_{st}) and firm fixed effects (φ_i). Standard errors are clustered at the firm level. The coefficient β_1 corresponds to the model prediction on herding behavior in firms' product-design decisions in Proposition 3 and the coefficient β_2 corresponds to the impact of ATT in Proposition 4. The estimation results are presented in Table 8, with Column 1 focusing on monetization SDKs and Columns 2–4 on SDKs related to customer engagement.

Consistent with the model's prediction, the changes in firms' product design positively load on their peers' product-design choices from the previous period across different features. Additionally, the coefficient on the interaction term, β_2 for ATT × $X_{-i,t-1}$, is negative. This indicates that the herding behavior emerges from data sharing, and ATT weakens this channel.

Using the simulated data from Section 5, we run the same regressions, and the last column of Table 8 reports the model-implied coefficients. The model-generated cross-firm herding in

product-design choices is in line with that observed in data. Moreover, the model also generates a weakened herding behavior after the ATT shock, in line with our empirical findings. Therefore, even though the product-design dynamics are not targeted when we calibrate the model, the model generates patterns that closely resemble those observed in the data.

7 Conclusion

One firm's data can be used simultaneously by other firms and reveals information about other firms' customers. Such non-rival nature and externalities make data a uniquely productive asset. This paper uncovers a network of inter-firm data conduits: firms collect data on their customers and share data with one another for customer profiling, which is critical for enhancing firms' operational efficiency, improving customer engagement, and supporting further data collection in a self-reinforcing cycle. Disruptions to the underlying software infrastructure have a significant negative impact on firms' operational and stock-market performances. Firms' own disclosures corroborate the economic significance of the data-sharing infrastructure.

We document that data sharing drives strong comovements in operational, financial, and stock-market performances among data-connected firms. We identify firms' advertising strategy as one channel through which shared data synchronize firms' behavior. Motivated by these empirical findings, we develop a dynamic network model of data economy that captures the interconnected dynamics of data collection, sharing, and utilization, providing insights into the economic implications of policy interventions. In addition, our model reveals a novel feature of data-driven firms that is supported by evidence: their product-design decisions exhibit herding—an out-of-sample validation of the model's mechanism, as these moments are not targeted in the calibration.

Importantly, our model provides a framework for valuing data as a networked asset. The value of a firm's data is the present value of the cash flows that the data generates both at the firm where it originates and at the other firms that receive them through the sharing network; the valuation thus embeds the network structure of data sharing, the propagation of data's value through higher-order externalities, and the self-perpetuating data growth that cross-firm propagation fuels over multiple time horizons. The same framework quantifies the network externality of data spillovers and identifies systemically important firms—those whose data is disproportionately valuable to the network relative to their own valuations. These firms hold critical positions in the data-sharing network and thus disproportionately influence the entire data economy.

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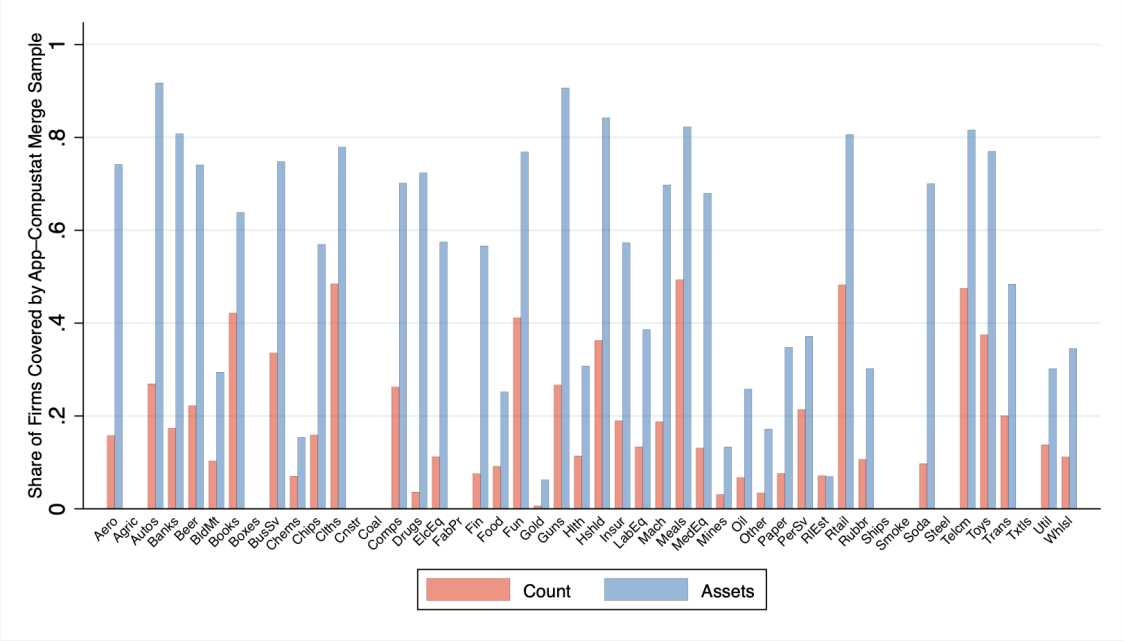
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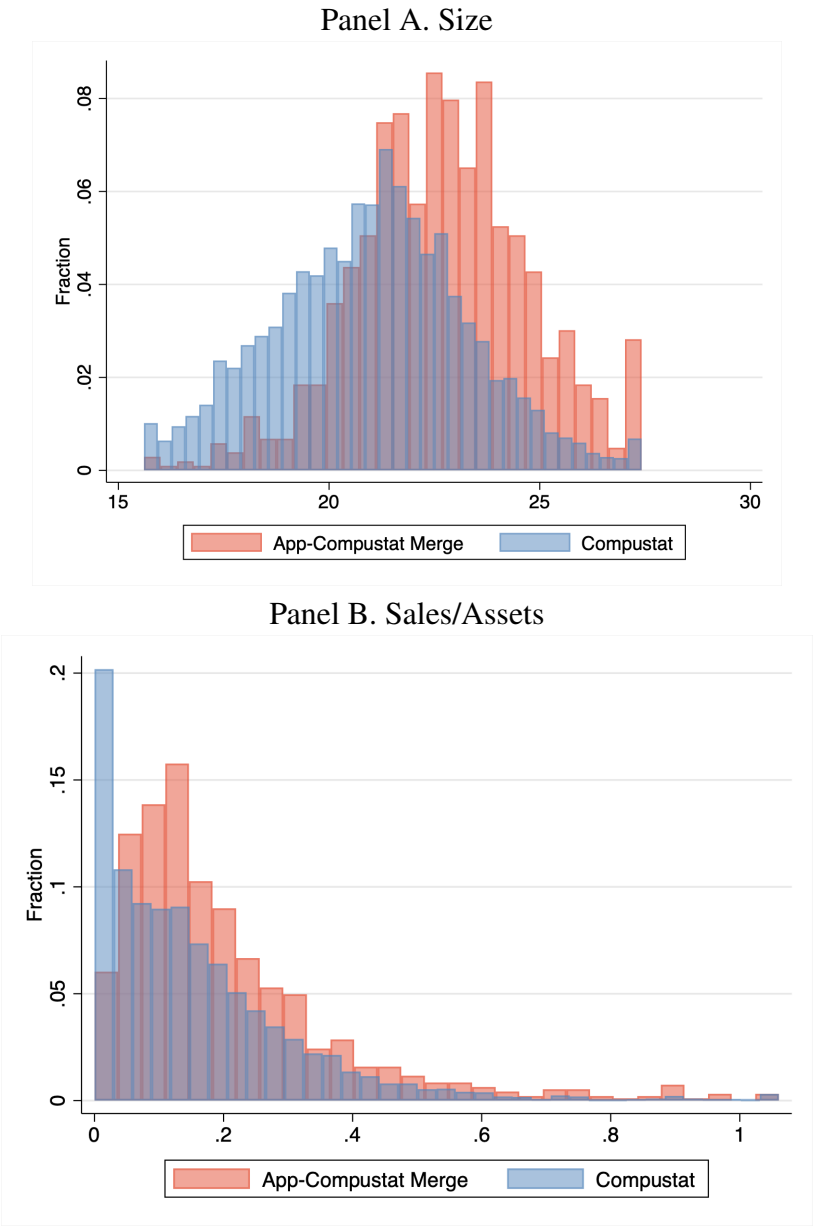
A Additional Figures and Tables

FIGURE A.1: **Firms in Our Sample vs. Compustat Universe by Industry**



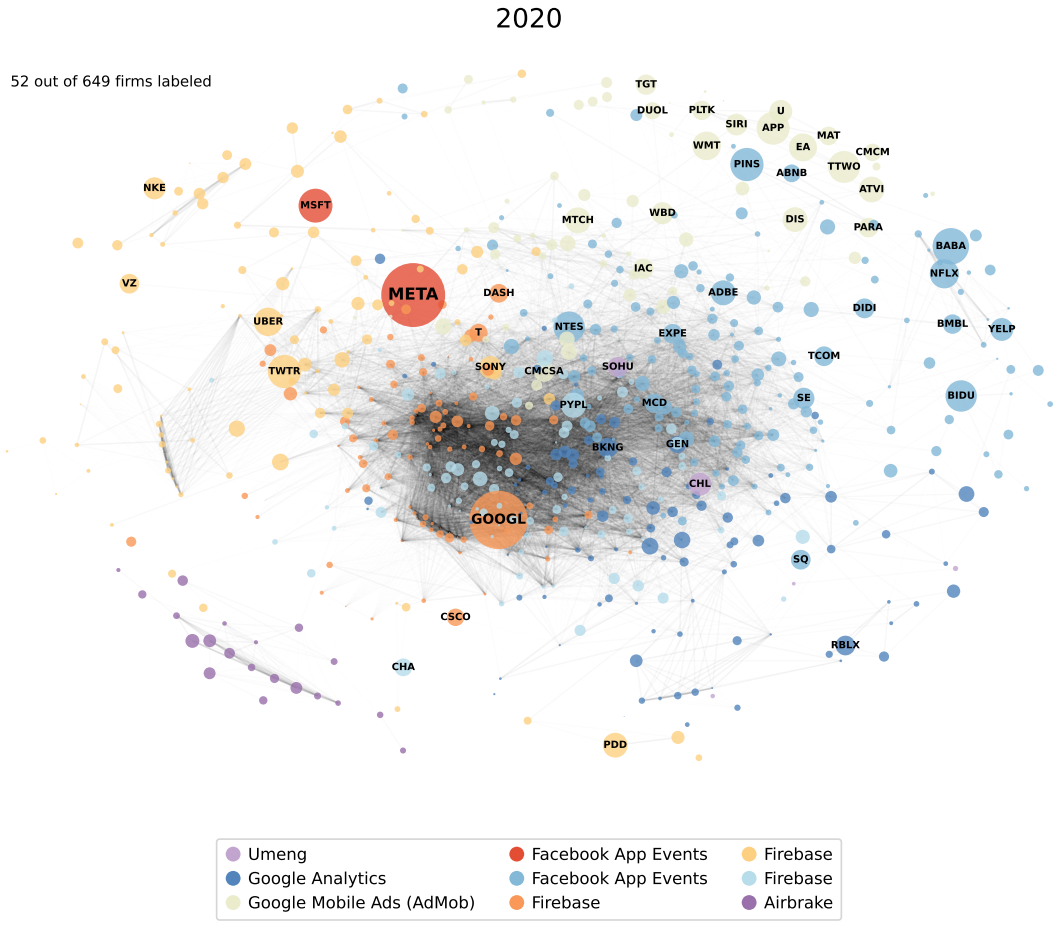
NOTE.—Figure A.1 reports the share of Compustat firms covered by our sample within each Fama–French 48 industry, measured by firm count and total assets.

FIGURE A.2: **Firms in the Data Network vs. Compustat Universe: Size and Asset Turnover**



NOTE.—Figure A.2 compares firms in our sample ("App-Compustat Merge") with those in the broader Compustat universe ("Compustat"). We focus on firm size, proxied by log(assets) (Panel A) and the sales/assets ratio (Panel B).

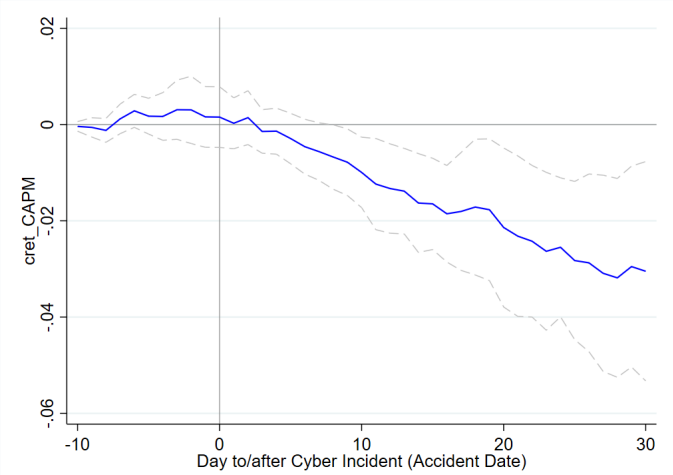
FIGURE A.3: **The Data-Sharing Network**



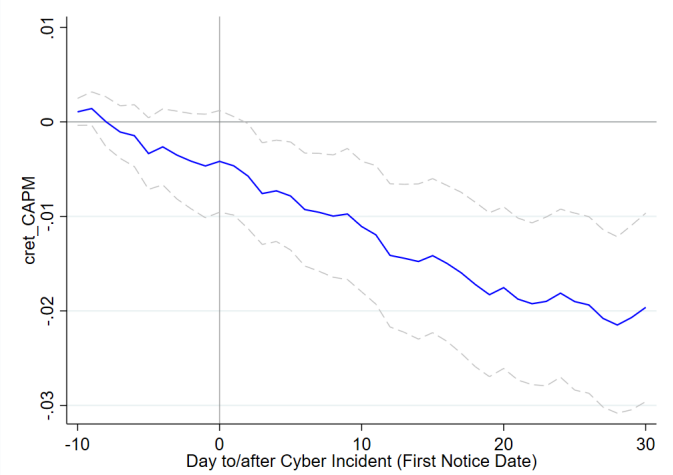
NOTE.—Figure A.3 visualizes the network of firms connected in the data space using the Kamada-Kawai Algorithm and average data connectedness in 2020. Each node represents a firm, and the size of the node corresponds to the firm's size, measured by the square root of average MAU in 2020. For readability, we only include firm pairs with data connectedness greater than 0.7, which includes 649 unique firms (63% of firms in our sample). Edge color darkens proportionally with the strength of the data connection between two firms. We label the stock tickers of firms with on average more than 6.5 million MAU in 2020. Firms are grouped into different colors ("community") based on the Louvain Community Detection Algorithm, and in the legends below the graph, we label the most popular (widely installed) SDK within each community.

FIGURE A.4: **Cyberattack Spillover Effects: Event Studies**

Panel A. Event Date: Incident Date



Panel B. Event Date: Notice Date



NOTE.—Figure A.4 presents event studies examining the cross-firm spillover effects of major cyber events. Panels A and B use the incident dates and notice dates, respectively, as the event dates. Both subfigures display the cumulative abnormal returns of firms exposed to the affected firm through the mobile data-sharing network, using CAPM as the benchmark model. Major cyber events are defined as those that expose over 10 million records. The sample includes all exposed firms. The dashed lines report 95% confidence intervals for event-day portfolio returns, with standard errors clustered at the cyber-event level.

TABLE A.1: **Data Capital of US Public Firms: A Segment-Based Analysis**

Panel A: Data-revenue and data-CAPEX shares, 2014–2024

	EW mean	VW (aggregate)	% obs > 0	% obs $\geq 50\%$	Firm-years
Data-reliance score ≥ 70*:*					
Revenue share, all public firms	30.5%	19.9%	34.3%	30.3%	62,582
Revenue share, app-Compustat merge	43.8%	29.6%	52.3%	43.2%	9,887
CAPEX share, all public firms	27.8%	12.6%	30.0%	27.7%	49,457
CAPEX share, app-Compustat merge	39.8%	21.7%	45.6%	39.5%	7,079
Data-reliance score ≥ 80*:*					
Revenue share, all public firms	22.7%	13.2%	26.0%	22.7%	62,582
Revenue share, app-Compustat merge	33.4%	20.2%	41.5%	33.1%	9,887
CAPEX share, all public firms	20.7%	10.5%	22.4%	20.7%	49,457
CAPEX share, app-Compustat merge	30.5%	18.5%	35.0%	30.5%	7,079

Panel B: Data-revenue shares by Fama–French 48 industry (selected)

	Consumer-facing / information			Upstream	
	EW mean	VW		EW mean	VW
Banks (depository institutions)	98.7%	99.3%	Oil	2.0%	0.3%
Business services	83.1%	86.7%	Mining	1.6%	0.0%
Computers	63.0%	47.8%	Agriculture	0.4%	0.1%
Publishing	41.9%	54.9%	Steel	0.3%	0.1%
Entertainment	33.7%	41.9%	Chemicals	0.1%	0.0%
Telecom	30.3%	28.0%	Paper	0.0%	0.0%
Retail	24.5%	11.4%	Coal	0.0%	0.0%
Finance (non-depository institutions)	24.3%	59.9%	Gold	0.0%	0.0%

NOTE.—Table A.1 benchmarks the data capital of US public firms using Compustat business-segment data, 2014–2024. A business segment is classified as data-reliant if its industry's data-reliance score is at least 70 (Panel A also reports the stricter cutoff of 80); a firm's data-revenue (data-CAPEX) share is the fraction of its revenue (CAPEX) generated by data-reliant segments, winsorized at 1% in both tails. "EW mean" reports the average share across firm-years; "VW (aggregate)" reports aggregate data-reliant revenue (CAPEX) divided by aggregate revenue (CAPEX). The columns "% obs > 0" and "% obs $\geq 50\%$" report the fractions of firm-years with positive and majority data-revenue shares, respectively; because the classification operates at the segment level, single-segment firms have shares of zero or one, which makes these two fractions similar. Panel B reports data-revenue shares for selected Fama–French 48 industries. Internet Appendix C.1 details the construction and robustness.

TABLE A.2: **Data-SDK Usage and Revenues: Data-Reliant versus Other Firms**

	log (sales)			
	(1)	(2)	(3)	(4)
Data-SDK count (std.)	0.254***	0.013		
	(3.90)	(0.55)		
Data-SDK share (std.)			0.162***	0.040*
			(3.14)	(1.67)
R&D/assets (std.)	0.258***	-0.034	0.265***	-0.035
	(2.75)	(-0.33)	(2.76)	(-0.35)
Investment/assets (std.)	0.098	0.039	0.126**	0.037
	(1.50)	(1.02)	(2.02)	(0.97)
Sample firms	data-reliant	other	data-reliant	other
Other firm controls (standardized)	Y	Y	Y	Y
Industry-by-quarter FE	Y	Y	Y	Y
Observations	4,194	14,094	4,194	14,094
R-sq	0.848	0.856	0.842	0.856

NOTE.—Table A.2 regresses log quarterly sales on the standardized number of data-sharing SDKs installed on the firm's apps (Columns 1–2) and on the standardized share of data-sharing SDKs in the firm's total SDK usage (Columns 3–4), on the firm-quarter panel of our sample firms from 2014 onward. Data-reliant firms (Columns 1 and 3) are those whose segment revenue comes entirely from data-reliant segments under the stricter cutoff of 80 (Panel A of Table A.1); other firms (Columns 2 and 4) are the remaining firms with non-missing segment data. The standardized coefficients on R&D/assets and investment/assets are displayed to benchmark data-sharing SDK usage against intangible and physical capital; the remaining standardized firm controls (total assets, long-term debt/assets, PP&E/assets, cash/assets) are included but not shown. All columns include industry-by-quarter (Fama–French 48 by year-quarter) fixed effects. Sales are winsorized at 0.5% in both tails. Standard errors are clustered by firm; t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

TABLE A.3: **Comovement in App Performance: Full Results**

	downloads		DAU	
	(1)	(2)	(3)	(4)
data connectedness	0.026***	0.024***	0.026***	0.023***
	(7.56)	(7.25)	(6.78)	(6.26)
ATT × data connectedness	-0.025***	-0.025***	-0.024***	-0.023***
	(-6.63)	(-6.74)	(-5.81)	(-5.75)
mobile user (0/1)		0.001		0.023***
		(0.79)		(7.44)
app category		0.008***		0.010***
		(4.85)		(5.64)
product horizontal		0.009***		0.010***
		(4.62)		(5.81)
product vertical		0.001		-0.002
		(0.36)		(-0.95)
supply chain (0/1)		-0.000		-0.001
		(-0.57)		(-1.60)
technology		-0.001		0.001
		(-1.42)		(0.56)
common analyst (0/1)		0.005***		0.005***
		(4.95)		(4.06)
geography		0.003		0.002
		(1.57)		(0.93)
ATT × mobile user (0/1)		0.010***		-0.004
		(4.06)		(-1.00)
ATT × app category		0.000		-0.003
		(0.00)		(-0.93)
ATT × product horizontal		-0.001		-0.005*
		(-0.29)		(-1.91)
ATT × product vertical		0.003		0.011*
		(0.72)		(1.88)
ATT × supply chain (0/1)		0.000		0.001
		(0.20)		(0.70)
ATT × technology		0.000		-0.002
		(0.38)		(-1.34)
ATT × common analyst (0/1)		-0.002		-0.001
		(-1.39)		(-0.81)
ATT × geography		0.002		0.000
		(0.77)		(0.12)
Firm i#ATT FE	Y	Y	Y	Y
Firm j#ATT FE	Y	Y	Y	Y
Observations	1,401,082	1,401,082	1,399,426	1,399,426
R-sq	0.066	0.068	0.113	0.114

NOTE.—Table A.3 shows the relationship between firm data connectedness and the comovement of app performance. Each observation represents a firm pair at a point in time. For each pair, the comovement of app performance is measured as the correlation between their quarterly log(downloads) and between their log(DAU), calculated separately for the periods before and after ATT (2021Q2). The data on app performance spans from 2014Q3 to 2023Q2. In even-numbered columns, we control for a comprehensive set of pairwise firm linkages, as well as interaction terms between these linkages and the ATT indicator, which equals one for periods after ATT. We include firm-by-time fixed effects (θ_{it} and ι_{jt}) and double-clustered standard errors by firm-i and -j. Time is defined relative to ATT. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

TABLE A.4: **Comovement in Firm Financial Performance: Full Results**

	earnings growth		sales/assets	
	(1)	(2)	(3)	(4)
data connectedness	0.002***	0.002**	0.005***	0.004***
	(2.82)	(2.33)	(3.64)	(2.79)
ATT × data connectedness	-0.003***	-0.003***	-0.003**	-0.003**
	(-3.16)	(-2.88)	(-2.08)	(-2.07)
mobile user (0/1)		0.000		0.000
		(0.20)		(0.50)
app category		0.002**		0.005***
		(2.00)		(4.36)
product horizontal		0.005***		0.016***
		(3.98)		(9.25)
product vertical		0.002		0.007***
		(1.12)		(3.37)
supply chain (0/1)		-0.000		0.001
		(-1.16)		(1.08)
technology		-0.000		-0.001*
		(-0.61)		(-1.69)
common analyst (0/1)		-0.001		0.005***
		(-1.08)		(6.64)
geography		-0.002***		0.009***
		(-3.98)		(4.24)
ATT × mobile user (0/1)		0.000		0.003***
		(0.31)		(4.74)
ATT × app category		-0.002**		0.007***
		(-2.05)		(3.17)
ATT × product horizontal		-0.002		-0.006***
		(-1.25)		(-2.73)
ATT × product vertical		-0.004*		0.001
		(-1.80)		(0.16)
ATT × supply chain (0/1)		0.000		0.000
		(0.21)		(0.33)
ATT × technology		0.000		0.001
		(0.52)		(1.13)
ATT × common analyst (0/1)		0.001		-0.003***
		(0.81)		(-2.75)
ATT × geography		0.001		-0.005*
		(0.70)		(-1.73)
Firm i#ATT FE	Y	Y	Y	Y
Firm j#ATT FE	Y	Y	Y	Y
Observations	1,379,592	1,379,592	1,231,716	1,231,716
R-sq	0.005	0.005	0.184	0.186

NOTE.—Table A.4 shows the relationship between firm data connectedness and the comovement of financial performance. Each observation represents a firm pair at a specific point in time. For each firm pair, the comovement of financial performance is measured as the correlation between their quarterly earnings growth and asset turnover (sales/assets), calculated separately for the periods before and after the introduction of ATT (2021Q2). The data on firms' financial performance spans from 2014Q3 to 2023Q2. In even-numbered columns, we include controls for a comprehensive set of pairwise firm connections, as well as interaction terms between these connections and the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects (θ_{it} and ι_{jt}) and double-clustered standard errors by firm-i and firm-j. Time is defined relative to ATT. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

TABLE A.5: **Comovement in Stock Returns: Full Results**

	raw return		return-CAPM		return-DGTW	
	(1)	(2)	(3)	(4)	(5)	(6)
data connectedness	0.004***	0.002***	0.005***	0.003***	0.003***	0.002***
	(6.52)	(3.64)	(6.71)	(4.35)	(6.25)	(3.76)
ATT × data connectedness	-0.002***	-0.002***	-0.003***	-0.003***	-0.002**	-0.002***
	(-3.00)	(-3.05)	(-3.86)	(-3.95)	(-2.53)	(-2.67)
mobile user (0/1)		0.001***		0.002***		0.001*
		(4.51)		(4.22)		(1.77)
app category		0.008***		0.009***		0.004***
		(8.60)		(7.89)		(5.25)
product horizontal		0.022***		0.030***		0.017***
		(22.26)		(21.54)		(19.58)
product vertical		0.002*		0.001		0.007***
		(1.70)		(0.96)		(4.26)
supply chain (0/1)		0.000		0.000		0.000
		(0.60)		(0.90)		(1.31)
technology		0.001***		0.002***		0.001
		(3.53)		(3.98)		(1.45)
common analyst (0/1)		0.009***		0.012***		0.009***
		(17.03)		(16.69)		(17.28)
geography		0.004***		0.004***		0.003***
		(4.50)		(3.90)		(3.66)
ATT × mobile user (0/1)		-0.001***		-0.002***		-0.000
		(-4.61)		(-4.78)		(-1.03)
ATT × app category		-0.001		-0.002		-0.001
		(-0.67)		(-1.50)		(-1.07)
ATT × product horizontal		0.001		-0.001		-0.001
		(0.55)		(-0.86)		(-1.29)
ATT × product vertical		0.001		0.003*		-0.002
		(0.68)		(1.69)		(-1.25)
ATT × supply chain (0/1)		0.000		0.000		0.000
		(0.36)		(1.00)		(1.10)
ATT × technology		0.000		-0.000		0.000
		(0.60)		(-0.09)		(0.26)
ATT × common analyst (0/1)		0.001		0.002*		0.002***
		(1.29)		(1.87)		(2.69)
ATT × geography		-0.002		-0.002		0.000
		(-1.45)		(-1.15)		(0.26)
Firm i#Time FE	Y	Y	Y	Y	Y	Y
Firm j#Time FE	Y	Y	Y	Y	Y	Y
Observations	5,748,100	5,748,100	5,645,298	5,645,298	4,993,880	4,993,880
R-sq	0.442	0.449	0.097	0.110	0.022	0.028

NOTE.—Table A.5 shows the relationship between firm data connectedness and return comovement. Each observation represents a firm pair at a specific point in time. For each firm pair, return comovement is calculated as the correlation between firms' monthly returns over non-overlapping 12-month windows, defined relative to the introduction of ATT in April 2021, for raw returns, CAPM abnormal returns, and DGTW-adjusted returns. In even-numbered columns, we include controls for a comprehensive set of firm-pair connections, along with interaction terms between these connections and the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects (θ_{it} and ι_{jt}) and double-clustered standard errors by firm-i and firm-j. Time is defined relative to ATT. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

TABLE A.6: **Performance Comovement Test with Placebo Connectedness**

	downloads		sales/assets		return-CAPM	
	(1)	(2)	(3)	(4)	(5)	(6)
media SDK connectedness	0.003**		−0.001***		0.001	
	(2.03)		(−5.60)		(0.54)	
ATT × media SDK connectedness	−0.002		0.002**		0.003	
	(−0.83)		(2.58)		(1.50)	
message SDK connectedness		0.001		−0.002***		0.000
		(0.94)		(−3.90)		(−0.53)
ATT × message SDK connectedness		−0.000		0.002**		−0.001
		(−0.03)		(2.13)		(−0.53)
Other linkages	Y	Y	Y	Y	Y	Y
ATT × other linkages	Y	Y	Y	Y	Y	Y
Firm i#ATT FE	Y	Y	Y	Y	Y	Y
Firm j#ATT FE	Y	Y	Y	Y	Y	Y
Observations	1,401,082	1,401,082	1,231,716	1,231,716	343,202	950,718
R-sq	0.067	0.067	0.186	0.186	0.100	0.137

NOTE.— Table A.6 shows the relationship between placebo connectedness and performance comovement. We construct placebo connectedness following the definition of Equation (1), replacing the top 50 data-sharing SDKs with the top 20 media SDKs (in odd columns) and message SDKs (in even columns). The outcome variables are comovement based on downloads (Columns 1–2), sales-to-assets (Columns 3–4), and CAPM-adjusted returns (Columns 5–6), respectively. In all columns, we include controls for a comprehensive set of firm-pair connections, along with interaction terms between these connections and the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects (θ_{it} and ι_{jt}) and double-clustered standard errors by firm-i and firm-j. Time is defined relative to ATT. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

TABLE A.7: **Cyberattack Spillover Effects on App Performances: Full Results**

	log(downloads)		log(DAU)	
	(1)	(2)	(3)	(4)
cyber event × high exposure	−0.116***	−0.107***	−0.148***	−0.137***
	(−6.60)	(−5.31)	(−7.28)	(−5.75)
cyber event × mobile user (0/1)		−0.003		−0.000
		(−0.19)		(−0.01)
cyber event × app category		−0.020		−0.032*
		(−1.25)		(−1.82)
cyber event × product horizontal		−0.003		−0.005
		(−0.20)		(−0.26)
cyber event × product vertical		0.026		0.029
		(0.70)		(0.63)
cyber event × technology		−0.025*		−0.031*
		(−1.85)		(−1.96)
cyber event × supply chain (0/1)		−0.006		−0.010
		(−0.70)		(−0.86)
cyber event × common analyst (0/1)		0.000		0.006
		(0.01)		(0.34)
cyber event × geography		−0.001		0.008
		(−0.04)		(0.27)
Firm controls	Y	Y	Y	Y
Firm#Event FE	Y	Y	Y	Y
Event-specific relative quarter FE	Y	Y	Y	Y
Observations	81,373	81,373	81,373	81,373
R-sq	0.955	0.955	0.957	0.957

NOTE.—Table A.7 presents the cross-firm spillover effects of major cyber events using a stacked difference-in-differences (DiD) specification in 16-quarter event windows. Major cyber events are defined as those that result in the exposure of over 10 million records. A comprehensive list of these events and their summaries can be found in Appendix Table B.1. For each firm k involved in a major cyber event, we define a peer firm i's exposure to the event as:

$$\text{Exposure}_{ik} = \frac{\sum_P \rho_{ik}^{\text{data},P} DAU_k^P}{\sum_P \sum_j \rho_{ij}^{\text{data},P} DAU_j^P}$$

where j represents any other firm connected to firm i within the data space, and P represents platforms, taking values from {iOS, Android}. A firm k is considered an important peer if $\text{Exposure}_{ik} \geq 0.01$, corresponding to the 75th percentile of the exposure distribution. Firms with $\text{Exposure}_{ik} < 0.01$ are considered as control firms. Each regression includes the following firm-level controls: firm size (log of assets), long-term debt, tangible assets, cash, R&D expenditure, and capital expenditures, with the latter five normalized by total assets. Additionally, we control for firm×event fixed effects and event-specific relative quarter fixed effects. Standard errors are double-clustered by event and firm. t-statistics are provided in parentheses. Statistical significance at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively.

TABLE A.8: **Cyberattack Spillover Effects on App Performances: Placebo Connectedness**

	log(downloads)		log(DAU)	
	(1)	(2)	(3)	(4)
cyber event × high exposure (media)	−0.003		−0.003	
	(−1.40)		(−1.24)	
cyber event × high exposure (msg)		−0.004		−0.002
		(−1.20)		(−0.63)
cyber event × other linkages	N	Y	N	Y
Firm controls	Y	Y	Y	Y
Firm#Event FE	Y	Y	Y	Y
Event-specific relative quarter FE	Y	Y	Y	Y
Observations	194,427	194,427	194,427	194,427
R-sq	0.958	0.958	0.962	0.962

NOTE.—Table A.8 reports the cross-firm spillover effects of major cyber events, using placebo connectedness to define firms exposed to these events. Placebo connectedness is constructed following the definition in Equation (1), substituting the top 50 data-sharing SDKs with the top 20 media SDKs (in odd-numbered columns) and message SDKs (in even-numbered columns). The specifications parallel those in Table 5, with even-numbered columns additionally controlling for the interactions between the cyber-event indicator and other firm linkages. All regressions control for firm characteristics, firm×event fixed effects, and event-specific relative-quarter fixed effects. Standard errors are double-clustered by event and firm. t-statistics are provided in parentheses. Statistical significance at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively.

TABLE A.9: **Cyberattack Spillover Effects on Financial Performances**

	earnings growth		sales/assets	
	(1)	(2)	(3)	(4)
cyber event × high exposure	−0.196**	−0.194**	−0.004*	−0.005**
	(−2.84)	(−2.78)	(−1.81)	(−2.26)
cyber event × other linkages	N	Y	N	Y
Firm controls	Y	Y	Y	Y
Firm#Event FE	Y	Y	Y	Y
Event-specific relative quarter FE	Y	Y	Y	Y
Observations	81,039	81,039	77,267	77,267
R-sq	0.073	0.073	0.937	0.937

NOTE.—Table A.9 presents the cross-firm spillover effects of major cyber events on peer firms' financial performances, extending the specification of Table 5. The outcome variables are two-quarter-ahead earnings growth over the past 12 months (Columns 1–2) and sales-to-assets (Columns 3–4). The sample of major cyber events and the event windows are the same as in Table 5. A firm is considered highly exposed to an event if $\text{Exposure}_{i\tau} \geqslant 0.01$, corresponding to the 75th percentile of the exposure distribution; firms with $\text{Exposure}_{i\tau} < 0.01$ serve as controls. All columns include forward firm-level controls: one-quarter-ahead firm size (log of assets), long-term debt to assets, and cash to assets, as well as investment to assets. Even-numbered columns add interactions of the cyber-event indicator with the full set of non-data firm linkages. All specifications include firm×event fixed effects and event-specific relative-quarter fixed effects. Standard errors are double-clustered by event and firm. t-statistics are provided in parentheses. Statistical significance at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively.

B Additional Information on Variable Construction

B.1 Additional data sources

Additional app metrics. We obtain quarterly app category and user overlap data from Apptopia. To construct the overlap in app category between firm pairs ("*app category*"), we create a binary vector recording a firm's presence in 27 iOS app categories and compute two firms' cosine similarity. Mobile user overlap is a binary variable, defined as 1 if any app from firm i shares users with any app from firm j in Apptopia, and 0 otherwise ("*mobile user*"). Apptopia derives this metric from 1.5 billion user reviews, each linked to a unique store user ID, which allows identification of audience overlap when a user reviews multiple apps.

Text-based Network Industry Classifications (TNIC). Horizontal and vertical industry linkages are obtained from the Hoberg-Phillips Data Library (Hoberg and Phillips, 2010, 2016; Frésard et al., 2020). Horizontal industry linkages ("*product horizontal*") are constructed using pairwise firm similarity scores derived from text analysis of 10-K product descriptions. Vertical industry linkages ("*product vertical*") are based on product vocabularies from the Bureau of Economic Analysis (BEA) Input-Output tables, combined with firms' 10-K product descriptions. Both measures are available at an annual frequency.

FactSet Revere. Supply chain relationships are constructed from FactSet Revere Supply Chain Relationships datasets. This data is collected by FactSet based on companies' relationship information from primary public sources such as SEC 10-K annual filings, investor presentations, and press releases. For each firm pair, we create an indicator variable that equals one ("*supply chain*") if the firm pair has customer-supplier relationships. We also construct a measure of geographical overlap using the FactSet Revere Geographic Revenue Exposure datasets, which provide revenue breakdowns by geography and business segment. The data is organized into a four-level geographic classification (Super-Region, Region, Area, Country). Our geographic overlap measure ("*geography*") is based on the third level, Area, which includes 29 distinct areas. We compute cosine similarity between firm pairs using their revenue distribution vectors across these areas. Both measures are created at an annual frequency.

USPTO. We construct firms' technology proximity using data on their patent applications from the USPTO. For each firm's patent portfolio, we follow the methodology of Jaffe (1986) and Bloom, Schankerman, and Van Reenen (2013) to compute technology proximity. Specifically, we first calculate the share of a firm's patents in each technology class and then compute the cosine similarity between firm pairs ("*technology*"). This measure is available at an annual frequency.

Cyberattack events. We obtain information on cyberattack events from Advisen. This dataset covers more than 90,000 cyber events between 2000 and 2023, collected from publicly verifiable sources, including government websites, keyword-based searches, and official court and litigation sources. We identify 22 major cyberattack events that involve at least 10 million exposed records and list them in the table at the end of this section.

Financial datasets. Firms that share sell-side analysts may exhibit performance comovement (Ali and Hirshleifer, 2020). Using analyst coverage data from I/B/E/S, we construct a binary variable equal to one if a firm pair shares at least one common analyst, and zero otherwise ("*common ana-*

lyst"). Quarterly data on firm accounting variables is from Compustat and data on stock prices and market capitalization from CRSP. Data on asset pricing factors is from Ken French's library.

TABLE B.1: **Major Cyberattack Events**

Company Name	**Exposed Records**	**Date of Accident**	**Date of Notice**	**Case Type**	**Case Description**
Baidu (China) Co., Ltd.	2 billion	13/05/2017	14/05/2017	Data – Malicious Breach	The DU Caller app, developed by Baidu's subsidiary, illegally stored users' personal data and secretly transferred contacts to its servers, which were hacked, exposing 2 billion phone numbers.
Marriott Int'l Inc	500 million	08/09/2018	19/11/2018	Data – Malicious Breach	On 8/9/2018, Marriott discovered an unauthorized attempt to access, encrypt, and remove data from its Starwood database. By 19/11/2018, Marriott believed data from up to 500 million guests had been compromised, including personal details for 327 million guests, with payment card information exposed for some.
Microsoft Corporation	250 million	28/12/2019	29/12/2019	Data – Unintentional Disclosure	Microsoft exposed call center data for nearly 250 million customers through several unsecured cloud servers, which was discovered by security researcher Bob Diachenko after the databases were indexed by the BinaryEdge search engine. The data spanned 14 years of Microsoft Customer Service and Support (CSS) records, which contained customer email and IP addresses, support agent emails, and internal notes. Microsoft secured the data by December 31, after being alerted on December 29.
Equifax Information Services of Puerto Rico Inc.	243 million	29/07/2017	12/09/2017	Privacy – Unauthorized Contact or Disclosure	On July 29, 2017, Equifax discovered a breach in its servers that exposed sensitive personal information, including the names, Social Security numbers, birth dates, and addresses of Michael W. Tomlin and Marilyn Tomlin. Equifax created a website for individuals to check if their data was compromised, with reports suggesting the breach affected over 100 million people. This incident resulted in Equifax violating the Fair Credit Reporting Act (FCRA).
Equifax Inc.	243 million	29/07/2017	20/09/2017	Phishing, Spoofing, Social Engineering	Software engineer Nick Sweeting created a fake version of Equifax's breach information site, equifaxsecurity2017.com, highlighting how easily the site could be impersonated. Several posts from Equifax's Twitter account mistakenly directed users to Sweeting's site, which received around 200,000 hits before being blacklisted by major browsers like Chrome, Firefox, and Safari. Equifax later deleted the incorrect links and apologized for the confusion.
Equifax Inc.	146 million	13/05/2017	30/07/2017	Data – Malicious Breach	In 2017, Equifax experienced a significant cybersecurity breach caused by criminals exploiting a vulnerability in the Apache Struts framework (CVE-2017-5638), affecting US, Canadian, and UK consumers. The attack occurred from mid-May to July 2017 and compromised names, Social Security numbers, birth dates, addresses, and in some cases, driver's license numbers. Equifax was notified of the vulnerability in March 2017 but failed to patch it until July 29, after detecting suspicious network activity. Initially, 145.5 million Americans were identified as affected, with an additional 2.4 million US victims later identified whose names and partial driver's license information were stolen. Credit card details of 209,000 consumers and personal dispute documents of 182,000 were also accessed. In February 2020, US authorities charged four Chinese military officers for the breach, alleging they sought Equifax's sensitive consumer data and trade secrets through the exploited vulnerability.
Capital One Financial Corp.	106 million	22/03/2019	19/07/2019	IT – Configuration/Implementation Errors	The breach was discovered on July 17, 2019, when a GitHub user alerted Capital One about a potential data theft, which the bank confirmed on July 19. Paige A. Thompson, an employee at a cloud computing company that provided data services to Capital One, was arrested for the breach after posting about it on GitHub. She exploited a misconfigured web application firewall to steal data from Capital One's servers. The breach impacted 106 million people, compromising transaction data, credit scores, payment history, balances, and for some, linked bank accounts and Social Security numbers.
Chex Systems Inc	100 million	24/09/2015	21/06/2016	Privacy – Unauthorized Contact or Disclosure	On September 24, 2015, Mission Bank sent Nicholas A. George a letter refusing to open a deposit account, citing information from a consumer report obtained from Chex Systems, Inc. (Chex). George later obtained his ChexSystems report on February 6, 2016, discovering that the Academy Bank trade line inaccurately reflected his liability for the account. This incorrect reporting harmed George by causing embarrassment, inconvenience, and annoyance. Due to its size and large consumer database, Chex's actions violated the Fair Credit Reporting Act (FCRA), harming the hundreds of millions of consumers for whom it holds banking history data.

Google LLC	53 million	07/11/2018	10/12/2018	IT – Configuration/Implementation Errors	On December 10, 2018, Google disclosed a second bug in the Google+ API that potentially exposed the private data of 52.5 million users. Discovered during internal tests, Google stated there was no evidence that third parties had exploited the bug. The issue, caused by a software update, affected Google+ APIs between November 7 and November 13, 2018, when it was fixed. As a result, Google moved the shutdown of Google+ for consumers from August 2019 to April 2019. The bug in the Google+ People API allowed apps to access profile data, including names, emails, and birthdays, which users had marked as private. More sensitive information, such as passwords and financial data, was not affected. Google has since notified impacted users.
T-Mobile US, Inc.	50 million	19/08/2021	07/10/2021	Data – Malicious Breach	Edward Mendez was a victim of SIM-swapping attacks on August 19 and September 12, 2021, with a loss of nearly $240,000 in cryptocurrency. The employee who granted the hacker access had bypassed the 'text-message notification' protocol that notifies all other members under the same account when there is a change to an account. The hackers also disabled two-factor authentication and accessed Mendez's Coinbase account, changing his password and deleting related emails. The breach exposed sensitive information, including Social Security numbers, phone numbers, addresses, and driver's license details. The Kansas attorney general reported that over 335,000 Kansas residents could be affected by the T-Mobile data breach.
T-Mobile US Inc	50 million	16/08/2021	18/08/2021	Data – Malicious Breach	The breach was detected after the attacker reported the incident to Motherboard. On August 16, T-Mobile confirmed the breach, which affected 7.8 million current postpaid customers and over 40 million records of former or prospective customers who applied for credit. The company claimed that the stolen data included personal information such as names, birthdates, Social Security numbers, and driver's license/ID numbers, but not bank, payment data, or passwords. Additionally, the names, phone numbers, and account PINs of around 850,000 prepaid users were exposed. T-Mobile quickly shut down the access point used in the attack. As a consequence, the company is offering two years of free identity theft protection via McAfee and advising postpaid customers to change their PINs while also providing account takeover protection.
T-Mobile USA, Inc.	50 million	17/08/2021	24/08/2021	Data – Malicious Breach	On August 17, 2021, T-Mobile discovered that a bad actor had illegally accessed unencrypted personal information, which included names, driver's license numbers, phone numbers, addresses, government identification numbers, Social Security numbers, dates of birth, and T-Mobile account PINs.
Chegg Inc	40 million	29/04/2018	19/09/2018	Data – Malicious Breach	Chegg, Inc., a US-based education technology company, plans to reset passwords for over 40 million users after discovering a security breach that dates back to April 29, 2018. The breach was detected on September 19, 2018, and involved unauthorized access to a company database containing user data for chegg.com and related brands, such as EasyBib. Hackers may have accessed user information, including names, email addresses, shipping addresses, usernames, and hashed passwords, although Chegg did not specify the hashing algorithm used. Social Security numbers and financial data were not compromised. The breach caused Chegg's stock price to drop by 10 percent.
T-Mobile US Inc	37 million	25/11/2022	05/01/2023	Data – Malicious Breach	On January 19, 2023, T-Mobile reported a cyberattack that exposed data from approximately 37 million postpaid and prepaid customer accounts. The breach was detected on January 5, 2023, when T-Mobile identified unauthorized data access through a single Application Programming Interface (API). The API did not expose sensitive information such as payment card details, Social Security numbers, or passwords, but did allow access to customer data including names, billing addresses, emails, phone numbers, birth dates, account numbers, and plan details.
Taobao	21 million	14/10/2015	04/02/2016	Data – Malicious Breach	From October 14 to 16, 2015, a group of hackers attempted to access over 20 million active user accounts on Taobao, Alibaba Group's e-commerce platform, using rented space on Alibaba's AliCloud services. Of the 99 million accounts involved, 20.59 million had matching passwords. The hackers aimed to acquire these accounts for order manipulation and sale to scammers. However, the attack did not involve a direct breach of Taobao. Instead, hackers used account information from non-Taobao platforms to find matching credentials. The hack was stopped a month later by Chinese authorities after website admins detected suspicious activity on the platform.
Morgan Stanley	14 million	21/02/2020	10/07/2020	Data – Malicious Breach	In 2019, Morgan Stanley replaced certain computer servers in local branch offices that stored information on encrypted disks, which may have contained personal data. During an inventory, Morgan Stanley was unable to locate these encrypted disks, leading to a data breach. The incident, which occurred on February 21, 2020, compromised personally identifiable information, including Social Security numbers, affecting 14,256,250 individuals.

Twitter Inc	13 million	07/02/2020	07/02/2020	Data – Malicious Breach	On February 7, 2020, the official Facebook Twitter account was briefly taken over by the hacking group OurMine. The incident lasted less than 30 minutes, during which a tweet was sent to Facebook's 13.4 million followers, stating: "Hi, we are OurMine. Well, even Facebook is hackable but at least their security better than Twitter," and offering "security services" to improve account protection. The breach was not a result of compromised Facebook or Twitter systems, but rather due to a third-party marketing platform used to manage social media. A Twitter spokesperson confirmed the issue, stating the compromised accounts were quickly locked. Facebook later confirmed in a tweet that the issue had been resolved and access restored.
Blackbaud Inc	13 million	07/02/2020	01/05/2020	Data – Malicious Breach	In May 2020, Blackbaud, Inc. was targeted in a sophisticated ransomware attack. The breach, which began on February 7, 2020, and lasted intermittently until May 20, 2020, compromised backup files for clients using Blackbaud's Raiser's Edge/NXT system. While the hackers did not access encrypted credit card information, bank account details, Social Security numbers, or login credentials, they did obtain contact information, demographic data, and donation histories. Blackbaud paid an undisclosed ransom after evidence showed the stolen data was destroyed, and it is believed the compromised data was not misused or publicly shared. However, further investigation revealed that more unencrypted data, including bank account information and Social Security numbers, may have been accessed. As of September 2020, the Identity Theft Resource Center reported that 536 organizations and nearly 13 million people were impacted.
Quest Diagnostics Inc	12 million	01/08/2018	14/05/2019	Data – Malicious Breach	On June 3, 2019, Quest Diagnostics revealed that a data breach potentially exposed the personal, financial, and medical information of approximately 11.9 million patients. The breach occurred through a billing collections vendor, American Medical Collection Agency (AMCA), which provides services to Optum360, a Quest contractor. An unauthorized user had access to AMCA's system from August 1, 2018, to March 30, 2019. The compromised data included credit card numbers, bank account information, medical details, and Social Security numbers, though lab results were not exposed. As of May 31, 2019, AMCA estimated that 11.9 million Quest patients were affected. AMCA has yet to provide full details about the breach, and Quest has been unable to verify all of the information.
MGM Resorts International	11 million	07/07/2019	21/02/2020	Data – Malicious Breach	On or around July 7, 2019, an unauthorized individual accessed MGM Resorts International's computer network and stole customer data, which included personal information such as names, addresses, driver's license and passport numbers, military IDs, phone numbers, emails, and dates of birth. A subset of this data was initially shared on a closed internet forum but was later fully exposed on a hacking forum in February 2020, affecting over 10.6 million MGM guests. This breach left customers vulnerable to phishing attacks and SIM-swapping schemes. MGM notified affected customers and government agencies on September 5, 2019, about two months after the breach, stating that it had no evidence the data had been misused; the incident became widely known only after the data surfaced on the hacking forum in February 2020.
Laboratory Corp of America Holdings	10 million	01/08/2018	14/05/2019	Data – Malicious Breach	On May 14, 2019, Laboratory Corporation of America Holdings (LabCorp) was notified by its vendor, Retrieval-Masters Creditors Bureau, Inc., operating as American Medical Collection Agency (AMCA), of unauthorized activity on AMCA's web payment page. The breach occurred between August 1, 2018, and March 30, 2019. LabCorp immediately ceased sending collection requests to AMCA and halted pending requests. AMCA, which serves as an external collection agency for LabCorp and other healthcare companies, stored data for approximately 7.7 million LabCorp consumers, according to LabCorp's initial SEC filing. The compromised data included personal information such as names, addresses, dates of birth, and payment details (credit card or bank account information). No laboratory results, test orders, Social Security numbers, or insurance details were exposed. LabCorp subsequently reported 10,251,784 affected individuals to the US Department of Health and Human Services.
Chipotle Mexican Grill Inc	10 million	24/03/2017	25/04/2017	Data – Malicious Breach	On April 25, 2017, Chipotle disclosed a data breach caused by credit card-stealing malware that infected the payment processing system in most of its 2,250 restaurants. The malware collected cardholder information, including names, card numbers, expiration dates, and verification codes, during transactions between March 24 and April 18, 2017. Chipotle has since removed the malware. The breach affected tens of millions of customers, including 1,798 New Jersey residents.

B.2 Examples of product-design SDKs

TABLE B.2: **Description of Example Product-Design (Functionality) SDKs**

SDK Name	Category	# Installations	Introduction
AliPay	Payment	23571	The cross-border app payment solution provides convenient, safe, and reliable payment services to third-party applications. This payment solution is applicable to wireless devices (including mobiles and tablet computers) supported by Android or iOS system.
Stripe	Payment	8435	The Stripe SDK allows you to quickly build a payment flow in your app. Stripe provides powerful and customizable UI elements that you can use out-of-the-box to collect your users' payment details.
Nimbus	Security	35884	The Nimbus SDK handles both requesting the ad and rendering the impression — all with a lightning-fast server-to-server connection — making it the easiest way to integrate with the Nimbus exchange. The SDK is customizable. You can choose to use the rendering function, the requesting function, or both.
Okta	Security	27663	Okta connects any person with any application on any device. It's an enterprise-grade, identity management service, built for the cloud, but compatible with many on-premises applications. With Okta, IT can manage any employee's access to any application or device.
Zendesk Support	Customer Support	4032	The SDK provides the following UIs for both Support and Guide to embed customer service features in an app: Help Center Overview - Lets the user access articles in your Zendesk Guide knowledge base and, optionally, submit a ticket; Help Center Article - Lets the user view a specific help center article; Request - Lets the user view, update, and submit tickets to your customer service team; Request List - Lets the user view a list of their tickets.
Helpshift	Customer Support	2278	The Helpshift SDK allows your support team to provide in-app help in the form of searchable, native FAQs and direct, two-way messaging to end users.
Appirater	Reviews & Feedback	30384	Appirater is a class that you can drop into any iPhone app (iOS 4.0 or later) that will help remind your users to review your app on the App Store.
iRate	Reviews & Feedback	26894	iRate is a library to help you promote your iPhone and Mac App Store apps by prompting users to rate the app after using it for a few days.

TABLE B.3: **Summary Statistics: Additional Variables**

Panel A. Pairwise variables

	mean	sd	p10	p25	p50	p75	p90	count
Pairwise connections								
data connectedness	0.171	0.19	0.00	0.00	0.11	0.29	0.45	1,401,082
mobile user (0/1)	0.036	0.19	0.00	0.00	0.00	0.00	0.00	1,401,082
app category	0.157	0.24	0.00	0.00	0.00	0.27	0.52	1,401,082
product horizontal	0.015	0.03	0.00	0.00	0.00	0.02	0.05	1,401,082
product vertical	0.003	0.00	0.00	0.00	0.00	0.00	0.01	1,401,082
technology	0.014	0.08	0.00	0.00	0.00	0.00	0.00	1,401,082
supply chain (0/1)	0.008	0.09	0.00	0.00	0.00	0.00	0.00	1,401,082
common analyst (0/1)	0.063	0.24	0.00	0.00	0.00	0.00	0.00	1,401,082
geography	0.300	0.43	0.00	0.00	0.00	0.90	0.99	1,401,082

Panel B. Firm-level variables

	mean	sd	p10	p25	p50	p75	p90	count
Δ payment SDK	0.008	0.28	0.00	0.00	0.00	0.00	0.00	20,344
Δ security SDK	0.005	0.14	0.00	0.00	0.00	0.00	0.00	20,344
Δ customer support SDK	0.001	0.08	0.00	0.00	0.00	0.00	0.00	20,344
Δ review & feedback SDK	0.002	0.07	0.00	0.00	0.00	0.00	0.00	20,344
L1.payment SDK (peers)	4.407	1.42	2.96	4.45	4.91	5.15	5.32	20,344
L1.security SDK (peers)	4.442	1.43	2.97	4.49	4.95	5.20	5.36	20,344
L1.customer support SDK (peers)	3.865	1.30	2.17	3.80	4.28	4.59	4.80	20,344
L1.review & feedback SDK (peers)	4.144	1.36	2.62	4.14	4.60	4.88	5.05	20,344
L1.payment SDK	2.023	2.06	0.00	0.00	2.00	4.00	5.00	20,344
L1.security SDK	0.827	0.84	0.00	0.00	1.00	1.00	2.00	20,344
L1.customer support SDK	0.139	0.39	0.00	0.00	0.00	0.00	1.00	20,344
L1.review & feedback SDK	0.333	0.54	0.00	0.00	0.00	1.00	1.00	20,344
L1.size	22.789	2.02	20.22	21.37	22.75	24.13	25.51	18,596
L1.long-term debt/assets	0.269	0.23	0.01	0.08	0.23	0.39	0.57	18,596
L1.tangibles/assets	0.206	0.22	0.01	0.04	0.12	0.31	0.57	18,596
L1.cash/assets	0.182	0.18	0.02	0.05	0.11	0.25	0.47	18,596
L1.R&D/assets	0.009	0.02	0.00	0.00	0.00	0.01	0.03	18,596
L1.investments/assets	0.021	0.03	0.00	0.00	0.01	0.03	0.05	18,596

NOTE.—Table B.3 reports summary statistics for additional variables used in the empirical analysis. Panel A summarizes pairwise data connectedness and other measures of firm linkages. Panel B reports firm-quarter statistics for changes in functionality-SDK usage, lagged SDK usage by the firm and its data-connected peers, and lagged firm characteristics. Δ denotes the quarterly change, and L1 denotes a one-quarter lag.

B.3 Additional summary statistics

Other statistics. Table B.3 provides additional summary statistics. Panel A lists all firm linkages. The app-category similarity and similarity in geographical distribution of business segments exhibit relatively large means of 0.157 and 0.300, respectively. Additionally, 0.8% of firm pairs have customer-supplier linkages, and 6.3% of firm pairs share common analysts. In Panel B of Table B.3, we report the summary statistics for firm-level variables, including changes in a firm's SDK usage and the level of peer firms' SDKs, categorized by SDK functionality. For example, the average firm's apps use up to 2.023 unique payment SDKs, and this number changes by 0.008 per quarter on average. In terms of firm characteristics, the average firm has a long term debt ratio of 26.9%, an asset tangibility of 20.6%, and a cash-to-asset ratio of 18.2%.

C The Economic Value of Data Assets and Data-Sharing SDKs

This appendix provides details on variable construction and results for analysis in Section 2 and how we calibrate the parameter ζ (cash flow per unit of data) for our model in Section 4. Section C.1 describes the segment-based measures of the importance of data assets. Section C.2 discusses the calibration of ζ. Section C.3 summarizes the Bureau of Economic Analysis (BEA) national accounts of data assets. Section C.4 documents why mobile applications, rather than the web, are the primary environment for data collection and cross-firm data sharing. Section C.5 details the classification of major SDK updates and the associated event-study design. Section C.6 describes the construction of the disclosure evidence and reports additional results.

C.1 Data reliance across Compustat business segments

We measure firms' reliance on data as a productive asset using business-segment information from Compustat Historical Segments over 2014–2024. Our data-reliance score measures how intensively a segment's industry uses customer data as a core operating input—transforming customer interactions into structured data and deploying it for targeting, segmentation, personalization, and recommendation in order to acquire, retain, and monetize customers.

Compustat retains up to ten segments per firm-year.[1] A large language model (LLM) is instructed to assign each NAICS industry code a score from 0 to 100: 0–10 when customer-data use is incidental; 11–40 when data supports the business but is not central to value creation; 41–70 when data is important and routinely used in operations or decision-making; and 71–100 when the business model is data-centric and data is a core input into production or service delivery.[2] A business segment is labeled as "data-reliant" if the score of its industry is at least 70, and we use a more conservative threshold of 80 for all regressions that involve this segment classification.

A firm's data-revenue (data-CAPEX) share in a given year is the fraction of its total revenue (capital expenditure) generated by data-reliant segments, winsorized at 1% in both tails.[3] In Table A.1, we report that 34.3% of firm-year observations have positive data-reliant revenue and 30.3% derive at least 50% of their revenue from data-reliant segments. The app-owning firms rely more on data: more than half of firm-years (52.3%) have positive data-reliant revenue and 43.2% derive more than half of revenues from data-reliant segments.

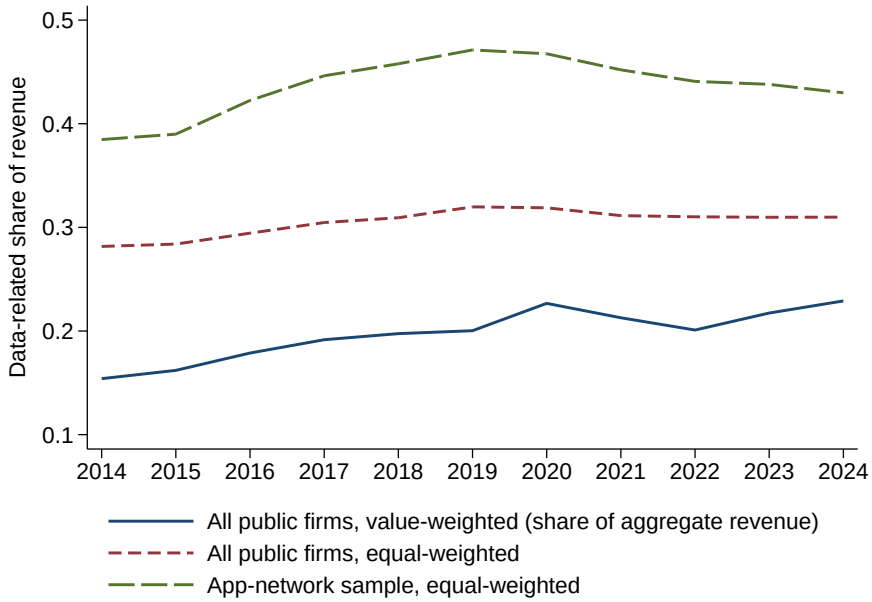
The magnitudes reported in Table A.1 are robust along three dimensions. First, under the stricter score cutoff of 80, data-reliant segments account for 13.2% of aggregate revenue (22.7% equal-weighted). Second, in the untabulated results, we use the Compustat fundamentals revenue as the denominator rather than the segment-aggregated revenue, the data-reliant revenue share is averaged at 22.9% over 2014–2024 and is 24.3% in 2024 under the baseline threshold of our data reliance score at 70. Third, also at the baseline cutoff of 70, excluding the financial industries (banks, insurance, trading, and real estate)—for which the scores classify customer data as a core

[1] Under accounting rules, firms must separately report an operating segment if it accounts for at least 10% of firm revenue, profit or loss, or total assets; additional segments must be reported until reportable segments account for at least 75% of total firm revenue. Compustat segment data therefore covers the economically material parts of firms' operations, although smaller operating segments may be aggregated or omitted.

[2] We instruct GPT-5.1 to assign the scores based on the NAICS industry title and the Census NAICS description.

[3] We compute the shares using both segment-aggregated and Compustat-fundamentals totals as denominators and obtain similar results; we report the former unless otherwise stated.

FIGURE C.1: **Data-Revenue Share of US Public Firms, 2014–2024**



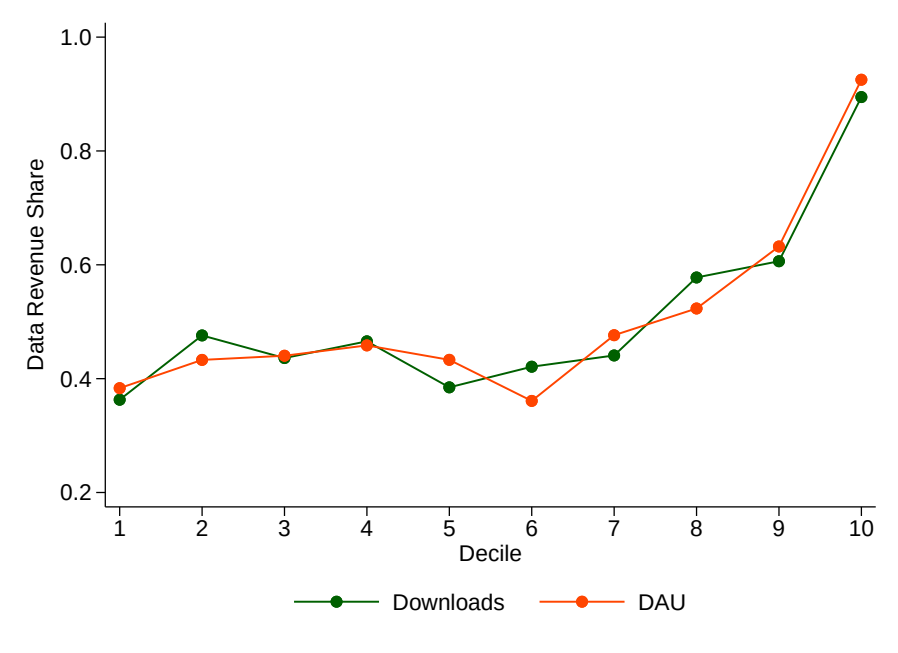
NOTE.—Figure C.1 plots the share of revenue generated by data-reliant business segments among US public firms from 2014 to 2024. A Compustat business segment is classified as data-reliant if its NAICS industry's data-reliance score is at least 70. The figure reports three series: the value-weighted share for all public firms, calculated as aggregate data-reliant revenue divided by aggregate segment revenue; the equal-weighted share for all public firms, calculated as the average firm-level share; and the equal-weighted share for firms in our app-network sample. The difference between the value- and equal-weighted series for all public firms indicates that smaller firms tend to have higher data-reliant revenue shares. The aggregate share dips in 2021–2022, immediately following the introduction of ATT.

operating input—the aggregate data-revenue share remains 16.6% (22.6% equal-weighted).

Figure C.1 plots the share of public-firm data revenue by year. The aggregate share climbs through 2020, dips in 2021–2022 immediately following the introduction of ATT (April 2021), and resumes growing thereafter. The dip corroborates, in aggregate revenue data, the disruption of data-driven activity that we identify at the firm level throughout the paper.

Validation of the classification. Two checks indicate that the scores capture genuine data reliance. First, the industry ranking that the scores produce (Panel B of Table A.1) is an intuitive one: data-revenue shares are largest where data is plausibly a core input—banking, business services, computers, publishing, entertainment, telecommunications, and retail—and at or near zero in industries producing physical or commodity goods. Because the language model scores each industry name in isolation, with no industry-level target to fit, recovering this intuitive pattern is an out-of-sample check of the classification rather than a mechanical artifact. The close alignment with the industries that the BEA identifies as the largest contributors to data investment—professional, scientific, and technical services; finance and insurance; health care; and information—provides further external validation (Section C.3). Second, we validate the classification using an independent firm-level outcome: firms' app user bases. Figure C.2 sorts firms into deciles of downloads and daily active users (downloads/DAU) and plots the average share of total (Compustat-fundamentals)

FIGURE C.2: **Data-Reliant Revenue as a Share of Total Revenue, by Downloads/DAU Decile**



NOTE.—Figure C.2 reports the average data-revenue share across deciles of firms' mobile downloads and daily active users (DAU). Data-revenue share is the fraction of total Compustat revenue generated by segments whose NAICS industries have a data-reliance score of at least 70. All variables are winsorized at 1% in both tails.

revenue that the classifier attributes to data-reliant segments (data-reliance score of at least 70). The share rises from about 0.36–0.38 in the bottom decile to 0.89–0.92 in the top decile—roughly a threefold increase, concentrated in the top two deciles—and the downloads and DAU sorts track each other closely. The user-base measures come from an entirely separate data source and play no role in the segment classification, yet the firms that the classifier identifies as relying most on data are precisely the firms with the largest app user bases, where data should play the biggest role.

Data-reliance of private firms. Our analysis is confined to publicly listed firms as we rely on Compustat accounting data. In Figure C.3, we report that in the Infogroup Historical Business Database of establishment-level records, more than 80% of establishments and roughly 60–70% of employment and sales in data-reliant sectors are from private firms. Here, data-reliant sectors are identified at the six-digit NAICS level using the same LLM-based data-reliance scores that underlie our business segment classification. Therefore, our valuation of data as a networked asset in Section 4.5 likely understates the aggregate importance of data.

C.2 Data-driven cash flow calibration for the model

We discuss the calibration of ζ—cash flow generated by one unit of data (proxied by DAU)—in our model in Section 4.3. We measure each firm's data-driven free cash flow from Compustat

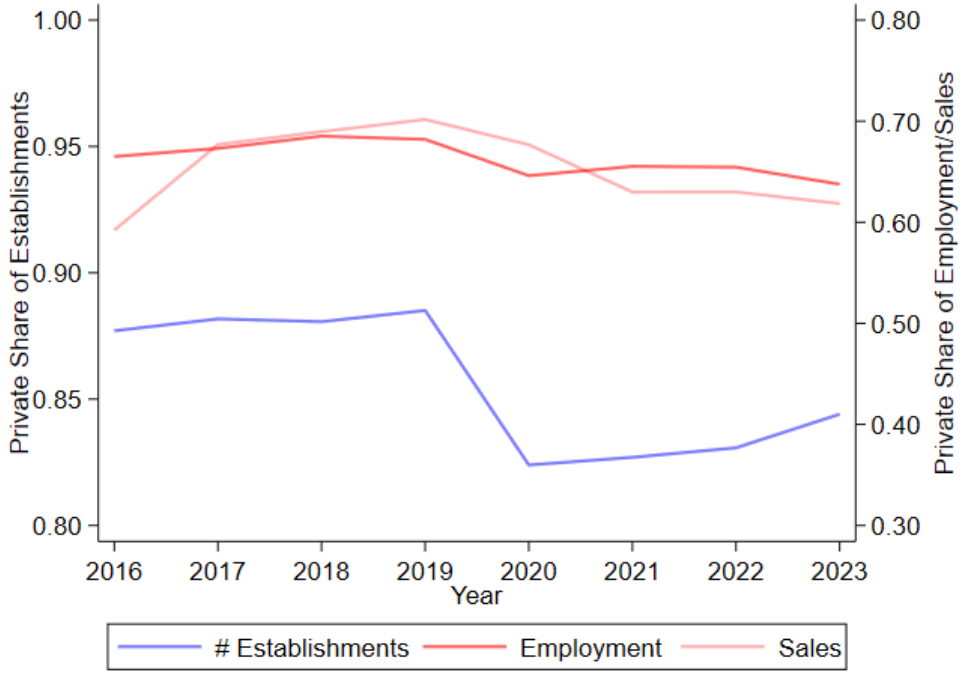


FIGURE C.3: **The Importance of Private Firms in Data-Reliant Sectors**

NOTE.—The figure reports the shares of establishments, employment, and sales accounted for by private firms (relative to public firms) in US data-reliant sectors over 2016–2023. The underlying data comes from the Infogroup Historical Business Database, which compiles establishment-level records from business directories, public records, company-provided information, and ongoing verification; employment and sales are estimated and should be interpreted with caution. Data-reliant sectors are six-digit NAICS industries with a data-reliance score of at least 80, using the LLM-based scores that underlie the segment classification in the paper.

Historical Segments. Annual data-driven free cash flow is

$$FCF_{it}^{\text{data}} = Revenue_{it}^{\text{data}} - COGS_{it}^{\text{data}} - SG\&A_{it}^{\text{data}} - CapEx_{it}^{\text{data}}.$$

Segment-level cost items are missing for roughly one quarter of the firm-years in our sample: the unfilled measure is available for only 2,776 of the 3,702 firm-years. Dropping these firm-years would both reduce the sample substantially and condition the analysis on cost-data availability. We therefore fill missing cost items in two steps. First, within each Fama–French 48 industry-year cell, we compute an industry free-cash-flow margin as the ratio of aggregate data-driven free cash flow to aggregate data-driven revenue, using firm-years for which both quantities are observed and requiring at least five such firm-years per cell. Second, for firm-years with observed data revenue but missing cost items, we impute data-driven free cash flow by applying the corresponding industry-year margin to the firm's own data revenue.

Next, we estimate the following firm-year regression:

$$FCF_{it}^{\text{data}} = \alpha_t + \zeta^{\text{ann}}\, DAU_{it}^{\text{mil}} + X_{it}'\Pi + \varepsilon_{it},$$

where FCF_{it}^{data} is measured in millions of dollars per year, DAU_{it}^{mil} is DAU measured in millions

of users, α_t denotes year fixed effects, and X_{it} includes total assets, cash and short-term investments, net property, plant, and equipment, and long-term debt, each winsorized at the 0.5% level. Standard errors are clustered by firm. The estimated coefficient implies that one additional million DAU is associated with \$28.983 million of annual cash flow. Because the model's time unit is a quarter, we set $\zeta = 28.983/4 = 7.246$. The magnitude is consistent with disclosed platform economics. Our estimate implies about \$29 of annual data-driven free cash flow per user. In 2021, Meta's annual revenue per user was about \$41 worldwide and about \$214 in the United States and Canada, and its free-cash-flow margin that year was about one third (\$38.4 billion of free cash flow on \$117.9 billion of revenue), so its disclosed free cash flow per user was on the order of \$13 per year worldwide and \$70 per year in the United States and Canada.

C.3 The BEA national accounts of data assets

In preparation for implementing the new standards in the 2025 System of National Accounts, the U.S. Bureau of Economic Analysis (BEA) has published estimates for *own-account data and databases* as a distinct fixed-asset class. "Own-account" is national-accounts language for how the asset is produced—data assets built in-house rather than purchased from outside. The term is silent on whose behavior the data describes and on whether the data is subsequently shared across firms; the BEA figures thus size the data-asset class as a whole, whereas our paper studies the cross-firm, networked layer of this new category of productive asset.

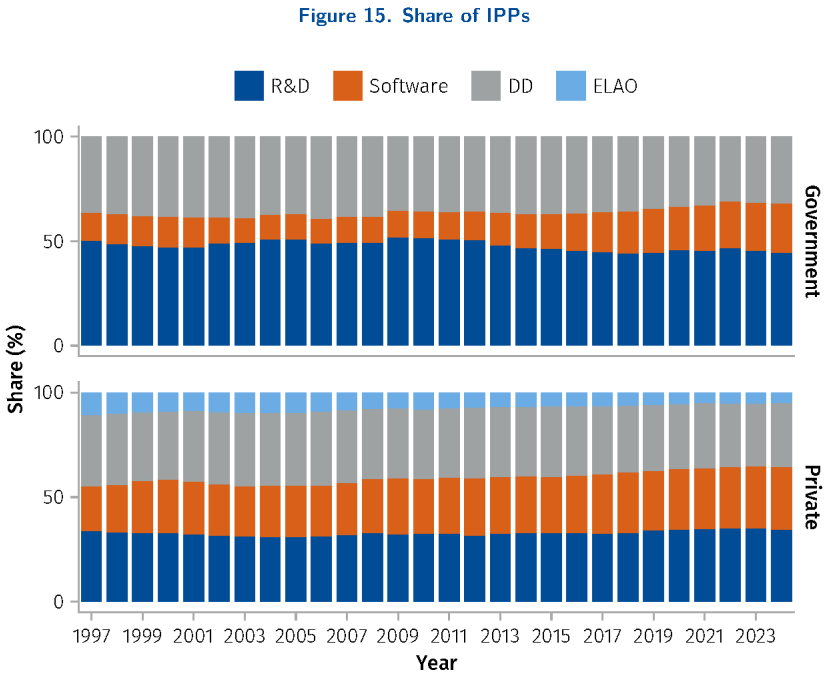
The BEA estimates that US investment in data assets was \$875 billion in 2024, with a current-cost net stock of \$2.0 trillion; capitalizing the new asset class raises measured annual US GDP growth by 7 basis points over 2002–2024 (Santiago Calderón, 2026).[4] Private data investment amounts to 11.5–16% of nonresidential fixed investment and roughly one third of investment in intellectual-property products (IPP)—the same order of magnitude as R&D and software. Figure C.4 reproduces the composition of IPP investment from Santiago Calderón (2026); data assets also account for 22.6% of the private IPP net stock in 2024.[5]

Two features of the BEA estimates are worth noting. First, the estimates are cost-based: own-account data assets are valued at the sum of production costs. As we discuss in the context of intangible-asset valuation, cost-based methods do not capture the positive network externalities of data—the value created when one firm's data informs other firms through the sharing network. The BEA figures are therefore likely to understate the importance of data capital; our network-augmented valuation framework in Section 4 is designed precisely to price these spillovers. Second, the estimates cover only own-account production; data acquired through market transactions falls outside them. That market-transacted channel can be sized directly, and it is an order of magnitude smaller: receipts of the US data-processing industry on data-related products were \$21.6 billion in 2012 on the direct Census measure and \$78.1 billion in 2017 on a broader survey-based measure (Rassier, Kornfeld, and Strassner, 2019), and US firms spent about \$19.2 billion in 2018

[4] In the BEA's fixed-asset accounting, the net stock is built by the perpetual-inventory method—the cumulative value of past investment less cumulative depreciation—and the current-cost net stock revalues those items at current end-of-year prices rather than at their original acquisition prices.

[5] The gap between the investment share (roughly one third of IPP investment) and the 22.6% net-stock share is a depreciation effect. The BEA assigns data and databases a geometric depreciation rate of one third per year, versus roughly 10–16% for R&D. Since the net stock scales inversely with the depreciation rate, a dollar of R&D investment generates roughly two to three times as much surviving capital as a dollar of data investment.

FIGURE C.4: **BEA: Data and Databases versus Other Intellectual-Property Products**



NOTE.—Figure C.4 reproduces Figure 15 of Santiago Calderón (2026): the composition of gross fixed capital formation in intellectual-property products (IPPs) for the government (top panel) and private (bottom panel) sectors, 1997–2024. "DD" (gray) is the BEA's new data-and-databases asset class (own-account); "ELAO" is entertainment, literary, and artistic originals. In the private sector, investment in data and databases is of the same order of magnitude as R&D and software throughout the sample—roughly one third of IPP investment.

on third-party audience data and related solutions (IAB Data Center of Excellence and Winterberry Group, 2018)—against $875 billion of own-account data investment in 2024. Data purchased in markets is thus small relative to data produced in-house, consistent with our premise that shared data infrastructure, rather than market transactions, is the primary channel for inter-firm data flows.

Finally, we highlight that the industries that the BEA identifies as the largest contributors to data investment—professional, scientific, and technical services; finance and insurance; health care; and information—line up closely with the industry ranking from our segment-based benchmark (Panel B of Table A.1), an external validation of the LLM-based classification.

C.4 Where is data collected: Apps versus the web

Apps host the majority of digital engagement. As we emphasize in Section 2, 60–80% of digital engagement occurs inside native apps. First, cross-platform measurement indicates that mobile devices account for roughly 70% of total digital time (desktop for 30%),[6] and that 85–90% of mobile time is spent inside native apps rather than mobile browsers.[7] Using the midpoint (87.5%), native

[6]See Comscore's *2017 U.S. Cross-Platform Future in Focus* (Comscore, 2017).

[7]US adults spent 88% of mobile-internet time within apps in 2020 (Wurmser, 2020).

apps account for $70\% \times 87.5\% \approx 61\%$ of total digital time. Second, as an alternative, normalize total mobile engagement to 100 units: 87.5% of mobile time in apps yields 87.5 units of native-app engagement and 12.5 units of mobile-browser engagement. Similarweb reports an approximately even desktop–mobile split within web traffic,[8] implying another 12.5 units of desktop-browser activity, so that native apps account for $87.5/(87.5 + 12.5 + 12.5) \approx 78\%$ of combined app and browser engagement. Engagement intensity amplifies this dominance: session lengths are materially longer in apps, and high-dwell activities (social media, video, gaming, fintech) are app-centric, so browser-based tracking captures not only a minority of traffic but a disproportionately smaller share of behavioral depth. Figure C.5 illustrates the app–web split for major banks.

Why cookies cannot substitute for SDK-mediated sharing. Third-party cookies provided a widely used but less effective means of cross-site identification than SDKs. Because cookies are scoped to individual web domains, advertising-technology firms had to reconcile their separate identifiers through "cookie syncing," which produced incomplete matches and fragmented user profiles; coverage was further limited as users deleted or blocked cookies; and Safari's Intelligent Tracking Prevention and Firefox's Enhanced Tracking Protection sharply reduced the reach of cookie-based tracking by blocking or partitioning third-party cookies. Mobile SDKs, by contrast, (i) historically observed a common, persistent device-level identifier (Apple's IDFA and Google's GAID) across apps, enabling consistent cross-app profiling; (ii) operate continuously in the background, whereas web tracking requires an active browser session; and (iii) access data types unavailable on the web, such as GPS location, device sensors, and contacts. The remaining alternative—bilateral firm-to-firm data agreements—is rare and non-standardized.

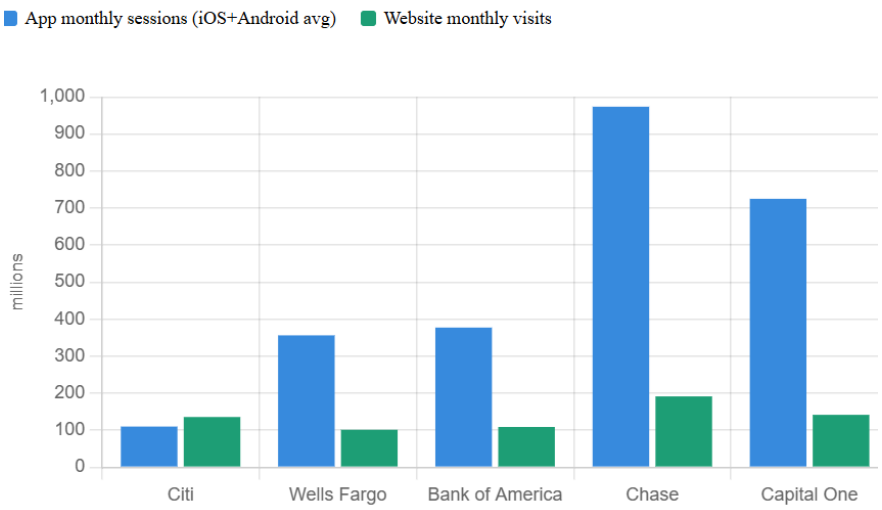
Market-based evidence from the literature. Mobile apps are also value-relevant to investors, precisely through their data-collection role. Wu (2023) builds a market-based measure of app value—the firm-specific component of the three-day stock return around each mobile-app release—and estimates values that are economically large (a mean of roughly $120 million and a median of roughly $24 million per app). Higher app values predict, over a five-year horizon, lower firm-specific risk and substantial growth in sales, assets, employment, and market share. Both effects are concentrated in apps that collect user data: firms releasing data-collecting apps see about a 6.7% five-year reduction in idiosyncratic volatility, versus 2.4% for apps without data collection, and the risk reduction weakens after the California Consumer Privacy Act restricted data collection.

C.5 Data-sharing SDK updates: Classification and analysis

This appendix details the analyses of major SDK updates summarized in Section 2. Firms' installation and removal of SDKs are their own decisions, so evidence based on firms' SDK usage—including the revenue regressions in Table A.2—cannot be interpreted causally. Major updates released by the SDK providers, by contrast, apply to the providers' entire customer base and are exogenous to any individual firm's decisions. We trace their impact in two designs: a firm-level event study of stock returns and an app-level difference-in-differences analysis of customer engagement.

[8] Similarweb reports that mobile devices accounted for 53.8% of US website traffic in May 2026, with desktop and tablet accounting for the remainder; its traffic measure combines desktop and mobile-web activity and does not include usage within native apps (Similarweb, 2026).

FIGURE C.5: **App versus Web Traffic: Banks as an Example**



NOTE.—Figure C.5 decomposes major US banks' digital traffic into native-app and web. Source: Similarweb.

SDK release histories and exposure. Starting from the top 50 data-sharing SDKs—the same set used to construct the data-connectedness measure—we collect each SDK's full version-release history (version string, release date, and the developer's release note) from the SDKs' published changelogs. This yields 1,961 version records; restricting to releases with a valid release date before ATT (April 26, 2021) leaves 1,031 version records covering the 22 SDKs with dated changelog histories, which form the classification universe. An app is exposed to an SDK's major update if it has that SDK installed on the update date; a firm is exposed if at least one of its apps is exposed.

Classifying major updates. A *version-series* rule identifies major-version releases: version strings whose leading integer increments with all subsequent components equal to zero (x, $x.0$, $x.0.0$, ...), which yields 47 of the 1,031 releases. We then organize each SDK's release history into major-version series, read the release notes at each series boundary, and, when the boundary release documents no material change, read the subsequent updates within the series. A release is flagged when its notes document a material restriction of the SDK's data-collection, data-sharing, or analytics capability. In total, 18 releases are flagged.

The 18 releases occur on 15 calendar dates, as AdMob and Google Ad Manager were updated together on three dates. Converting calendar dates to trading dates yields 14: myTarget's July 3, 2020 release shifts to July 6, a trading day following a market holiday, and then coincides with another event. Hundreds of apps are exposed to each update. The focus on negative updates reflects our sample period, in which rising consumer awareness and regulatory pressure in the United States and Europe led SDK providers to issue privacy-protecting updates, and it is precisely these updates that compromise firms' access to consumer data and thereby shock the cross-firm data flows we study. In both designs, identification compares firms (apps) exposed to a data-restricting update with contemporaneous benchmarks that are not exposed: factor-model expected returns in the stock-return analysis and matched control apps observed over the same months in the app-level analysis.

Below we provide examples of data-restricting SDK updates. Most are privacy-related releases that restrict the data an SDK can collect, often in response to regulatory requirements. Two of the 18 updates, quoted from the providers' release notes, are:

- *MoPub v5.0.0 (14 May 2018)*: "General Data Protection Regulation (GDPR) update ... to obtain and manage consent from users in the EEA ... Required for receiving personalized ads." The SDK now requires user consent before personalized advertising can be served.
- *Facebook Audience Network v5.10.0 (6 July 2020)*: "We've released a Limited Data Use feature to give businesses more control over how their data is used in our systems and better support them with their California Consumer Privacy Act (CCPA) compliance efforts." The update allows businesses to restrict Facebook's use of customer data to comply with CCPA.

Two features of these negative updates matter for identification. First, the restriction is set by the SDK provider and applies to its entire user base, so an individual firm's exposure does not reflect its own contemporaneous choice—which is precisely the variation we exploit. Second, neither release is driven by ATT: these updates respond to GDPR, CCPA, and platform policy more broadly, and all flagged releases occur before ATT takes effect. The SDK-update evidence is therefore independent of the ATT experiment rather than a restatement of it.

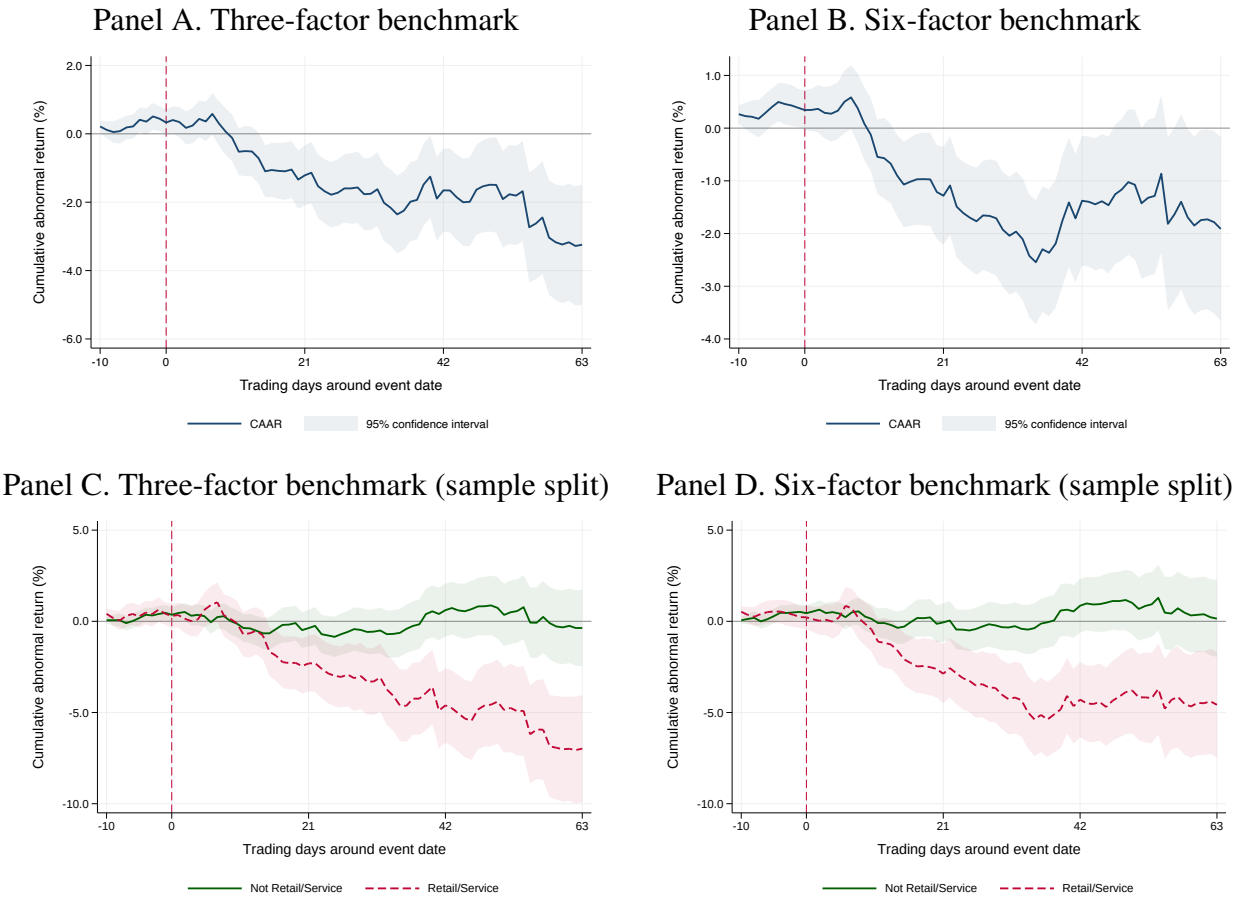
Firm-level evidence on stock returns. We conduct a standard stock-return event study: the event indicator equals one on a date if the firm experiences any of the 18 negative updates in at least one of its apps. Abnormal returns are benchmarked against the Fama–French three- and six-factor models over a $[-21, +63]$-trading-day event window (three months after the event), with factor loadings estimated over the preceding 252 trading days. The sample comprises 608 exposed public firms and 1,534 firm-event observations. Figure C.6 plots the average cumulative abnormal returns under the three- and six-factor benchmarks; the negative drift is consistent in sign across the factor models. Averaged across firm-events, the cumulative abnormal return over the three-month post-event window is -3.2% (three-factor benchmark), and -7.0% in the retail and consumer-services subsample, both statistically significant at 1%. The stock market thus assigns value to the data-sharing capability embedded in firms' apps: when an SDK's capacity to collect and share data is curtailed, the valuations of the firms relying on that SDK decline.

App-level evidence on customer engagement. Our second empirical design is more granular: a stacked difference-in-differences analysis of the monthly iOS app-performance panel, where the unit of observation is the app-month. For each negative update month and public-firm subgroup (a "stack"), treated apps are those exposed to the focal update in that month; control apps belong to the same subgroup, are not treated by the focal update, and have no SDK event of their own inside the window, and they are matched to the pre-period DAU distribution of the treated apps (20 quantile bins). The event window is ± 6 months. We index apps by i, update events (stacks) by τ, and the month relative to the update by t, and estimate the dynamic specification

$$\ln \mathrm{DAU}_{i\tau t} = \alpha + \sum_{\substack{k=-6 \\ k\neq 0}}^{6} \beta_k \left(\textit{Treated}_{i\tau} \times \mathbf{1}[\, t = k \,] \right) + \theta_{i\tau} + \delta_{m(\tau,t)} + \varepsilon_{i\tau t},$$

where $\theta_{i\tau}$ are app-by-event fixed effects, $\delta_{m(\tau,t)}$ are calendar-month fixed effects, and standard errors are clustered by app. Table C.1 reports the average post-update effects, obtained by replacing

FIGURE C.6: **Stock-Market Reaction to Negative Major Updates of Data-Sharing SDKs**



NOTE.—Figure C.6 plots the average cumulative abnormal returns of exposed public firms around negative (data-restricting) major updates of the top 50 data-sharing SDKs, with 95% confidence intervals. A firm is exposed to an update if the updating SDK is installed on at least one of its apps on the update date. The sample comprises 608 exposed firms and 14 negative major-update dates (1,534 firm-event observations). Abnormal returns are computed against the Fama–French three-factor model (Panels A and C) and six-factor model (Panels B and D), with factor loadings estimated over the preceding 12 months. Panels A and B plot the cumulative average abnormal returns across all exposed firms; Panels C and D split the sample into retail and consumer-service firms (SIC codes 5200–5999 and 7000–8999) and all other exposed firms. Firm-level cumulative abnormal returns are winsorized at the 0.5% level in both tails within each event day; the displayed window covers $[-10, +63]$ trading days around the update, with each firm's cumulative return rebased to zero ten trading days before the event.

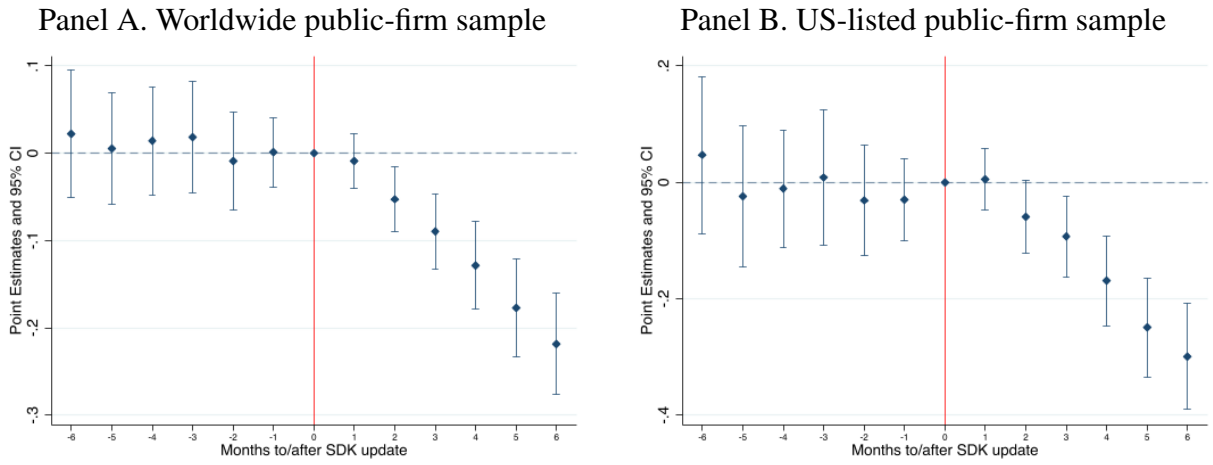
the event-time interactions with a single interaction of $\mathit{Treated}_{i\tau}$ and a post-update indicator and includes app-by-event and relative-event-month fixed effects. For US-listed firms' treated apps, downloads decline by 18% (-0.200 log points, $t = -6.9$) and $\ln$ DAU by 12% (-0.126, $t = -3.5$) relative to matched controls; the worldwide sample gives 14% and 11%, respectively. Figure C.7 shows no pre-trends—all six pre-update coefficients are statistically indistinguishable from zero in both samples—and a steady decline of $\ln$ DAU that begins one to two months after the update and cumulates to roughly 26% below matched controls in the US-listed sample (20% worldwide) by month six. Both the table and the figure restrict the sample to apps whose average pre-update DAU is above the median. We focus on these apps because SDK-mediated data sharing is more eco-

TABLE C.1: **App Performance around Negative Major SDK Updates**

	US public firms		WW public firms	
	(1) log(downloads)	(2) log(DAU)	(3) log(downloads)	(4) log(DAU)
SDK update × treated	-0.200***	-0.126***	-0.156***	-0.112***
	(-6.89)	(-3.54)	(-9.20)	(-5.54)
Event × app FE	Y	Y	Y	Y
Relative-month FE	Y	Y	Y	Y
Observations	66,458	66,458	158,001	158,001
R-sq	0.960	0.971	0.951	0.965

NOTE.—Table C.1 reports stacked difference-in-differences estimates of the effect of a negative major SDK update on treated apps' downloads and DAU, for the above-median-DAU subsample. The coefficient of interest is the interaction of treated-app status with the post-update indicator. The unit of observation is the app-month. Columns 1–2 restrict the sample to apps owned by US-listed public firms; Columns 3–4 use the broader worldwide public-firm sample, with control apps matched on public-firm status and on pre-period DAU. All specifications include app-by-event and relative-event-month fixed effects. Standard errors are clustered by app; t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Figure C.7 presents the event-time dynamics of DAU.

FIGURE C.7: **App-Performance Dynamics around Negative Major SDK Updates**



NOTE.—Figure C.7 is the dynamic counterpart of Table C.1: it reports the stacked difference-in-differences coefficients, β_k, for app performance, measured as log(DAU), around negative major SDK updates, for the above-median-DAU subsample of apps. The vertical line denotes the reference month, $k = 0$. Error bars show 95% confidence intervals. Panel A uses the broader worldwide public-firm sample; Panel B restricts the sample to apps owned by US-listed public firms.

nomically meaningful when there is sufficient user activity for cross-app profiling, advertisement targeting, and attribution (tracking customer's conversion or purchase).[9]

Note that app users are typically unaware that an SDK update has occurred, so the estimates

[9]The estimates are similar in sign and statistical significance, though smaller in magnitude, in the full sample of treated and matched control apps: downloads decline by 12% (-0.131, $t = -4.8$) and $\ln$ DAU by 7% (-0.076, $t = -2.2$) for US-listed firms' apps, and by 11% (-0.115, $t = -6.6$) and 6% (-0.063, $t = -2.8$) in the worldwide sample.

do not reflect users reacting to the updates. Instead, a negative release reduces the efficiency of the firm's data collection, the firm's knowledge of its customers deteriorates, and customer engagement declines. One channel operates through user acquisition. Many of the SDKs are attribution and advertising tools, including AppsFlyer, Adjust, AdMob, MoPub, and Facebook Audience Network. They use shared device identifiers, such as the IDFA, to target install advertisements toward users who resemble a firm's existing high-value users and to identify which advertisements generate each install. When a negative update restricts identifier access—for example, through GDPR consent requirements—targeting becomes less precise and installs become harder to attribute to specific campaigns; the cost of acquiring users rises, campaigns become less effective, and firms scale them back. Fewer new users arrive, so downloads fall and DAU declines. Another channel operates through product design. Analytics SDKs, such as Flurry, Firebase Analytics, and Amplitude, provide firms with information about how users interact with the app—which features improve retention, where users drop out, and what content different user segments respond to—and pooled cross-app data makes these insights more informative. Firms use these insights to improve app experience and product design. When data access is reduced, firms' ability to improve app experience and product design deteriorates.

C.6 Firm disclosures on data sharing and the ATT shock

This appendix details what firms themselves communicated to investors and the public about the importance of SDK-mediated data sharing. Absent a salient event, firms have limited opportunities to discuss the SDKs embedded in their apps, so we organize the evidence around ATT. We accordingly collect firm-level evidence over 2020–2022—the three-year window around the policy's enforcement in April 2021—for the app-owning publicly listed firms in our sample, searching for discussions triggered by ATT as well as statements of reliance on third-party data more broadly.

The evidence comes from three sources. First, we examine SEC 10-K and 10-Q filings, where firms are most likely to discuss developments that they view as financially material. Second, we examine earnings calls, which provide more detailed and timely accounts of the forces affecting firm performance, including management's responses to analysts' questions. These two sources provide the most reliable evidence but may miss relevant cases: firms are not required to discuss every operational issue, and earnings-call coverage depends partly on which questions sell-side analysts ask. Third, we search open-web sources, including firms' privacy documents, news coverage, and industry analysis. These sources are less formal but offer broader coverage and often identify product-level details—such as the specific SDKs embedded in an app. Below we summarize the results and then provide more details on the three data sources.

Each unit of text—a filing section, an earnings-call excerpt, a privacy-policy snapshot, or a web page—is evaluated by a large language model using a structured prompt. The model identifies whether the text contains a concrete business effect related to ATT or the IDFA and assigns a confidence score. It then classifies the primary impact channel using an eleven-category taxonomy spanning data inputs (signal availability, measurement accuracy, targeting precision), financial outcomes (advertising revenue, overall revenue, customer-acquisition costs, return on advertising spend, guidance), and strategic responses (investments in identity infrastructure and changes in market positioning). The model takes a conservative approach: it assigns a direction only when the text explicitly attributes an effect to ATT or the IDFA; forward-looking, preparatory, or otherwise ambiguous statements are coded as uncertain. It also extracts verbatim any quantitative estimate

linked to the change and distinguishes realized effects from projections. Table C.2 summarizes coverage, relevance, and timing across the sources; four results stand out.

First, detection is substantially higher in less formal channels: 9.2% of firms explicitly mention ATT or the IDFA in regulatory filings, compared with 16.1% in earnings calls, 31.6% in firms' own privacy policies, and 59.3% on the open web. Broader dependence on third-party data is disclosed by 24.9% of firms in regulatory filings, far below the corresponding shares of 81.2% in privacy policies and 92.1% on the open web. The formal disclosure rates are thus conservative lower bounds.

Second, firms frequently quantify the impact, providing a direct indication of its economic magnitude. Across the three sources, 373 disclosure units contain an explicit quantitative estimate linked to ATT or the IDFA, of which 274 describe realized effects; examples include Stitch Fix's reported $42.6 million revenue decline in its MD&A, Criteo's estimate of a $60 million privacy-related impact on an earnings call, and press reports that Oracle's advertising business fell from $2 billion to $300 million. The direction of the evidence is nearly unanimous: 99.1% of the relevant filing sections describe negative consequences. Among disclosures that quantify revenue impacts, the median reported effect is a 15.0% decline in revenue among percentage-based estimates and a $30.0 million decline among dollar-based estimates.

Third, the firms reporting these effects extend well beyond the advertising and technology sectors. They include a mortgage lender (Rocket Companies), a major consumer-goods company (Procter & Gamble), retailers (Home Depot, Williams-Sonoma, and Qurate), healthcare firms (SmileDirectClub, Eargo, and Labcorp), fintech firms (PayPal and Remitly), and a commercial real estate REIT (SL Green). Exposure is thus determined less by conventional industry boundaries than by firms' reliance on consumer-facing apps and paid digital customer acquisition. The evidence remains, of course, confined to the app-owning segment of the public-firm universe and does not speak to firms without a meaningful consumer-facing digital presence.

Fourth, discussion increases sharply after ATT across all sources, and its nature changes at that point: before enforcement, firms describe ATT as a prospective risk; afterward, they report realized effects, often with explicit dollar estimates. This timing points to a common shock rather than a gradual rise in privacy concerns. The policy disrupted an infrastructure built around the IDFA and therefore affected many firms simultaneously through the same data-sharing channel.

SEC 10-K and 10-Q filings. We collect all 10-K and 10-Q filings submitted by our sample firms between 2020Q1 and 2022Q4 from EDGAR. A rule-based parser extracts the Risk Factors and Management Discussion & Analysis sections from 11,588 filings, producing 18,353 sections that are each evaluated in full. Because relevant disclosures are often embedded within longer discussions, a single qualifying sentence is sufficient for a section to be classified as relevant. We do not require firms to mention Apple explicitly, since formal filings often describe the underlying economic exposure in broader terms. Instead, a relevant disclosure must jointly identify a dependence on third-party data, a disruption to that access, and a resulting business consequence. General compliance costs associated with GDPR or CCPA are excluded. Explicit references to Apple, iOS, ATT, or the IDFA are distinguished from implicit disclosures using the model's confidence score.

Of the 955 relevant filing sections, 560 appear in Risk Factors and 395 in MD&A. Table C.3 presents illustrative examples, with realized dollar impacts from MD&A in Panel A and the breadth of affected industries in Panel B. The formal disclosure rates likely understate the true extent of exposure. Firms generally name another company's policy in a 10-K or 10-Q only when it is espe-

TABLE C.2: **Disclosure Evidence: Coverage, Relevance, and Timing across Sources**

	Units		Firms covered		Timing	
Source	**Total**	**Relevant**	**Counts**	**Share**	Pre	Post
SEC 10-K/10-Q: third-party-data lens	18,353	955	233	24.9%	344	611
SEC 10-K/10-Q: explicit ATT/IDFA	18,353	325	86	9.2%	103	222
Earnings calls: ATT/IDFA	15,758	1,144	161	16.1%	250	894
Privacy policies: ATT/IDFA	2,608	577	298	31.6%	170	407
Privacy policies: broad data sharing	2,608	1,922	765	81.2%	631	1,291
Open web: ATT/IDFA	11,977	1,632	611	59.3%	200	1,072
Open web: broad data sharing	11,977	5,378	950	92.1%	615	3,365

NOTE.—Units are filing sections (SEC, 2020Q1–2022Q4), transcript snippets (earnings calls, 2020–2022), firm-year privacy-policy snapshots, and web pages. Firm rates are relative to the firms each source covers: 936 (SEC), 997 (calls), 1,031 (open web), and 942 (privacy policies). "Post" is 2021Q2 and later; for privacy policies, timing uses the snapshot year (2020 pre; 2021–2022 post). Web pages without a reliable date are excluded from the timing split. Units with an explicit quantitative estimate tied to the shock: 76 filing sections (63 realized, from 31 firms), 107 call snippets (72 realized, 45 firms), and 190 web pages (139 realized, 120 firms).

cially important to their consolidated results; more often, they describe the underlying economic risk in general terms—dependence on mobile identifiers, restrictions on attribution, or changes in platform policies—without naming Apple, ATT, or the SDKs through which the data is transmitted. Indeed, the term "SDK" appears in only 11 of the 955 relevant sections, even though many disclosures describe mechanisms that operate through SDK-mediated data sharing. The classification procedure is also deliberately conservative, requiring evidence of data dependence, a disruption to that access, and a resulting business impact.

Earnings conference calls. Because the universe of earnings-call transcripts over 2020–2022 is vast, we first identify potentially relevant passages using keywords related to ATT, the IDFA, iOS privacy, tracking, attribution, and opt-in rates. This screening yields at least one flagged passage for most of our sample firms. The language model then evaluates each passage to determine whether it concerns ATT-related changes and whether it contains quantitative evidence of their business effects. Unlike formal filings, earnings calls allow managers and analysts to discuss specific causes directly. We therefore apply a narrower relevance criterion: a passage must explicitly refer to Apple, iOS, ATT, or the IDFA, arise in a mobile-app context, and describe a business consequence. We exclude recurring false positives, including discussions of Safari cookies, web-cookie deprecation, and generic references to the App Store. The post-enforcement discussion peaks in 2021Q4, the first full quarter with realized effects in reported results. Table C.4 presents examples.

Other public sources. The open-web pipeline runs in two stages so that the full-text reading is applied only to pages worth reading. Stage 1 queries a web-search API for each firm, combining the company name with the ATT keywords and broader data-sharing terms, and a lightweight model scores every returned link on relevance and credibility. Stage 2 has a stronger model read the full content of each selected page and apply the shared classifier, returning two relevance scores—one

TABLE C.3: **SEC Filing Disclosures (Selected)**

Firm	Industry	Filing	Disclosure
Panel A. Quantified impacts (verbatim extracts)			
Poshmark	Electronic shopping and mail-order houses	10-Q, 2021Q3	Marketing expense rose to "46% of revenue from 28%" partly due to Apple's privacy changes (realized).
IAC	Other information services	10-K, 2021Q4	"$55.5 million to $77.7 million" projected reduction in search monetization from platform policy changes.
Stitch Fix	Clothing stores	10-Q, 2022Q1	"$42.6 million or 8.0% decrease" in revenue, tied to iOS-driven marketing disruption (realized).
SciPlay	Software publishers	10-Q, 2022Q2	User-acquisition spend rose "$11.8 million"; firm cites iOS user-acquisition uncertainty as a margin risk (realized).
Unity Software	Software publishers	10-Q, 2022Q2	Operate Solutions "revenue declined to $158.5 million from $182.9 million," as ATT opt-in limited personalized-advertising monetization (realized).
Quotient Technology	Advertising and related services	10-Q, 2022Q3	"$65.5 million, or 48% decrease" in revenue as partner data access tightened (realized).
Panel B. Industry breadth			
Home Depot	Building material and supplies dealers	10-K, 2020Q4	Third-party data for personalized customer experience limited by privacy rules and platform changes.
Rocket Companies	Activities related to credit intermediation	10-K, 2020Q4	Third-party data signals for digital marketing disrupted by Apple and Google mobile-OS policy changes.
Williams-Sonoma	Home furnishings stores	10-K, 2020Q4	Depends on third-party behavioral data for targeted, personalized marketing; disruptions to that access would impair its marketing effectiveness.
Blue Apron	Direct selling establishments	10-Q, 2021Q3	Elimination of tracking impairs understanding of customer behavior and marketing execution.
Revolve Group	Electronic shopping and mail-order houses	10-K, 2021Q4	Apple's ATT framework disrupted third-party marketing data, "resulting in higher costs to acquire and retain customers."
Match Group	Other information services	10-K, 2022Q4	Apple and Google restrictions on advertising identifiers, including Apple's IDFA, impair marketing measurement and targeting.

NOTE.—Each row is one classified filing section (the highest-confidence section per firm; relevance confidence between 0.85 and 1.0). Panel A quotes the classifier's verbatim extraction; Panel B paraphrases its structured summary of the filing text. Industry is defined at the four-digit NAICS level.

for the narrow ATT/IDFA lens and one for broad third-party-data dependence—along with a source genre and a content date. After cleaning, the pipeline classifies 11,977 pages covering 1,031 firms.

As a systematic record of firms' own data practices, we also collect archived privacy policies for 2020–2022, following Ramadorai, Uettwiller, and Walther (2025). For each firm, we identify the URL of its official privacy-policy page and retrieve historical versions from the Internet

TABLE C.4: **Quantified ATT/IDFA Impacts Stated on Earnings Calls (Selected)**

Firm	Industry	Period	Stated magnitude (verbatim extract)
Unity Software	Software publishers	2021Q1	Guidance of "approximately $30 million to revenues or just over 3%" headwind for the year.
SmileDirectClub	Medical equipment and supplies manufacturing	2021Q3	Cost-per-lead "up over 100% on a quarter-over-quarter basis," attributed to Apple's iOS update (realized).
Snap	Other information services	2021Q4	"Missed the lower end of our guidance by $3 million" due to changes to iOS ad tracking (realized).
1stdibs.com	Electronic shopping and mail-order houses	2022Q1	"Approximately $3 million to $6 million drag to GMV in the quarter" (realized).
AppLovin	Software publishers	2022Q1	Before iOS 14.5 "75% of users had IDFA"; "we now live in a world where about 65% don't share, 35% do" (realized).
Gaia	Motion picture and video industries	2022Q1	Member-acquisition cost of "$8.2 million or 39% of revenues" amid iOS privacy headwinds (realized).
Meta	Other information services	2022Q1	Estimated that "the impact of iOS overall as a headwind on our business in 2022 is on the order of $10 billion."
Poshmark	Electronic shopping and mail-order houses	2022Q1	Apple channels represent "about 1/3 of our advertising spend and over 70% of our GMV," concentrating privacy exposure (realized).
QVC Group (Zulily)	Electronic shopping and mail-order houses	2022Q1	iOS privacy changes among factors driving "a 37% decline in traffic in Q4" (realized).

NOTE.—Each row is one classified snippet with an explicit quantitative estimate; quotes are verbatim extractions from the transcript. Figures labeled "realized" describe effects already reflected in reported results; the remainder are forward guidance. Industry is defined at the four-digit NAICS level.

Archive's Wayback Machine, querying one archived snapshot per firm per calendar year so that the text reflects the policy as it stood during 2020, 2021, and 2022, respectively. This procedure yields at least one snapshot for 942 firms with a recorded web domain, and 2,608 firm-year snapshots enter the classification (853 in 2020, 884 in 2021, and 871 in 2022). The extracted text is passed to the same two-lens classifier used for the open-web pages.

The disclosure rates in Table C.2, which use a conservative relevance threshold of 0.3, are robust to stricter thresholds: at a score of 0.7 or above, the open-web shares are 57.5% (broad third-party-data involvement) and 18.7% (ATT-specific), and the privacy-policy shares are 63.2% and 10.7%, respectively. Firms' own privacy disclosures account for the largest share of ATT-relevant pages (39.7%), followed by news articles (27.4%).

This approach adds two types of evidence that rarely appear in formal investor disclosures (Table C.5). First, firms' product-level documents identify the specific data infrastructure on which their apps rely: Coca-Cola's App Store privacy disclosure states that its app "includes an AppsFlyer SDK integration," and Labcorp's app privacy policy explicitly lists collection of the IDFA. Such details are unlikely to appear in a 10-K, but they provide direct evidence from the firms themselves about their participation in SDK-mediated data sharing. Second, press coverage records econom-

TABLE C.5: **Open-Web Examples (Selected)**

Firm	Industry	Source type	Finding
Coca-Cola	Beverage manufacturing	Own privacy disclosure	App Store privacy label: "This version of the App includes an AppsFlyer SDK integration."
Labcorp	Medical and diagnostic laboratories	Own privacy disclosure	App privacy policy discloses collection of "advertising IDs associated with your device (Apple's Identifier for Advertising (IDFA))."
T-Mobile	Wired and wireless telecommunications carriers	Own privacy / product pages	Documents that the IDFA is collected and shared as a mobile advertising ID; the firm's ad product excludes iOS data post-ATT.
Nielsen	Other professional, scientific, and technical services	News / legal analysis	ATT-related disruption to cross-app measurement associated with a stock drop of "more than 25 percent."
Oracle	Software publishers	News coverage	Advertising revenue "declined to about $300 million in fiscal year 2024, down from $2 billion in 2022."
Poshmark	Electronic shopping and mail-order houses	News coverage	"Apple's privacy changes and IDFA made mobile advertising less effective. . . marketing spend was nearly 45% of revenue."

NOTE.—Each row is one classified firm–page pair from the second stage of the open-web pipeline. Industry is defined at the four-digit NAICS level.

ically meaningful outcomes that firms may not disclose through formal channels: Nielsen's stock price fell by more than 25% as ATT weakened its ability to measure activity across apps. Agreement across sources further strengthens the evidence; Poshmark's estimate of higher marketing costs, for instance, appears at a similar magnitude in its 10-Q, its earnings call, and press coverage.

D Performance Comovement: Robustness Checks

D.1 Data connectedness measurement and regression specifications

We demonstrate that the results on performance comovement are robust to alternative measures of data connectedness and regression specifications. For each robustness test, we plot in Figure D.1 the estimated coefficients of data connectedness (left panel) and the coefficients on the interaction between data connectedness and ATT indicator variable (right panel) from Equation (2), focusing on correlations based on three metrics of firms' performances—app downloads, sales-to-assets, and CAPM-adjusted returns—shown from top to bottom panels.

Alternative measures of data connectedness. We consider seven variants of the data connectedness measure. First, we compute data connectedness using only the top 20 data-sharing SDKs, ranked by cumulative installations. In the main text, we consider the top 50 data-sharing SDKs. Second, motivated by the concern that the DAU-weighted measure may partly capture firms' reliance on app traffic rather than data connectivity alone, we construct an extensive-margin variant: data connectedness computed from binary SDK-presence vectors—an SDK either is or is not installed in the firm's apps—without DAU weighting. Third, we incorporate the popularity of SDKs

into the construction of the data connectedness measure by extending S_{jt} to $\widetilde{S_{jt}} = \Omega_t \times S_{jt}$, where Ω_t is a diagonal matrix whose diagonal element ω_{it} represents the market share of SDK i:

$$\Omega_t = \begin{pmatrix} \omega_{1t} & 0 & \cdots & 0 \\ 0 & \omega_{2t} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \omega_{Kt} \end{pmatrix}.$$

The resulting *popularity-weighted* ("PW") data connectedness is defined as

$$\rho_{ijt}^{\text{data, PW}} = \frac{\widetilde{S_{it}}' \widetilde{S_{jt}}}{|\widetilde{S_{it}}| \cdot |\widetilde{S_{jt}}|}.$$

Fourth, we incorporate heterogeneity in the data collection intensity of each app by modifying the elements of the SDK usage vector S_{ikt}. Specifically, we redefine $s_{iakt} = m_{iat} \times d_{iakt}$ (in the main text) as $s_{iakt} = m_{iat} \times d_{iakt} \times n_{iat}$, where n_{iat} is the number of unique data types collected by app a based on its privacy labels (following the definition in Bian et al. (2021)).[10]

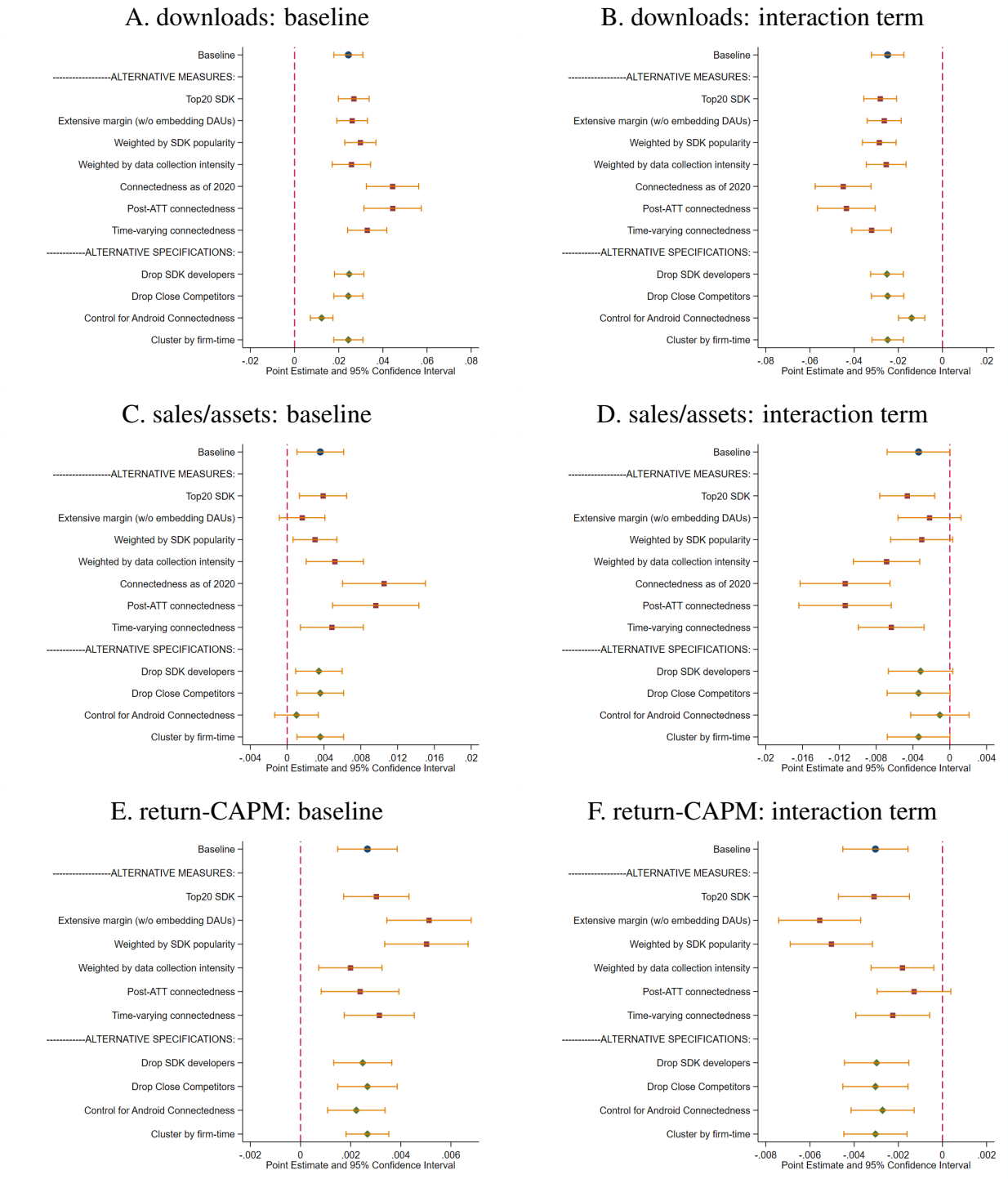
The next three variants address the concern that our results depend on how to time-average ρ_{ijt}^{data}. In the main text, we consider the pre-ATT average values of ρ_{ijt}^{data}. If the data network is relatively stable, we would expect the results to hold even when averaging data connectedness in alternative time periods rather than in the pre-ATT period. For the fifth variant of data connectedness measure, we use the average quarterly data connectedness in 2020 (the year before ATT was introduced). In the sixth variant, we use the average in the post-ATT period. Lastly, we compute averages for the pre-ATT and post-ATT periods separately so the left-side outcome variable is matched in time subscript ($\in \{$*pre-ATT*, *post-ATT*$\}$) with the right-side data connectedness.

Alternative specifications. We consider four alternative regression specifications: (1) excluding eight firms that are data-sharing SDK providers—AppLovin, Twitter, Google, Meta, Yandex, Adobe, Unity, and Amazon; (2) dropping close competitors, defined as pairwise connection in product space (Hoberg–Phillips horizontal product-space overlap) above the sample median; (3) controlling for an analogous Android-based connectedness measure and its interaction with the ATT indicator; and (4) clustering standard errors at both the firm-i-by-time and firm-j-by-time levels. Note that the second specification is motivated by the potential concern that data-sharing relationships between direct competitors may be confounded by strategic considerations or contractual limitations. The third specification reflects that ATT is an iOS-specific policy: because firms often deploy similar SDKs across platforms—the iOS- and Android-based connectedness measures have a correlation of 0.30—Android connectedness is not an untreated placebo network, and controlling for it isolates the iOS network exposure most directly affected by ATT.

Robustness analysis results. In summary, we consider a total of 11 variant regressions, including 7 data connectedness measures in the baseline regression specification and 4 regression specifications using the baseline data connectedness measure, so, with 3 outcome variables representing firms' app, financial, and stock-market performances, respectively, we have a total of 33 sets of

[10] An example of Facebook's privacy label is available at `https://apps.apple.com/us/app/facebook/id284882215`.

FIGURE D.1: **Robustness Checks**



NOTE.—Figure D.1 presents the estimated baseline coefficients (left panel) and the coefficients on the interaction term (right panel) from Equation (2) across various alternative measures and specifications. From top to bottom, the correlation is based on downloads, sales-to-assets, and CAPM-adjusted returns, respectively.

TABLE D.1: **Comovement in App Performance: Controlling for Social-SDK Connectedness**

	downloads		DAU	
	(1)	(2)	(3)	(4)
data connectedness	0.026***	0.024***	0.026***	0.024***
	(7.49)	(7.22)	(6.67)	(6.36)
ATT × data connectedness	-0.025***	-0.025***	-0.024***	-0.023***
	(-6.57)	(-6.58)	(-5.73)	(-5.66)
social SDK connectedness	0.001	-0.000	0.004***	0.003**
	(0.56)	(-0.16)	(2.95)	(2.19)
ATT × social SDK connectedness	-0.001	-0.001	-0.003*	-0.003
	(-0.67)	(-0.54)	(-1.81)	(-1.54)
Other linkages	N	Y	N	Y
ATT × other linkages	N	Y	N	Y
Firm i#ATT FE	Y	Y	Y	Y
Firm j#ATT FE	Y	Y	Y	Y
Observations	1,401,082	1,401,082	1,399,426	1,399,426
R-sq	0.066	0.067	0.113	0.114

NOTE.—Table D.1 adds social-SDK connectedness—constructed like data connectedness in Equation (1) but from firms' overlap in the popular social-media SDKs—and its ATT interaction to the app-performance comovement specification of Table 2. Each observation represents a firm pair at a point in time. The outcome is the pairwise correlation in quarterly log(downloads) (Columns 1–2) and log(DAU) (Columns 3–4), calculated separately for the periods before and after the introduction of ATT (2021Q2). In even-numbered columns, we control for the full set of pairwise firm connections and their interactions with the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects and double-clustered standard errors by firm-i and firm-j. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

robustness results. The estimates of coefficient of data connectedness (interaction term with ATT) are almost always positive (negative) and statistically significant at the 5% level. These findings suggest that our results are not driven by any factors above that motivate our robustness analysis.

D.2 Social-media SDK connectedness

Although our baseline measure is constructed only from data-sharing SDKs—so that social-media SDKs, such as social-media login and social-media sharing kits, do not enter it—several major data-sharing SDKs are operated by social-media platforms. Firm pairs that are jointly integrated with these platforms may then be exposed to common shocks to the platforms' ecosystems, including ATT itself, that do not operate through inter-firm data flows. To test this hypothesis, we construct a *social-SDK connectedness* measure by applying the cosine-similarity formula in Equation (1) to firms' overlap in the most popular social-media SDKs—components supplied by social-media platforms for social login, sharing, and engagement. We re-estimate Equation (2) adding social-SDK connectedness and its ATT interaction as controls alongside our data connectedness measure and its ATT interaction (Tables D.1, D.2, and D.3). As in the baseline analysis, even-numbered columns control for the full set of other firm linkages and their ATT interactions.

For customer engagement and financial performance, the estimates on data connectedness are essentially unchanged when the social measure is added: the coefficient for the download comovement remains at our baseline estimate of 0.024 (similarly for DAU), and the coefficients

TABLE D.2: **Comovement in Financial Performance: Controlling for Social-SDK Connectedness**

	earnings growth		sales/assets	
	(1)	(2)	(3)	(4)
data connectedness	0.002***	0.002**	0.005***	0.004***
	(2.96)	(2.49)	(3.76)	(2.87)
ATT × data connectedness	-0.003***	-0.003***	-0.004**	-0.004**
	(-3.31)	(-3.05)	(-2.02)	(-2.02)
social SDK connectedness	0.000	-0.000	-0.000	-0.001**
	(0.35)	(-0.14)	(-0.56)	(-2.28)
ATT × social SDK connectedness	-0.002***	-0.001***	0.001*	0.001*
	(-3.40)	(-3.00)	(1.85)	(1.90)
Other linkages	N	Y	N	Y
ATT × other linkages	N	Y	N	Y
Firm i#ATT FE	Y	Y	Y	Y
Firm j#ATT FE	Y	Y	Y	Y
Observations	1,379,592	1,379,592	1,231,716	1,231,716
R-sq	0.005	0.005	0.184	0.186

NOTE.—Table D.2 adds social-SDK connectedness—constructed like data connectedness in Equation (1) but from firms' overlap in the popular social-media SDKs—and its ATT interaction to the financial-performance comovement specification of Table 3. Each observation represents a firm pair at a point in time. The outcome is the pairwise correlation in quarterly earnings growth (Columns 1–2) and asset turnover (Columns 3–4), calculated separately for the periods before and after the introduction of ATT (2021Q2). In even-numbered columns, we control for the full set of pairwise firm connections and their interactions with the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects and double-clustered standard errors by firm-i and firm-j. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

for earnings growth and asset turnover are equally stable, as are their ATT interactions. In addition, social-SDK connectedness itself is weak on these outcomes. For return comovement, social-SDK connectedness enters with a magnitude comparable to that of data connectedness (0.004 versus 0.003 for raw returns and 0.006 versus 0.005 for CAPM-adjusted returns), and both measures decline after ATT. The two measures, however, pick up close to orthogonal variation: when they enter jointly, the data-connectedness coefficient is essentially unchanged relative to the baseline, and both terms remain individually significant.

TABLE D.3: **Comovement in Stock Returns: Controlling for Social-SDK Connectedness**

	raw return		return-CAPM		return-DGTW	
	(1)	(2)	(3)	(4)	(5)	(6)
data connectedness	0.003***	0.001***	0.005***	0.002***	0.003***	0.002***
	(6.21)	(3.36)	(6.43)	(4.08)	(5.83)	(3.33)
ATT × data connectedness	-0.002***	-0.002***	-0.003***	-0.003***	-0.002**	-0.002**
	(-2.79)	(-2.94)	(-3.63)	(-3.82)	(-2.35)	(-2.54)
social SDK connectedness	0.004***	0.003***	0.006***	0.004***	0.005***	0.004***
	(7.17)	(4.99)	(7.28)	(5.22)	(7.48)	(5.90)
ATT × social SDK connectedness	-0.002***	-0.002***	-0.003***	-0.003***	-0.002***	-0.002***
	(-3.14)	(-3.03)	(-3.78)	(-3.59)	(-2.72)	(-2.62)
Other linkages	N	Y	N	Y	N	Y
ATT × other linkages	N	Y	N	Y	N	Y
Firm i#Time FE	Y	Y	Y	Y	Y	Y
Firm j#Time FE	Y	Y	Y	Y	Y	Y
Observations	5,742,144	5,742,144	5,639,376	5,639,376	4,988,256	4,988,256
R-sq	0.442	0.449	0.097	0.110	0.022	0.028

NOTE.—Table D.3 adds social-SDK connectedness—constructed like data connectedness in Equation (1) but from firms' overlap in the most popular social-media SDKs—and its ATT interaction to the return-comovement specification of Table 4. Each observation represents a firm pair at a specific point in time. Return comovement is calculated as the correlation between firms' monthly returns over non-overlapping 12-month windows, defined relative to the introduction of ATT in April 2021, for raw returns, CAPM abnormal returns, and DGTW-adjusted returns. In even-numbered columns, we control for the full set of pairwise firm connections and their interactions with the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects and double-clustered standard errors by firm-i and firm-j. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

D.3 Nonlinear effects and network topology

This appendix presents the results of nonlinear spillover effects of the data-sharing network in Table D.4 and details the construction of higher-order connectedness and local-density measures discussed in Section 3.1 and the corresponding estimation results. First, Table D.4 adds a squared data-connectedness term (and its ATT interaction) to the baseline specification for the nonlinear effects; the remaining tables involve the following measures about network topology.

Higher-order connectedness. In our model (see Section 4), the network adjacency matrix, Γ, summarizes the strength of bilateral connections, with the ij-th element capturing the pairwise data connectedness in our empirical setting. To measure indirect connections, we transform the pairwise data-connectedness matrix into a row-normalized network operator (i.e., a Markov transition matrix): we set the diagonal of $\{\rho_{ij}^{data}\}$ to zero and divide each row by its row sum, obtaining a matrix G with $G_{ij} = \rho_{ij}^{data} / \sum_k \rho_{ik}^{data}$, an empirical analogue of Γ. Setting the diagonal to zero removes self-links and ensures that the higher-order measures are not mechanically driven by own-node terms: the off-diagonal element $(G^2)_{ij} = \sum_{k \neq i,j} G_{ik} G_{kj}$ sums the length-two paths from firm i to firm j through third-party intermediaries, and higher powers G^m sum length-m paths—the standard representation of higher-order interconnections in the networks literature (Acemoglu, Carvalho, Ozdaglar, and Tahbaz-Salehi, 2012). Since G is asymmetric, each pair has two directed entries; the pair-level regressor averages $(G^m)_{ij}$ and $(G^m)_{ji}$ and is standardized. Because the regressions include the first-order term, the coefficient on the higher-order term identifies whether indirectly connected firms comove over and above their direct connectedness. Table D.5 uses

TABLE D.4: **Comovement in App Performance: Nonlinearity in Data Connectedness**

	downloads		DAU	
	(1)	(2)	(3)	(4)
data connectedness	0.044***	0.042***	0.055***	0.053***
	(7.45)	(7.18)	(7.92)	(7.64)
ATT × data connectedness	-0.043***	-0.042***	-0.053***	-0.052***
	(-6.71)	(-6.65)	(-7.10)	(-7.00)
data connectedness^2	-0.011***	-0.011***	-0.018***	-0.017***
	(-5.37)	(-5.21)	(-7.18)	(-7.03)
ATT × data connectedness^2	0.011***	0.010***	0.017***	0.017***
	(4.84)	(4.72)	(6.69)	(6.57)
Other linkages	N	Y	N	Y
ATT × other linkages	N	Y	N	Y
Firm i#ATT FE	Y	Y	Y	Y
Firm j#ATT FE	Y	Y	Y	Y
Observations	1,401,082	1,401,082	1,399,426	1,399,426
R-sq	0.067	0.068	0.113	0.114

NOTE.—Table D.4 adds a squared data-connectedness term and its ATT interaction to the app-performance comovement specification of Table 2. Each observation represents a firm pair at a specific point in time. The outcome is the pairwise correlation in quarterly log(downloads) (Columns 1–2) and log(DAU) (Columns 3–4), calculated separately for the periods before and after the introduction of ATT (2021Q2). Odd-numbered columns include only the (standardized) data-connectedness terms; even-numbered columns add the full set of pairwise firm connections and their interactions with the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects and double-clustered standard errors by firm-i and firm-j. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

the second-order term; Table D.6 replaces it with the average of the second- through fifth-order components—a truncated Neumann series that approximates the network-propagation operator of our model, whose higher powers are geometrically discounted (see the decomposition in (16)).

Local density. We measure how densely each firm is embedded in the data network with two node-level statistics computed on the fixed pre-ATT network. The local clustering coefficient is the share of a firm's neighbors that are themselves directly connected, computed on the network that keeps edges with data connectedness above 0.3, the 75th percentile of the distribution. The pair-level regressor in Table D.7 is the product of the two firms' clustering coefficients, standardized, and it enters in a full triple interaction with data connectedness and the ATT indicator. Note that with firm-by-time fixed effects, each firm's own clustering coefficient—and any additively separable combination of the two firms' coefficients, such as their sum or average—is already absorbed, so the specification controls for own-firm clustering by construction; the product is the natural non-separable, pair-level combination, capturing both firms being simultaneously embedded in a dense local neighborhood. The k-core number of a firm is the largest k for which the firm belongs to the network's k-core, that is, the maximal subgraph in which every node has at least k neighbors. The density regressor in Table D.8 is the standardized k-core number of firm i in each pair record, which enters interacted with data connectedness (and with the ATT indicator); its main effect is absorbed by the firm-by-time fixed effects.

TABLE D.5: **Comovement in App Performance: Higher-Order Data Connectedness**

	downloads		DAU	
	(1)	(2)	(3)	(4)
data connectedness	0.023***	0.022***	0.014***	0.013***
	(6.74)	(6.50)	(3.91)	(3.61)
ATT × data connectedness	-0.023***	-0.023***	-0.012***	-0.012***
	(-6.09)	(-6.09)	(-3.09)	(-3.03)
data connectedness (2nd order)	0.016***	0.015***	0.064***	0.063***
	(4.07)	(3.76)	(8.23)	(8.03)
ATT × data connectedness (2nd order)	-0.011**	-0.011**	-0.064***	-0.064***
	(-2.23)	(-2.23)	(-8.20)	(-8.18)
Other linkages	N	Y	N	Y
ATT × other linkages	N	Y	N	Y
Firm i#ATT FE	Y	Y	Y	Y
Firm j#ATT FE	Y	Y	Y	Y
Observations	1,401,082	1,401,082	1,399,426	1,399,426
R-sq	0.067	0.068	0.115	0.116

NOTE.—Table D.5 adds a second-order data-connectedness term—the direction-averaged, normalized linkage between indirectly connected firms constructed from the square of the row-normalized data-connectedness matrix, as described in this appendix—to the app-performance comovement specification of Table 2. Each observation represents a firm pair at a specific point in time. The outcome is the pairwise correlation in quarterly log(downloads) (Columns 1–2) and log(DAU) (Columns 3–4), calculated separately for the periods before and after the introduction of ATT (2021Q2). Odd-numbered columns include only the (standardized) first- and second-order data-connectedness terms; even-numbered columns add the full set of pairwise firm connections and their interactions with the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects and double-clustered standard errors by firm-i and firm-j. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

TABLE D.6: **Comovement in App Performance: Higher-Order Data Connectedness Averaging the Second through Fifth Orders**

	downloads		DAU	
	(1)	(2)	(3)	(4)
data connectedness	0.023***	0.022***	0.016***	0.015***
	(6.98)	(6.73)	(4.76)	(4.43)
ATT × data connectedness	-0.023***	-0.023***	-0.015***	-0.014***
	(-6.22)	(-6.24)	(-3.83)	(-3.78)
data connectedness (2nd-5th order avg)	0.017***	0.015***	0.066***	0.064***
	(4.09)	(3.79)	(8.19)	(8.00)
ATT × data connectedness (2nd-5th order avg)	-0.011**	-0.011**	-0.066***	-0.065***
	(-2.22)	(-2.21)	(-8.20)	(-8.17)
Other linkages	N	Y	N	Y
ATT × other linkages	N	Y	N	Y
Firm i#ATT FE	Y	Y	Y	Y
Firm j#ATT FE	Y	Y	Y	Y
Observations	1,401,082	1,401,082	1,399,426	1,399,426
R-sq	0.067	0.068	0.115	0.116

NOTE.—Table D.6 parallels Table D.5, replacing the second-order term with the average of the second- through fifth-order data-connectedness components (direction-averaged and normalized so that higher-order magnitudes remain bounded). Together with the first-order term, this averaged higher-order term is a truncated empirical approximation of the network-propagation operator in our model, whose higher powers are geometrically discounted (see the decomposition in (16)). Each observation represents a firm pair at a specific point in time. The outcome is the pairwise correlation in quarterly log(downloads) (Columns 1–2) and log(DAU) (Columns 3–4), calculated separately for the periods before and after the introduction of ATT (2021Q2). Odd-numbered columns include only the (standardized) first-order and averaged higher-order data-connectedness terms; even-numbered columns add the full set of pairwise firm connections and their interactions with the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects and double-clustered standard errors by firm-i and firm-j. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

TABLE D.7: **Comovement in App Performance: Local Clustering**

	downloads		DAU	
	(1)	(2)	(3)	(4)
data connectedness	0.016***	0.014***	0.015***	0.013**
	(3.58)	(3.26)	(2.88)	(2.54)
ATT × data connectedness	-0.012**	-0.012**	-0.011**	-0.011*
	(-2.37)	(-2.30)	(-1.99)	(-1.85)
data connectedness × local clustering	0.018***	0.019***	0.016**	0.017**
	(2.71)	(2.83)	(2.22)	(2.35)
ATT × data connectedness × local clustering	-0.024***	-0.025***	-0.019**	-0.020***
	(-3.15)	(-3.27)	(-2.43)	(-2.58)
local clustering	0.036***	0.035***	0.065***	0.064***
	(4.20)	(4.15)	(5.62)	(5.56)
ATT × local clustering	-0.045***	-0.045***	-0.073***	-0.073***
	(-4.52)	(-4.51)	(-5.74)	(-5.73)
Other linkages	N	Y	N	Y
ATT × other linkages	N	Y	N	Y
Firm i#ATT FE	Y	Y	Y	Y
Firm j#ATT FE	Y	Y	Y	Y
Observations	1,401,082	1,401,082	1,399,426	1,399,426
R-sq	0.067	0.068	0.113	0.115

NOTE.—Table D.7 estimates the triple interaction of the ATT indicator, data connectedness, and local clustering in the app-performance comovement specification of Table 2. Each firm's local clustering coefficient is computed on the pre-ATT data network as described in this appendix; the pair-level measure is the product of firm i's and firm j's coefficients, standardized. Each observation represents a firm pair at a specific point in time. The outcome is the pairwise correlation in quarterly log(downloads) (Columns 1–2) and log(DAU) (Columns 3–4), calculated separately for the periods before and after the introduction of ATT (2021Q2). Odd-numbered columns include only the (standardized) connectedness and clustering terms; even-numbered columns add the full set of pairwise firm connections and their interactions with the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects and double-clustered standard errors by firm-i and firm-j. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

TABLE D.8: **Comovement in App Performance: Local Network Density**

	downloads		DAU	
	(1)	(2)	(3)	(4)
data connectedness	0.010***	0.009***	0.004	0.003
	(2.95)	(2.73)	(1.12)	(0.86)
ATT × data connectedness	-0.010**	-0.010**	-0.000	0.000
	(-2.43)	(-2.39)	(-0.04)	(0.04)
data connectedness × local network density	0.035***	0.034***	0.048***	0.047***
	(4.61)	(4.50)	(5.26)	(5.14)
ATT × data connectedness × local network density	-0.033***	-0.033***	-0.052***	-0.052***
	(-3.78)	(-3.78)	(-4.81)	(-4.78)
Other linkages	N	Y	N	Y
ATT × other linkages	N	Y	N	Y
Firm i#ATT FE	Y	Y	Y	Y
Firm j#ATT FE	Y	Y	Y	Y
Observations	1,401,082	1,401,082	1,399,426	1,399,426
R-sq	0.067	0.068	0.113	0.114

NOTE.—Table D.8 interacts data connectedness with local network density—the standardized k-core number of firm i in each pair record, computed on the pre-ATT data network as described in this appendix—in the app-performance comovement specification of Table 2. The density main effect is absorbed by the firm-by-time fixed effects. Each observation represents a firm pair at a specific point in time. The outcome is the pairwise correlation in quarterly log(downloads) (Columns 1–2) and log(DAU) (Columns 3–4), calculated separately for the periods before and after the introduction of ATT (2021Q2). Odd-numbered columns include only the (standardized) data-connectedness and density-interaction terms; even-numbered columns add the full set of pairwise firm connections and their interactions with the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-time fixed effects and double-clustered standard errors by firm-i and firm-j. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

E The Advertising Channel

This appendix describes the construction of the advertising variables used in Section 3 and reports the estimation results in Tables E.1 through E.3.

Data sources and variables. Advertising spend is from Nielsen Ad Intel. Nielsen Ad Intel monitors advertising activity across traditional and digital media and reports estimated advertising expenditures. The data combines Nielsen's own and third-party sources under a common classification system; its digital coverage incorporates site- and device-level measurement using both virtual and human panel data. See Nielsen (2020) and Shapiro, Hitsch, and Tuchman (2021a) for details. We aggregate expenditures to the firm-month level, distinguish digital from non-digital advertising, and use the two series in logs. Advertising allocation is from Apptopia's advertising intelligence data, which cover advertisements hosted in mobile apps. For each firm and month, we construct two allocation variables: the share of advertising spend directed to app publishers and the share of advertising impressions delivered in apps.

Measuring advertising effectiveness. Following Shapiro et al. (2021b), we measure a firm's advertising effectiveness as the elasticity of its app performance with respect to its own advertising

TABLE E.1: **Comovement in Advertising Spend**

	ln(digital ad$)		ln(non-digital ad$)	
	(1)	(2)	(3)	(4)
data connectedness	0.006***	0.005***	0.001	0.001
	(3.79)	(3.39)	(1.63)	(1.07)
ATT × data connectedness	-0.005***	-0.004**	-0.000	-0.000
	(-2.80)	(-2.52)	(-0.29)	(-0.24)
Other linkages	N	Y	N	Y
ATT × other linkages	N	Y	N	Y
Firm i#ATT FE	Y	Y	Y	Y
Firm j#ATT FE	Y	Y	Y	Y
Observations	1,373,300	1,373,300	1,095,396	1,095,396
R-sq	0.227	0.227	0.051	0.051

NOTE.—Table E.1 shows the relationship between data connectedness and the comovement of advertising spend. Each observation represents a firm pair at a specific point in time. For each firm pair, comovement is measured as the rolling correlation between firm i's and firm j's monthly log advertising spend—digital (Columns 1–2) and non-digital (Columns 3–4)—from Nielsen Ad Intel, calculated relative to the introduction of ATT in April 2021. In even-numbered columns, we control for the full set of pairwise firm connections, as well as interaction terms between these connections and the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-ATT fixed effects ($\text{Firm}_i \times \text{ATT}$ and $\text{Firm}_j \times \text{ATT}$) and double-clustered standard errors by firm-i and firm-j. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

stock. For firm i in a rolling window w of 26 weeks, we estimate

$$\log Y_{i,t} = \beta_{i,w} \log\left(1 + A_{i,t}\right) + g(t) + \varepsilon_{i,t}, \qquad A_{i,t} = \sum_{\tau \le t} \delta^{\,t-\tau} s_{i,\tau}, \tag{E.1}$$

where $Y_{i,t}$ is firm i's weekly app downloads or MAU, $g(t)$ is a linear time trend, and the advertising stock, $A_{i,t}$, is a geometrically decaying sum of past weekly advertising spend, $s_{i,\tau}$, with a weekly carryover parameter $\delta = 0.80$ and a one-month lag between advertising and the outcome. The estimated elasticity, $\beta_{i,w}$, is firm i's advertising effectiveness in window w; the estimates are similar for carryover parameters between 0.80 and 0.90. We drop firm-windows with fewer than 26 weeks of data and remove noisy estimates—windows with near-collinear regressors, fewer than three residual degrees of freedom, an advertising stock equal to zero in more than 80% of the window, or negligible variation in the estimated elasticities—and winsorize the remaining estimates at the 1st and 99th percentiles and standardize them.

Regression specification. Tables E.1 through E.3 apply the pairwise comovement design of Section 3.1 to the advertising variables. For each firm pair, the outcome is the rolling correlation between firm i's and firm j's advertising variable, computed relative to the introduction of ATT in April 2021, and the explanatory variables are data connectedness and its interaction with the ATT indicator. Even-numbered columns control for the full set of other pairwise firm linkages and their ATT interactions. All specifications include firm-i-by-ATT and firm-j-by-ATT fixed effects, and standard errors are double-clustered by firm-i and firm-j.

TABLE E.2: **Comovement in Advertising Strategy**

	% ad to app publishers		% app ad impressions	
	(1)	(2)	(3)	(4)
data connectedness	0.041***	0.041***	0.033***	0.034***
	(4.49)	(4.40)	(3.23)	(3.25)
ATT × data connectedness	-0.040***	-0.040***	-0.029**	-0.031***
	(-3.92)	(-3.95)	(-2.56)	(-2.65)
Other linkages	N	Y	N	Y
ATT × other linkages	N	Y	N	Y
Firm i#ATT FE	Y	Y	Y	Y
Firm j#ATT FE	Y	Y	Y	Y
Observations	111,340	111,340	111,344	111,344
R-sq	0.071	0.071	0.058	0.058

NOTE.—Table E.2 shows the relationship between data connectedness and the comovement of firms' advertising strategy. Each observation represents a firm pair at a specific point in time. For each firm pair, comovement is measured as the rolling correlation between firm i's and firm j's monthly advertising-allocation variables—the share of advertising spend directed to app publishers (Columns 1–2) and the share of advertising impressions delivered in apps (Columns 3–4)—constructed from Apptopia data and calculated relative to the introduction of ATT in April 2021. In even-numbered columns, we control for the full set of pairwise firm connections, as well as interaction terms between these connections and the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-ATT fixed effects ($\text{Firm}_i \times \text{ATT}$ and $\text{Firm}_j \times \text{ATT}$) and double-clustered standard errors by firm-i and firm-j. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

TABLE E.3: **Comovement in Advertising Effectiveness**

	ad$ → downloads		ad$ → MAU	
	(1)	(2)	(3)	(4)
data connectedness	0.002**	0.001*	0.002**	0.002**
	(2.04)	(1.87)	(2.36)	(2.23)
ATT × data connectedness	-0.002**	-0.002**	-0.003**	-0.003**
	(-1.97)	(-1.99)	(-2.23)	(-2.31)
Other linkages	N	Y	N	Y
ATT × other linkages	N	Y	N	Y
Firm i#ATT FE	Y	Y	Y	Y
Firm j#ATT FE	Y	Y	Y	Y
Observations	649,278	649,278	649,278	649,278
R-sq	0.003	0.003	0.001	0.001

NOTE.—Table E.3 shows the relationship between data connectedness and the comovement of firms' advertising effectiveness. For each firm, advertising effectiveness is the own-advertising-stock elasticity of app performance, estimated in rolling 26-week windows by regressing log app performance—downloads (Columns 1–2) and MAU (Columns 3–4)—on the firm's own log advertising stock and a linear time trend, following Shapiro et al. (2021b); the advertising stock is a geometrically decaying sum of past weekly advertising spend (weekly carryover $\delta = 0.80$, one-month lag). Pairs enter when both firms' effectiveness is estimated on at least 26 weeks of data. Each observation represents a firm pair at a specific point in time; for each pair, comovement is measured as the rolling correlation between firm i's and firm j's estimated effectiveness, calculated relative to the introduction of ATT in April 2021. In even-numbered columns, we control for the full set of pairwise firm connections, as well as interaction terms between these connections and the ATT indicator, which equals one for periods after 2021Q2. We include firm-by-ATT fixed effects (Firm$_i$×ATT and Firm$_j$×ATT) and double-clustered standard errors by firm-i and firm-j. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

F Model Derivation and Proofs

F.1 Proof of Proposition 1

Note that the optimal choice of $x_{i,t}$ is characterized by the HJB equation,

$$\rho V^i\left(\delta_{i,t}, \{\delta_{j,t}\}_{j\neq i}\right) = \max_{x_{i,t}} \zeta\delta_{i,t} + g\left(x_{i,t}, \lambda_{i,t}\right) + V^i_{\delta_{i,t}}\left[\theta(\alpha\lambda_{i,t} + x_{i,t}) + \mu_\delta\delta_{i,t}\right] + \frac{1}{2}V^i_{\delta_{i,t}\delta_{i,t}}\delta^2_{i,t}\sigma^2_{i,\delta} + \sum_{j\neq i}\left[V^i_{\delta_{j,t}}\left[\theta(\alpha\lambda_{j,t} + x_{j,t}) + \mu_\delta\delta_{j,t}\right] + \frac{1}{2}V^i_{\delta_{j,t}\delta_{j,t}}\delta^2_{j,t}\sigma^2_{j,\delta}\right]. \tag{F.2}$$

Taking FOC with respect to $x_{i,t}$, we have

$$-g_x(x_{i,t}, \lambda_{i,t}) = V^i_{\delta_{i,t}}\theta\,. \tag{F.3}$$

F.2 Proof of Proposition 2

Value function conjecture. To solve for firm i's valuation, we conjecture that the firm's value function has the following functional form:

$$V^i(\bar{\delta}_t) = v_{i,0} + v_i^\top\bar{\delta}_t = v_{i,0} + \sum_{j=1}^N v_{i,j}\delta_{j,t}, \tag{F.4}$$

where $\bar{\delta}_t$ is the column vector of all firms' data stock, $\bar{\delta}_t = [\delta_{1,t}, \ldots, \delta_{N,t}]^\top$. Therefore, we obtain the following expressions for firm i's dependence on firm j's data $\forall j = 1, 2, \ldots, N$:

$$V^i_{\delta_{j,t}}(\bar{\delta}_t) = v_{i,j},$$
$$V^i_{\delta_{j,t}\delta_{j,t}}(\bar{\delta}_t) = 0.$$

We substitute these into the HJB equation to obtain:

$$\rho V^i(\bar{\delta}_t)dt = \max_{x_{i,t}}\left(g\left(x_{i,t}, \lambda_{i,t}\right) + \zeta\delta_{i,t}\right)dt + v_{i,i}\left[\theta\left(x_{i,t} + \frac{\alpha\kappa}{\sigma_\eta}D_{i,t}\right) + \mu_\delta\delta_{i,t}\right]dt + \sum_{j\neq i} v_{i,j}\left[\theta\left(x_{j,t} + \frac{\alpha\kappa}{\sigma_\eta}D_{j,t}\right) + \mu_\delta\delta_{j,t}\right]dt. \tag{F.5}$$

FOC (12) becomes

$$\frac{\zeta\phi_1}{(\phi_0 D_{i,t} - \phi_1 x_{i,t})} = \theta v_{i,i}, \tag{F.6}$$

where $\phi_0 = \phi_\lambda\kappa/\sigma_\eta$. If $v_{i,i} > 0$, this ensures that $\phi_0 D_{i,t} - \phi_1 x_{i,t} > 0$ and also gives

$$x_{i,t} = \frac{\phi_0}{\phi_1}D_{i,t} - \frac{\zeta}{\theta v_{i,i}}. \tag{F.7}$$

In what follows, we use $D_{i,t} = \xi\sum_{j=1}^N \gamma_{ij}\delta_{j,t}$, with $\xi = 1$ corresponding to the baseline model in Proposition 2.

Thus,

$$x_{i,t} = \frac{\phi_0}{\phi_1} D_{i,t} - \frac{\zeta}{\theta v_{i,i}} = \frac{\phi_0}{\phi_1} \xi (\sum_{j=1}^{N} \gamma_{ij} \delta_{j,t}) - \frac{\zeta}{\theta v_{i,i}}. \tag{F.8}$$

The cash flow from user activities is given by

$$g_{i,t} = \zeta \log(\frac{\zeta \phi_1}{\theta v_{i,i}}). \tag{F.9}$$

Substitute FOC into HJB

$$\begin{aligned} \rho V_{i,t} &= \zeta \delta_{i,t} + \zeta \log(\frac{\zeta \phi_1}{\theta v_{i,i}}) + \sum_{j=1}^{N} v_{i,j} \left[\theta x_{j,t} + \frac{\alpha \kappa}{\sigma_\eta} \theta D_{j,t} + \mu_\delta \delta_{j,t} \right] \\ &= \zeta \delta_{i,t} + \zeta \log(\frac{\zeta \phi_1}{\theta v_{i,i}}) + \sum_{j=1}^{N} v_{i,j} \left(\theta \frac{\phi_0}{\phi_1} D_{j,t} - \frac{\zeta}{v_{j,j}} + \frac{\alpha \kappa}{\sigma_\eta} \theta D_{j,t} + \mu_\delta \delta_{j,t} \right), \end{aligned} \tag{F.10}$$

where

$$D_{j,t} = \left(\sum_{k=1}^{N} \gamma_{jk} \delta_{k,t} \xi \right). \tag{F.11}$$

After substitution and simplification:

$$\rho V_{i,t} = \left(\zeta \delta_{i,t} + \sum_{j=1}^{N} v_{i,j} \left[\theta (\frac{\phi_0}{\phi_1} + \frac{\alpha \kappa}{\sigma_\eta}) \left(\sum_{k=1}^{N} \gamma_{jk} \delta_{k,t} \xi \right) + \mu_\delta \delta_{j,t} \right] \right) + A_i, \tag{F.12}$$

where

$$A_i = \left(\zeta \log(\frac{\zeta \phi_1}{\theta v_{i,i}}) + \sum_{j=1}^{N} v_{i,j} \left[-\frac{\zeta}{v_{j,j}} \right] \right). \tag{F.13}$$

This gives the constant term in the valuation

$$v_{i,0} = \frac{A_i}{\rho}. \tag{F.14}$$

There are two components in it. The first is the discounted value of all future cash flow from user activities. The second component is the present value of the change in data accumulation resulting from optimal product design.

Next, by comparing coefficients on LHS and RHS of N states $\delta_{j,t}$, we have N equations for N unknown $v_{i,j}$, $\quad \forall j = 1, 2, ..., N$. The valuation vector is

$$v_i = \begin{pmatrix} v_{i,1} \\ v_{i,2} \\ \vdots \\ v_{i,N} \end{pmatrix}$$

Γ is an $N \times N$ matrix of network linkages

$$\Gamma = \begin{pmatrix} \gamma_{11} & \gamma_{12} & \gamma_{13} & \cdots & \gamma_{1N} \\ \gamma_{21} & \gamma_{22} & \gamma_{23} & \cdots & \gamma_{2N} \\ \gamma_{31} & \gamma_{32} & \gamma_{33} & \cdots & \gamma_{3N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \gamma_{N1} & \gamma_{N2} & \gamma_{N3} & \cdots & \gamma_{NN} \end{pmatrix}$$

For (F.12), the coefficients on $\delta_{i,t}$ should be the same on both sides, therefore we have for $v_{i,i}$:

$$\rho v_{i,i} = \zeta + \theta(\frac{\phi_0}{\phi_1} + \frac{\alpha\kappa}{\sigma_\eta}) \left(\sum_{j=1}^{N} \gamma_{ji} v_{i,j} \xi \right) + \mu_\delta v_{i,i},$$

and for the coefficient of $\delta_{j,t}$ for $j \neq i$ we have

$$\rho v_{i,j} = \theta(\frac{\phi_0}{\phi_1} + \frac{\alpha\kappa}{\sigma_\eta}) \left(\sum_{k=1}^{N} \gamma_{kj} v_{i,k} \xi \right) + \mu_\delta v_{i,j}.$$

Define:

$$\beta = \theta(\frac{\phi_0}{\phi_1} + \frac{\alpha\kappa}{\sigma_\eta})\xi.$$

The above equations can be written as

$$(\rho - \mu_\delta) v_{i,i} = \zeta + \beta(\Gamma^\top v_i)_i \tag{F.15}$$

and

$$(\rho - \mu_\delta) v_{i,j} = \beta(\Gamma^\top v_i)_j.$$

Let's define

$$\hat{\rho} = \rho - \mu_\delta$$

Rearranging the above equations in a matrix form, we get:

$$(I\hat{\rho} - \beta\Gamma^\top) v_i = \zeta \mathbf{e}_i.$$

Solving for the valuation vector v_i, we obtain

$$v_i = (I\hat{\rho} - \beta\Gamma^\top)^{-1} \zeta \mathbf{e}_i. \tag{F.16}$$

Therefore, the valuation vector is given by

$$\begin{aligned} v_i &= (I\hat{\rho} - \beta\Gamma^\top)^{-1} \zeta \mathbf{e}_i \\ v_i^\top &= \mathbf{e}_i^\top (I - \frac{\beta}{\hat{\rho}}\Gamma)^{-1} \frac{\zeta}{\hat{\rho}} \\ &= \frac{\zeta}{\hat{\rho}} \mathbf{e}_i^\top \left(I + \frac{\beta}{\hat{\rho}}\Gamma + (\frac{\beta}{\hat{\rho}})^2\Gamma^2 + (\frac{\beta}{\hat{\rho}})^3\Gamma^3 + ... \right). \end{aligned}$$

And all elements of the vector are nonnegative. Therefore, the firm's valuation is

$$V_{i,t} = v_{i,0} + \frac{\zeta}{\hat{\rho}}\mathbf{e}_i^\top \left(I + \frac{\beta}{\hat{\rho}}\Gamma + (\frac{\beta}{\hat{\rho}})^2\Gamma^2 + (\frac{\beta}{\hat{\rho}})^3\Gamma^3 + ...\right)\bar{\delta}_t \tag{F.17}$$

$$= v_{i,0} + \eta\delta_{i,t} + \eta\mathbf{e}_i^\top \frac{\beta}{\hat{\rho}}(I + \frac{\beta}{\hat{\rho}}\Gamma + (\frac{\beta}{\hat{\rho}})^2\Gamma^2 + ...)\Gamma\bar{\delta}_t \tag{F.18}$$

$$= v_{i,0} + \eta\delta_{i,t} + \eta\mathbf{e}_i^\top \frac{\beta}{\hat{\rho}}(I - \frac{\beta}{\hat{\rho}}\Gamma)^{-1}\Gamma\bar{\delta}_t, \tag{F.19}$$

where the constant η is defined as

$$\eta = \frac{\zeta}{\hat{\rho}}. \tag{F.20}$$

From the valuation of firm i given by

$$V_{i,t} = v_{i,0} + \eta\delta_{i,t} + \eta\frac{\beta}{\hat{\rho}}\mathbf{e}_i^\top \sum_{k=0}^{\infty}\left(\frac{\beta}{\hat{\rho}}\Gamma\right)^k \left(\Gamma\bar{\delta}_t\right), \tag{F.21}$$

we derive the network-augmented Gordon growth formula:

$$V_{i,t} = v_{i,0} + \eta\delta_{i,t} + \eta\frac{\beta}{\hat{\rho}}\mathbf{e}_i^\top \left(\mathbf{I} + \frac{\beta}{\hat{\rho}}\Gamma + (\frac{\beta}{\hat{\rho}})^2\Gamma^2 + ...\right)\Gamma\bar{\delta}_t \tag{F.22}$$

$$= v_{i,0} + \eta\mathbf{e}_i^\top\mathbf{I}\bar{\delta}_t + \eta\frac{\beta}{\hat{\rho}}\mathbf{e}_i^\top \left(\Gamma + \frac{\beta}{\hat{\rho}}\Gamma^2 + (\frac{\beta}{\hat{\rho}})^2\Gamma^3 + ...\right)\bar{\delta}_t \tag{F.23}$$

$$= v_{i,0} + \eta\mathbf{e}_i^\top \left(\mathbf{I} + \frac{\beta}{\hat{\rho}}\Gamma + (\frac{\beta}{\hat{\rho}})^2\Gamma^2 + (\frac{\beta}{\hat{\rho}})^3\Gamma^3 + ...\right)\bar{\delta}_t \tag{F.24}$$

$$= v_{i,0} + \zeta\mathbf{e}_i^\top(\hat{\rho}\mathbf{I} - \beta\Gamma)^{-1}\bar{\delta}_t. \tag{F.25}$$

F.3 Proof of Corollary 1

Define $R_{i,t}$ as the undiscounted cumulative return of firm i. We have

$$dR_{i,t} = v_i^\top \frac{d\bar{\delta}_t}{V_{i,t}} = \mathbf{e}_i^\top(I - \frac{\beta}{\hat{\rho}}\Gamma)^{-1}\frac{\zeta}{\hat{\rho}}\frac{d\bar{\delta}_t}{V_{i,t}} = \frac{\zeta}{\hat{\rho}}\sum_{n=1}^{N}\mathbf{e}_i^\top \left(I - \frac{\beta}{\hat{\rho}}\Gamma\right)^{-1}\mathbf{e}_n\frac{d\delta_{n,t}}{V_{i,t}}. \tag{F.26}$$

We are interested in the correlation between $dR_{i,t}, dR_{j,t}$, that is

$$\rho_{ij} = \text{corr}(dR_{i,t}, dR_{j,t}) = \frac{\text{cov}(dR_{i,t}, dR_{j,t})}{\sqrt{\text{var}(dR_{i,t})\text{var}(dR_{j,t})}}. \tag{F.27}$$

In calculating covariance, drift terms vanish, and only the diffusion terms contribute. The diffusion term for firm i is $\sigma_{i,\delta}\delta_{i,t}dz_{i,t}$, where $dz_{i,t}$ are i.i.d. stochastic increments with $\text{cov}(dz_{i,t}, dz_{j,t}) = 0$ for $i \neq j$.

Denote $\mathbf{M} = (I - \frac{\beta}{\hat{\rho}}\Gamma)^{-1}$, then $\mathbf{M}$ is a matrix of size $N \times N$. We also denote $\mathbf{M}_i$ as a vector

consisting of i-th row elements of $\mathbf{M}$. That is, $\mathbf{M}_i$ is given by:

$$\mathbf{M}_i = \begin{pmatrix} M_{i,1} \\ M_{i,2} \\ \cdots \\ M_{i,N} \end{pmatrix},$$

where $M_{i,j}$ is the element in the i-th row and j-th column of $\mathbf{M}$. Therefore,

$$\text{corr}(dR_{i,t}, dR_{j,t}) = \frac{\text{cov}\left(\sum_{n=1}^{N} \mathbf{e}_i^\top \mathbf{M} \mathbf{e}_n \frac{d\delta_{n,t}}{V_{i,t}}, \sum_{n=1}^{N} \mathbf{e}_j^\top \mathbf{M} \mathbf{e}_n \frac{d\delta_{n,t}}{V_{j,t}}\right)}{\sqrt{\text{var}\left(\sum_{n=1}^{N} \mathbf{e}_i^\top \mathbf{M} \mathbf{e}_n \frac{d\delta_{n,t}}{V_{i,t}}\right) \text{var}\left(\sum_{n=1}^{N} \mathbf{e}_j^\top \mathbf{M} \mathbf{e}_n \frac{d\delta_{n,t}}{V_{j,t}}\right)}} \tag{F.28}$$

$$= \frac{\text{cov}\left(\sum_{n=1}^{N} \mathbf{e}_i^\top \mathbf{M} \mathbf{e}_n \sigma_{n,\delta} \delta_{n,t} dz_{n,t}, \sum_{n=1}^{N} \mathbf{e}_j^\top \mathbf{M} \mathbf{e}_n \sigma_{n,\delta} \delta_{n,t} dz_{n,t}\right)}{\sqrt{\text{var}\left(\sum_{n=1}^{N} \mathbf{e}_i^\top \mathbf{M} \mathbf{e}_n \sigma_{n,\delta} \delta_{n,t} dz_{n,t}\right) \text{var}\left(\sum_{n=1}^{N} \mathbf{e}_j^\top \mathbf{M} \mathbf{e}_n \sigma_{n,\delta} \delta_{n,t} dz_{n,t}\right)}}. \tag{F.29}$$

To further simplify it, we define $\mathbf{dz}$ as the vector of $\sigma_{n,\delta} \delta_{n,t} dz_{n,t}$:

$$\mathbf{dz} = \begin{pmatrix} \sigma_{1,\delta} \delta_{1,t} dz_{1,t} \\ \sigma_{2,\delta} \delta_{2,t} dz_{2,t} \\ \vdots \\ \sigma_{N,\delta} \delta_{N,t} dz_{N,t} \end{pmatrix}.$$

Thus, the summation terms in the numerator of (F.29) become:

$$\sum_{n=1}^{N} \mathbf{e}_i^\top \mathbf{M} \mathbf{e}_n \sigma_{n,\delta} \delta_{n,t} dz_{n,t} = \mathbf{M}_i^\top \mathbf{dz}, \quad \sum_{n=1}^{N} \mathbf{e}_j^\top \mathbf{M} \mathbf{e}_n \sigma_{n,\delta} \delta_{n,t} dz_{n,t} = \mathbf{M}_j^\top \mathbf{dz}.$$

Substitute the vector definitions into the covariance expression:

$$\text{cov}(\mathbf{M}_i^\top \mathbf{dz}, \mathbf{M}_j^\top \mathbf{dz}) = \mathbb{E}[(\mathbf{M}_i^\top \mathbf{dz})(\mathbf{M}_j^\top \mathbf{dz})] - \mathbb{E}[\mathbf{M}_i^\top \mathbf{dz}] \mathbb{E}[\mathbf{M}_j^\top \mathbf{dz}].$$

Since $\mathbf{dz}$ is a random vector with zero mean, we can express the covariance as:

$$\text{cov}(\mathbf{M}_i^\top \mathbf{dz}, \mathbf{M}_j^\top \mathbf{dz}) = \mathbf{M}_i^\top \text{cov}(\mathbf{dz}) \mathbf{M}_j = \mathbf{M}_i^\top \Sigma_z \mathbf{M}_j dt,$$

where $\Sigma_z dt = \text{cov}(\mathbf{dz})$ is the covariance matrix of $\mathbf{dz}$. And because $dz_{i,t}$ is i.i.d., Σ_z is a diagonal matrix,

$$\Sigma_z = \begin{pmatrix} c_1^2 & 0 & 0 & \cdots & 0 \\ 0 & c_2^2 & 0 & \cdots & 0 \\ 0 & 0 & c_3^2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & c_N^2 \end{pmatrix},$$

with each diagonal element c_i defined as:

$$c_i^2 = \sigma_{i,\delta}^2 \delta_{i,t}^2, \quad i = 1, 2, \ldots, N.$$

Similarly, the variance terms in the denominator of (F.29) can be simplified as:

$$\text{var}(\mathbf{M}_i^\top \mathbf{dz}) = \mathbf{M}_i^\top \Sigma_z \mathbf{M}_i dt,$$

$$\text{var}(\mathbf{M}_j^\top \mathbf{dz}) = \mathbf{M}_j^\top \Sigma_z \mathbf{M}_j dt.$$

Substitute the covariance and variance expressions into the correlation term:

$$\rho_{ij} = \text{corr}(dR_{i,t}, dR_{j,t}) = \frac{\mathbf{M}_i^\top \Sigma_z \mathbf{M}_j}{\sqrt{\mathbf{M}_i^\top \Sigma_z \mathbf{M}_i \cdot \mathbf{M}_j^\top \Sigma_z \mathbf{M}_j}}.$$

To proceed further we define the following quantity:

$$\|\mathbf{M}_i\| = \sqrt{\mathbf{M}_i^\top \Sigma_z \mathbf{M}_i} = \sqrt{\sum_{n=1}^{N} M_{i,n}^2 c_n^2}.$$

Then the correlation can be expressed as

$$\rho_{ij} = \frac{\mathbf{M}_i^\top \Sigma_z \mathbf{M}_j}{\|\mathbf{M}_i\| \|\mathbf{M}_j\|} \tag{F.30}$$

Next, we characterize the derivative $\frac{\partial \rho_{ij}}{\partial \gamma_{ij}}$. We use the quotient rule, differentiating the numerator and the denominator separately.

First note that, from matrix calculus, for $\mathbf{M} = (I - \frac{\beta}{\hat{\rho}}\Gamma)^{-1}$:

$$\frac{\partial \mathbf{M}}{\partial \gamma_{ij}} = \frac{\beta}{\hat{\rho}} \mathbf{M} E_{ij} \mathbf{M}, \tag{F.31}$$

where E_{ij} is the elementary matrix with 1 at (i, j) and 0 otherwise. This implies, for any $k, l \in \{1, 2, ..., N\}$:

$$\frac{\partial M_{k,l}}{\partial \gamma_{ij}} = \frac{\beta}{\hat{\rho}} (\mathbf{M} E_{ij} \mathbf{M})_{k,l} = \frac{\beta}{\hat{\rho}} M_{k,i} M_{j,l}. \tag{F.32}$$

Numerator. Now let us first focus on the numerator:

$$S = \mathbf{M}_i^\top \Sigma_z \mathbf{M}_j = \sum_n c_n^2 M_{i,n} M_{j,n}.$$

Expanding its derivative with respect to γ_{ij} using the product rule:

$$\frac{\partial S}{\partial \gamma_{ij}} = \sum_n \left(\frac{\partial M_{i,n}}{\partial \gamma_{ij}} M_{j,n} c_n^2 + c_n^2 M_{i,n} \frac{\partial M_{j,n}}{\partial \gamma_{ij}} \right).$$

We use (F.32) to calculate the derivatives:

$$\frac{\partial M_{i,n}}{\partial \gamma_{ij}} = \frac{\beta}{\hat{\rho}} M_{i,i} M_{j,n}, \tag{F.33}$$

$$\frac{\partial M_{j,n}}{\partial \gamma_{ij}} = \frac{\beta}{\hat{\rho}} M_{j,i} M_{j,n}. \tag{F.34}$$

Substituting these derivatives gives:

$$\frac{\partial S}{\partial \gamma_{ij}} = \sum_n \frac{\beta}{\hat{\rho}} c_n^2 \left(M_{i,i} M_{j,n}^2 + M_{j,i} M_{i,n} M_{j,n} \right).$$

Using the definition of $\|\mathbf{M}_j\|^2$ and S:

$$\frac{\partial S}{\partial \gamma_{ij}} = \frac{\beta}{\hat{\rho}} \left(M_{i,i} \|\mathbf{M}_j\|^2 + M_{j,i} S \right).$$

Denominator. We next calculate

$$\frac{\partial}{\partial \gamma_{ij}} (\|\mathbf{M}_i\| \|\mathbf{M}_j\|) = \|\mathbf{M}_j\| \frac{\partial \|\mathbf{M}_i\|}{\partial \gamma_{ij}} + \|\mathbf{M}_i\| \frac{\partial \|\mathbf{M}_j\|}{\partial \gamma_{ij}}.$$

Since $\|\mathbf{M}_i\| = \sqrt{\sum_n c_n^2 M_{i,n}^2}$, we have:

$$\frac{\partial \|\mathbf{M}_i\|}{\partial \gamma_{ij}} = \frac{1}{2\|\mathbf{M}_i\|} \frac{\partial}{\partial \gamma_{ij}} \left(\sum_n c_n^2 M_{i,n}^2 \right).$$

The derivative of the summation term is:

$$\frac{\partial}{\partial \gamma_{ij}} \left(\sum_n c_n^2 M_{i,n}^2 \right) = 2 \sum_n c_n^2 M_{i,n} \frac{\partial M_{i,n}}{\partial \gamma_{ij}}.$$

Substitute $\frac{\partial M_{i,n}}{\partial \gamma_{ij}} = \frac{\beta}{\hat{\rho}} M_{i,i} M_{j,n}$:

$$\frac{\partial \|\mathbf{M}_i\|}{\partial \gamma_{ij}} = \frac{\beta}{\hat{\rho}} \frac{M_{i,i}}{\|\mathbf{M}_i\|} \sum_n c_n^2 M_{i,n} M_{j,n} = \frac{\beta}{\hat{\rho}} \frac{M_{i,i} S}{\|\mathbf{M}_i\|}.$$

Similarly:

$$\frac{\partial \|\mathbf{M}_j\|}{\partial \gamma_{ij}} = \frac{1}{2\|\mathbf{M}_j\|} \frac{\partial}{\partial \gamma_{ij}} \left(\sum_n c_n^2 M_{j,n}^2 \right).$$

The derivative of the summation term is:

$$\frac{\partial}{\partial \gamma_{ij}} \left(\sum_n c_n^2 M_{j,n}^2 \right) = 2 \sum_n c_n^2 M_{j,n} \frac{\partial M_{j,n}}{\partial \gamma_{ij}}.$$

Substitute $\frac{\partial M_{j,n}}{\partial \gamma_{ij}} = \frac{\beta}{\hat{\rho}} M_{j,i} M_{j,n}$:

$$\frac{\partial \|\mathbf{M}_j\|}{\partial \gamma_{ij}} = \frac{\frac{\beta}{\hat{\rho}} M_{j,i}}{\|\mathbf{M}_j\|} \sum_n c_n^2 M_{j,n} M_{j,n} = \frac{\beta}{\hat{\rho}} M_{j,i} \|\mathbf{M}_j\|.$$

Therefore

$$\frac{\partial}{\partial \gamma_{ij}} (\|\mathbf{M}_i\| \|\mathbf{M}_j\|) = \frac{\beta}{\hat{\rho}} \left(\frac{M_{i,i} S \|\mathbf{M}_j\|}{\|\mathbf{M}_i\|} + M_{j,i} \|\mathbf{M}_i\| \|\mathbf{M}_j\| \right).$$

Substituting these results:

$$\frac{\partial \rho_{ij}}{\partial \gamma_{ij}} = \frac{\beta}{\hat{\rho}} \frac{\left(M_{i,i} \|\mathbf{M}_j\|^2 + M_{j,i} S\right) \|\mathbf{M}_i\| \|\mathbf{M}_j\| - S \cdot \left(\frac{M_{i,i} S \|\mathbf{M}_j\|}{\|\mathbf{M}_i\|} + M_{j,i} \|\mathbf{M}_i\| \|\mathbf{M}_j\| \right)}{\|\mathbf{M}_i\|^2 \|\mathbf{M}_j\|^2}.$$

Simplifying the numerator and factoring out $M_{i,i} \|\mathbf{M}_j\| / \|\mathbf{M}_i\|$, the numerator becomes:

$$\frac{M_{i,i} \|\mathbf{M}_j\|}{\|\mathbf{M}_i\|} \left(\|\mathbf{M}_i\|^2 \|\mathbf{M}_j\|^2 - S^2 \right).$$

Thus, the final expression for the derivative is:

$$\frac{\partial \rho_{ij}}{\partial \gamma_{ij}} = \frac{\beta}{\hat{\rho}} \frac{M_{i,i} \left(\|\mathbf{M}_j\|^2 \|\mathbf{M}_i\|^2 - S^2 \right)}{\|\mathbf{M}_i\|^3 \|\mathbf{M}_j\|}.$$

Since S is an inner product, we can apply the Cauchy-Schwarz inequality:

$$\left(\sum_{n=1}^{N} c_n^2 M_{i,n} M_{j,n} \right)^2 \leq \left(\sum_{n=1}^{N} c_n^2 M_{i,n}^2 \right) \left(\sum_{n=1}^{N} c_n^2 M_{j,n}^2 \right),$$

which simplifies to:

$$S^2 \leq \|\mathbf{M}_i\|^2 \|\mathbf{M}_j\|^2.$$

Thus, we obtain:

$$\|\mathbf{M}_i\|^2 \|\mathbf{M}_j\|^2 - S^2 \geq 0,$$

which implies:

$$\frac{\partial \rho_{ij}}{\partial \gamma_{ij}} > 0.$$

The inequality holds strictly because $\mathbf{M}_i$ and $\mathbf{M}_j$ are not collinear. Moreover, we see that when $\beta = 0$, the network effect disappears entirely. As β increases, it amplifies the overall network influence.

F.4 Proof of Proposition 3

For firm i, substituting the functional form of its valuation (F.4) into the HJB equation, then taking the FOC with respect to $x_{i,t}$, we get

$$x_{i,t} = \frac{\phi_0}{\phi_1} D_{i,t} - \frac{\zeta}{\theta v_{i,i}}, \tag{F.35}$$

where $v_{i,i}$ is the i-th element of vector v_i. $\phi_0 > 0, \phi_1 > 0$, so $x_{i,t}$ is increasing with $D_{i,t}$. And v_i is solved from (F.16). Recall that the law of motion of data capital is given by

$$d\delta_{i,t} = \left(\theta x_{i,t} + \alpha\theta \frac{\kappa}{\sigma_\eta} D_{i,t} + \mu_\delta \delta_{i,t}\right) dt + \sigma_{i,\delta}\delta_{i,t} dz_{i,t}. \tag{F.36}$$

Consequently, taking the differential of firm i's product-design choice $x_{i,t}$, we obtain

$$dx_{i,t} = \frac{\phi_0}{\phi_1}\left[\sum_{j=1}^{N} \gamma_{ij}\xi \left((\theta x_{j,t} + \alpha\theta \frac{\kappa}{\sigma_\eta} D_{j,t} + \mu_\delta \delta_{j,t})dt + \sigma_{j,\delta}\delta_{j,t} dz_{j,t}\right)\right]. \tag{F.37}$$

Therefore, for $\gamma_{ij} > 0$, we obtain

$$\frac{\partial^2 \left(\mathbb{E}_t[dx_{i,t}]/dt\right)}{\partial x_{j,t}\, \partial \xi} = \frac{\phi_0}{\phi_1}\, \gamma_{ij}\, \theta > 0.$$

F.5 Proof of Proposition 4

Recall that in Appendix F.2 we showed the firm's valuation takes the following form

$$V_{i,t} \equiv V^i(\bar{\delta}_t) = v_{i,0} + \zeta\, e_i^\top \left(\hat{\rho} I - \beta\Gamma\right)^{-1} \bar{\delta}_t.$$

Hence

$$V^i_{\delta_{i,t}}(\bar{\delta}_t) := \frac{\partial V^i(\bar{\delta}_t)}{\partial \delta_{i,t}} = \zeta\, \left[(\hat{\rho} I - \beta\Gamma)^{-1}\right]_{ii}. \tag{F.38}$$

Since

$$\beta = \theta\Big(\frac{\phi_0}{\phi_1} + \frac{\alpha\kappa}{\sigma_\eta}\Big)\,\xi, \quad \frac{\partial \beta}{\partial \xi} > 0,$$

differentiating (F.38):

$$\frac{\partial V^i_{\delta_{i,t}}(\bar{\delta}_t)}{\partial \xi} = \zeta\, \Big[(\hat{\rho} I - \beta\Gamma)^{-1}\Gamma(\hat{\rho} I - \beta\Gamma)^{-1}\Big]_{ii} \frac{\partial \beta}{\partial \xi} > 0.$$

Thus, a decrease in ξ lowers $V^i_{\delta_{i,t}}(\bar{\delta}_t)$. Recall from (12) we have

$$-g_x\big(x_{i,t}, \lambda_{i,t}\big) = \theta\, V^i_{\delta_{i,t}}(\bar{\delta}_t), \quad g_x < 0.$$

By the implicit-function theorem, $\partial x_{i,t}/\partial V^i_{\bar{\delta}_{i,t}}(\bar{\delta}_t) > 0$. Therefore, a lower $V^i_{\bar{\delta}_{i,t}}(\bar{\delta}_t)$ implies a lower $x_{i,t}$. For $\gamma_{ij} > 0$, the dynamics of (F.37) we have

$$\frac{\partial\left(\mathbb{E}_t[dx_{i,t}]/dt\right)}{\partial x_{j,t}} = \frac{\phi_0}{\phi_1}\,\gamma_{ij}\,\xi\,\theta, \qquad \frac{\partial^2\left(\mathbb{E}_t[dx_{i,t}]/dt\right)}{\partial x_{j,t}\,\partial \xi} = \frac{\phi_0}{\phi_1}\,\gamma_{ij}\,\theta > 0.$$